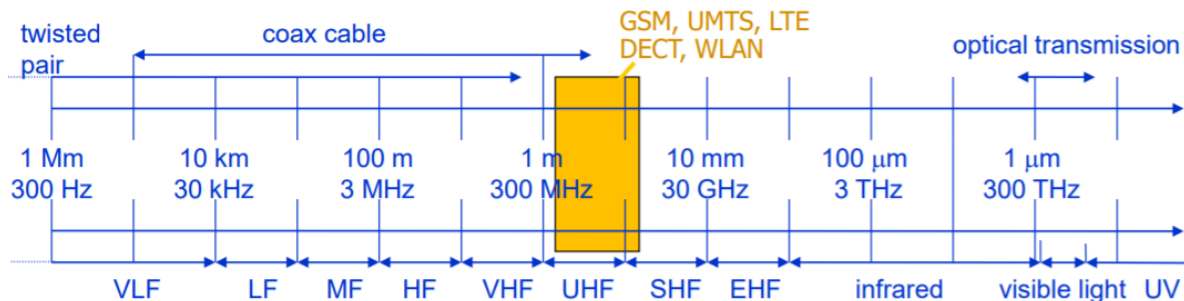


1. Wireless Transmission

1.1. Radio Frequencies and Signals

1.1.1. What are the wavelengths of the different band used by GSM and WLAN?



UHF = Ultra High Frequency: between 300MHz and 3GHz (1m - 10sm) Most of mobile communications take place here.

WLAN at 2,4 GHz band and 5 GHz band -12,5 cm and 6cm

GSM 1800 MHz band 16,7 cm, GSM 900 MHz band? 33,3 cm

What is the wavelength of the GSM 1800 MHz band? 16,7 cm

What is the wavelength of the GSM 900 MHz band? 33,3 cm

What is the wavelength of WLAN at 2,4 band? 12,5 cm

What is the wavelength of WLAN at 5 GHz band? 6 cm

Relation between frequency f and wave length λ : $\lambda = c / f$, with speed of light $c \cong 3 \times 10^8$ m/s
Important: the wave length highly influences the size of antennas!

1.1.2. Which bands typically provide better penetration of walls?

VHF/UHF-ranges for mobile radio (<3GHz) :

- simple & small antennas
- good propagation characteristics (limited reflections, small path loss, good penetration of walls)

The smaller the frequency, the better is the penetration of walls.

1.1.3. What's the disadvantage of using higher frequencies for communications? What's the advantage?

disadvantage: Limitations due to absorption by water and oxygen molecules (resonance frequencies) resulting in weather dependent fading, signal loss caused by heavy rainfall etc.

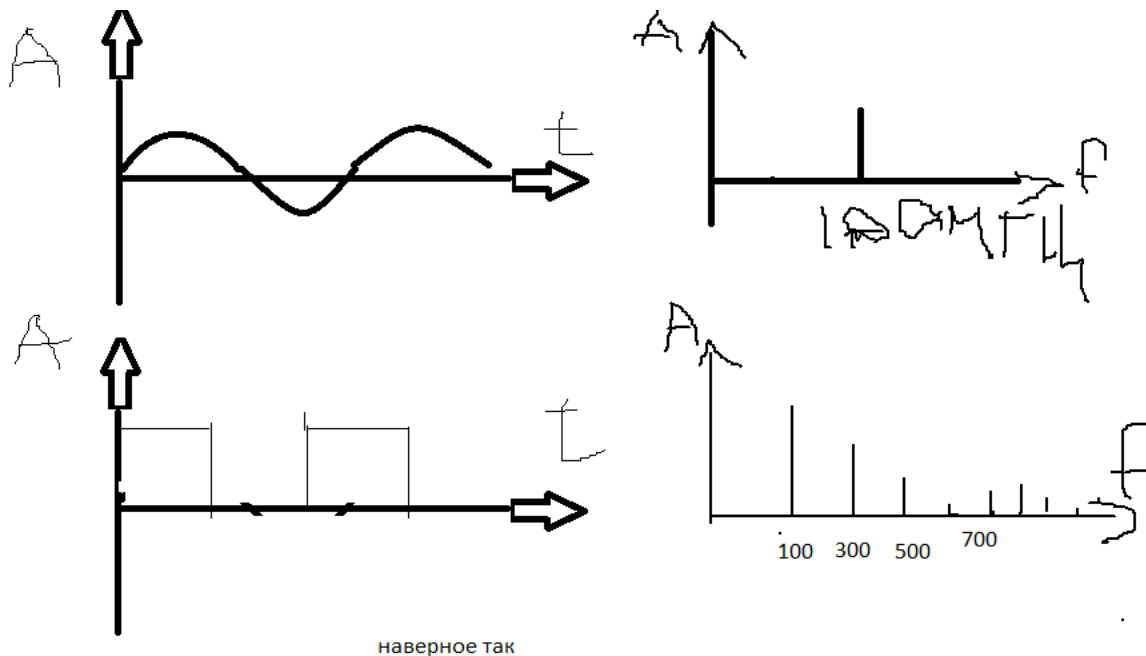
The closer we are to visible light, the more problems with obstacles arise.

advantage: VHF/UHF (very (30MHz-300MHz)/ultra (300MHz-3GHz) high frequency)

- ranges for mobile radio
- simple, small antennas

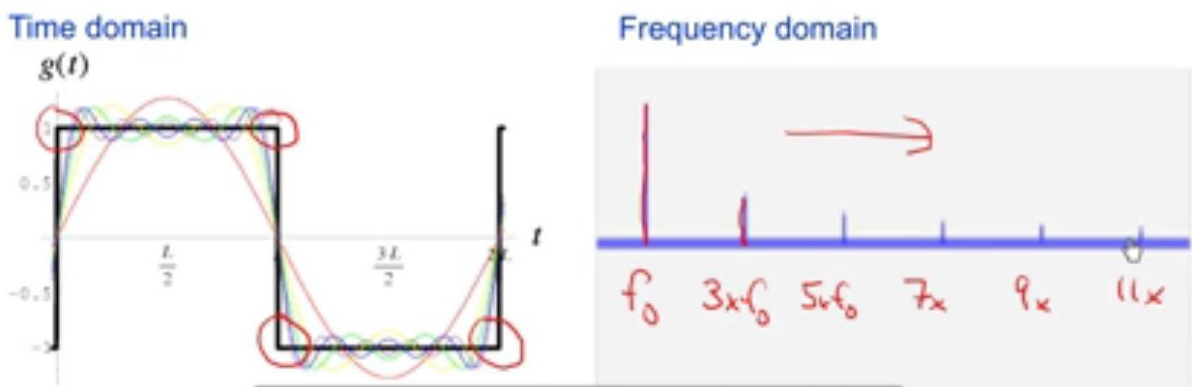
- good propagation characteristics (limited reflections, small path loss, penetration of walls)
- SHF (super high frequency (3GHz-30GHz)) and higher for directed radio links, satellite communication
 - small antenna, strong focus (size of antenna is proportional to the wavelength)
 - larger bandwidth available
 - no penetration of walls

1.1.4. Sketch the frequency spectrum of a 100 MHz sinus signal and of a symmetric rectangular signal with the same period.



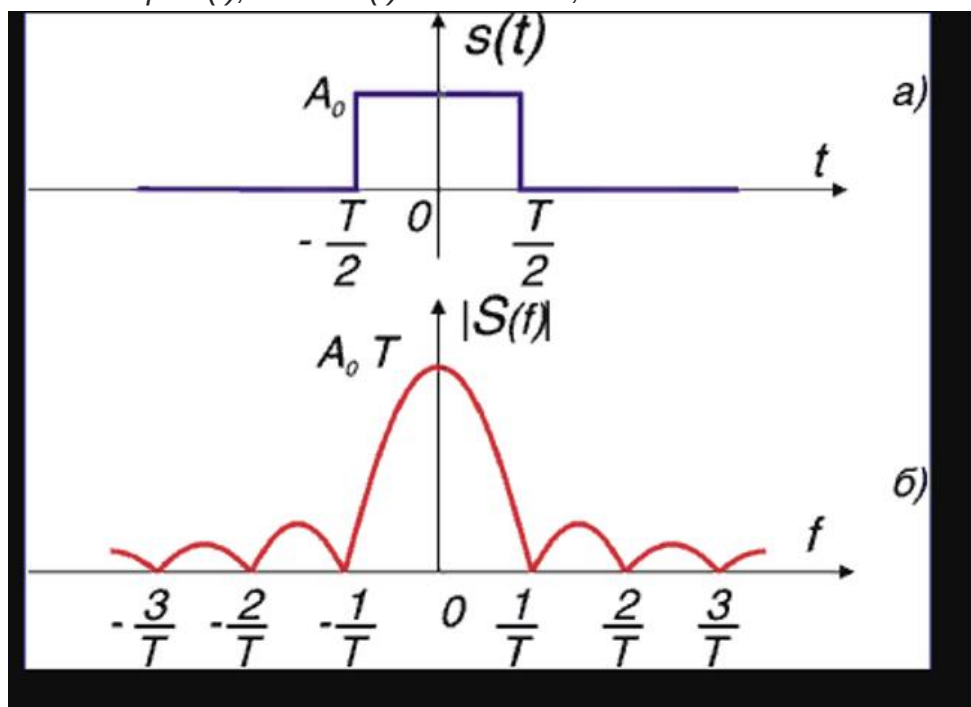
наверное так

Below there's a symmetric **rectangular signal** - it consists of many harmonics. For a perfect rectangle, harmonics will go up to infinity. Harmonics means we have a base frequency, and then it's divided by 3,5,7...(2nd picture below)



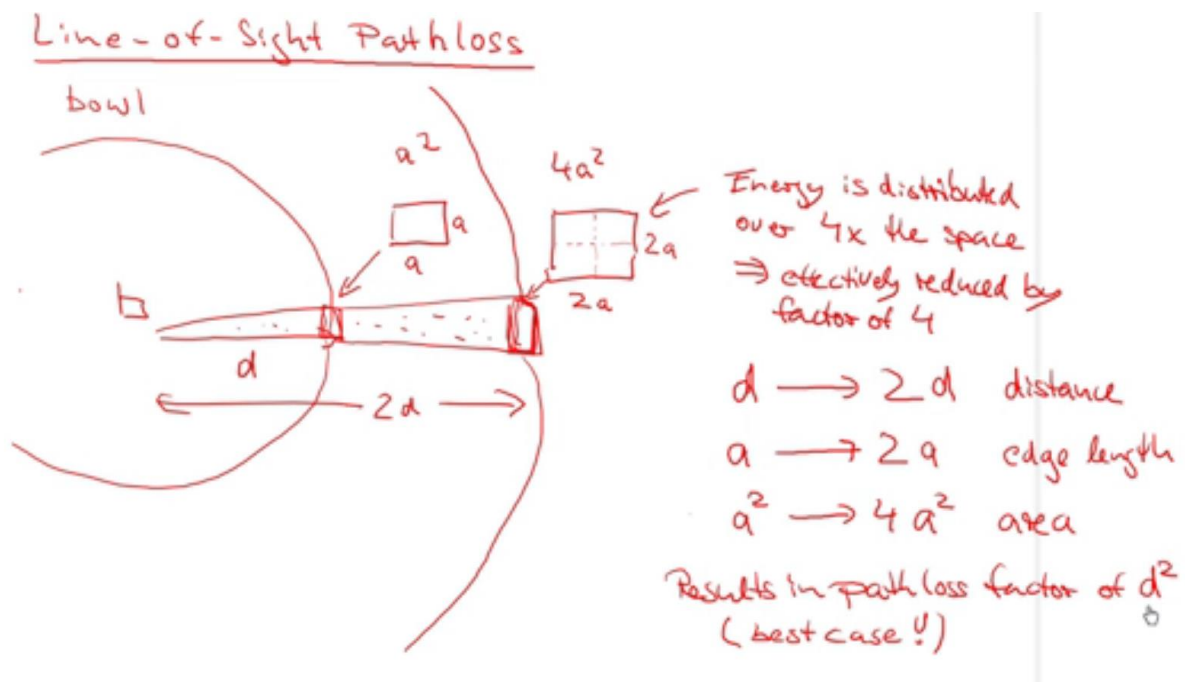
1.1.5. - How does a rectangular signal look in the frequency domain?

When we have a signal limited in duration, such as the rectangular pulse shown at the right, the frequency domain representation is no longer a delta function, but broadens into a continuous, limited spectrum. (The signal in frequency space is called a spectrum by analogy with an optical spectrum that spreads out the wavelengths.) Note the reciprocal relation: the narrower the pulse in the time domain, the wider the spectrum in the frequency domain. The area of the pulse, here unity, is also the height of the spectrum at zero frequency, another useful fact. The rectangular pulse is represented as the sum of two unit steps $u(t)$, where $u(t) = 0$ for $t < 0$, and $= 1$ for $t > 0$.



1.2. Signal propagation

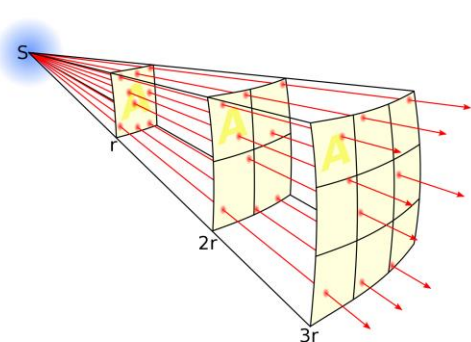
1.2.1. Why is a pathloss proportional to d^2 is assumed for line-of-sight connections? Why is d^2 considered as the best case?



Assuming an isotropic radiator and zero absorption in space, the sum of the energy at any radius of the bowl can be assumed the same independent of the size of the surface of the bowl itself. However, as the surface of the bowl is proportional to the square of its radius, the energy per any fixed area reduces proportional to the inverse squared distance.

or

Line-of-sight is an ideal case - we assume there's no loss over distance (the energy conservation law)



The density of signal is inversely proportional to the square of the distance from the source because the surface area of a sphere increases with the square of the radius. Thus the field intensity is inversely proportional to the square of the distance from the source.

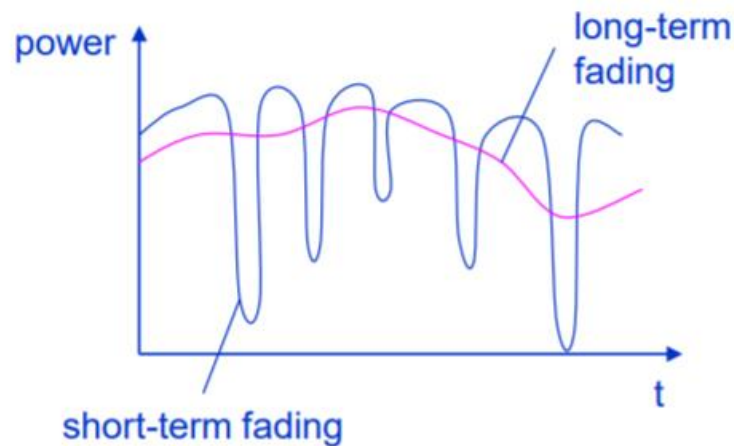
1.2.2. What are the reasons for fast fading? What is exactly happening?

Multiple transmission paths from a single sender result in overlap of the signal at the receiver

Due to mobility, channel characteristics change over time and location

- signal paths change
- different delay variations of different signal parts (frequencies)
- different phases of signal parts

results in quick changes in the power received (called short-term fading or fast fading)



// из тестов

Fast fading is going to happen in cases of Mobility of receiver or sender Mobility of objects in the transmission path. It is caused because of multiple transmission paths between sender and receiver. The higher is the number of paths, the faster is the fading.

What is happens during fast fading is that multiple transmission paths from a single sender result in overlap of the signal at the receiver

1.2.3. Why does fast fading depend on mobility? Why is fast fading frequency dependent?

Because fast fading is due to a shift in the phases between multiple paths which requires some dynamic behavior

Because the exact phase depends on the frequency -из тестов

Fast fading varies quickly with the frequency. Fast fading originates due to effects of constructive and destructive interference patterns which is caused due to multipath.

Signal can take many different paths between sender and receiver due to reflection, scattering, diffraction.

Time dispersion:

- signal is dispersed over time
- interference with "neighbor" symbols, Inter Symbol Interference (ISI)

The signal reaches a receiver directly and phase shifted

- distorted signal depending on the phases of the different parts

Effects of mobility – Fading

Channel characteristics change over time and location

- signal paths change
- different delay variations of different signal parts (frequencies) - different phases of signal parts
- quick changes in the power received. (fast fading)

1.2.4. What is the difference between fading due to simple phase shifts between arriving signals and inter symbol interference?

Two signals with different phases are summed. Worst case - 180 shift - the signal is cancelled out. That's the reason for fast fading. It can happen due to moving your head etc.

inter symbol interference happens due to interference with “neighbor” symbols (bouncing off walls => losing some energy => new path created). In a wireless channel, a signal can travel from the transmitter to the receiver via multiple delayed paths (echoes). If the delays between paths are large compared to a symbol period, the energy carried by these echoes can cause significant ISI to the succeeding symbols.

1.3. What are the factors of the received signal important for error-free reception?

good ratio of SINR?
good power of signal

or

There are two requirements for a successful transmission:

- the signal to interference & noise ratio SINR is above a threshold (this depends on the radio technology (modulation, coding), HW and signal processing capabilities of the receiver)
- the received signal strength S is above a threshold (this is defined by the sensitivity of the receiver)

1. What are the two major factors of the received signal that limit the quality of the reception?

1. Range limited systems (lack of coverage)

- Mobile stations located far away from BS (at cell border or even beyond the coverage zone)
- S at the receiver is too low (below receiver sensitivity) because the path loss between sender and receiver is too high
- S is too low
- No signal reception possible

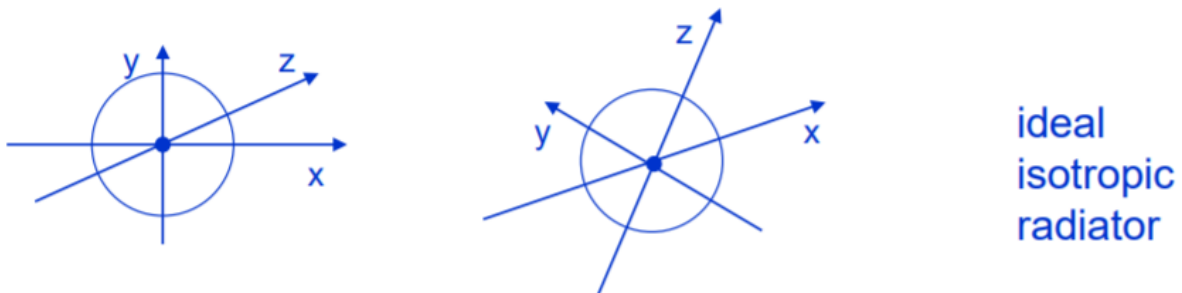
2. Interference limited systems (lack of capacity)

- Mobile station is within coverage zone
- S is sufficient, but too much interference I at the receiver
- SINR is too low
- No more resources / capacity left

1.4. Antennas

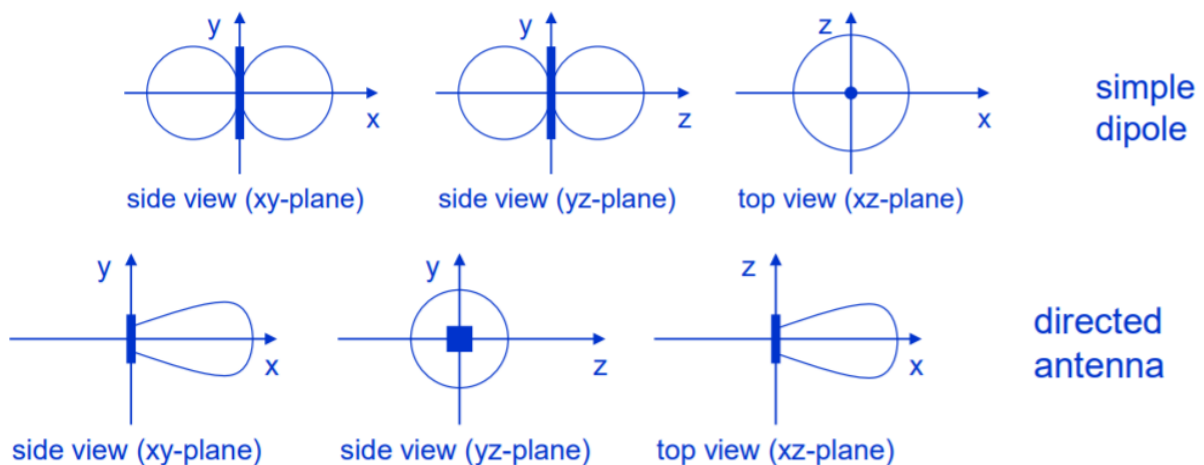
1.4.1. What are the characteristics of an isotropic radiator?

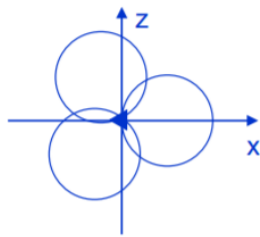
Emits same power in all directions, Is used as a reference for other antenna systems, Power is equally distributed in all three dimensions



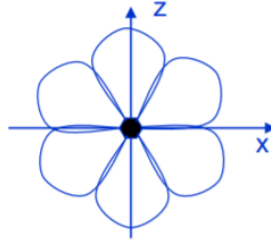
1.4.2. What directional effects show simple practical antennas?

Example: Radiation pattern of a simple Hertzian dipole





top view, 3 sector



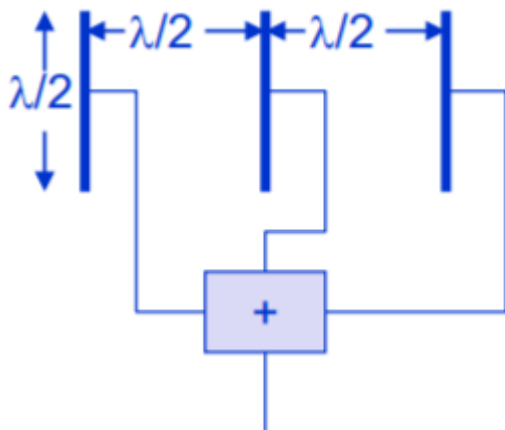
top view, 6 sector

sectorized
antenna

Antenna gain: maximum power in the direction of the main lobe compared to the power of an isotropic radiator (with the same average power)

1.4.3. How can you build a directional antenna from 2 simple $\lambda/2$ antennas? What are the prerequisites?

Separate the antenna by $\lambda/2$ and provide the same signal to both
diversity combining combine output power to produce gain cophasing needed to
avoid cancellation

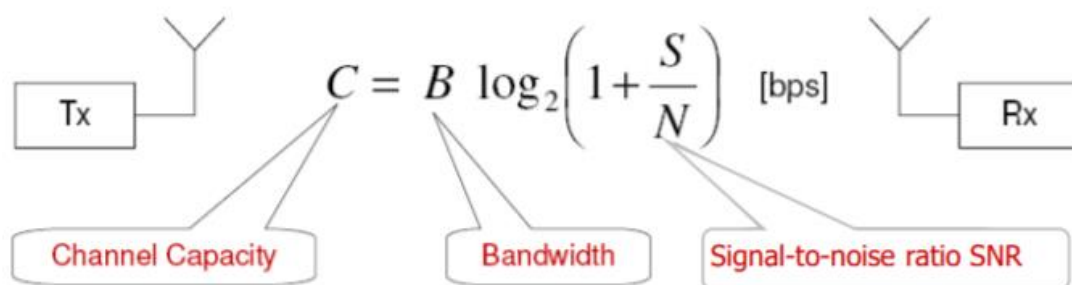


1.5. Channel Capacity – Signal vs. Interference

- 1.5.1. How do interference and the received signal strength determine the capacity of a channel? - How do interference and received signal strength determine the capacity of a channel? Are there approximations for this?

- Shannon (1950):
Channel Capacity $\leq$
Maximum mutual information between
sink and source

C. Shannon
Bell Labs Technical Journal, 1950



- For $S/N \gg 1$ (high signal-to-noise ratio), approximate

$$C \approx B \frac{1}{3} \frac{S}{N_{dB}} \quad [\text{bps}]$$

Observation: Bandwidth and S/N are reciprocal to each other

This means: with low bandwidth very high data rate is possible, provided S/N is high enough e.g. allowing the use of higher order modulation schemes
with high noise (low S/N) data communication is possible if bandwidth is large e.g. employing spread spectrum technology

- How can the spectral efficiency or capacity of a system be improved? Discuss the effects of different approaches. Is D2D communication a solution? Under which circumstances?

With low bandwidth very high data rate is possible provided

- S/N is high enough

- Example: higher order modulation schemes

With high noise (low S/N) data communication is possible if

- bandwidth is large

- Example: spread spectrum

So, we again rely on Shannon rule. We can change bandwidth used, increase the signal strength, or choose more throughput modulation scheme.

In the first case, if we increase bandwidth, we have to buy new frequency resource from the government, this is not always possible.

In the case, if we increase signal, we will have to increase the transmission power, which increase coverage area. If there are also other systems uses the same frequency, the interference level may increase

And in the last case, if we change modulation scheme, we may increase spectral efficiency and throughput, but we must be aware of errors level in channel.

D2D communication may be a solution, under these circumstances:

- *The distance between devices is low*
- *No interference with other Cellular users and D2D pairs*
- *Amount of users in Cell / D2D area is under the given threshold*

1.5.2. Why is interference – or more accurate its absence – as important as the signal itself?

For a high system capacity, the signal strength is as important as the absence of high levels of interference!

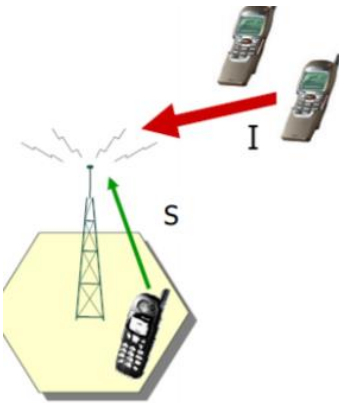
Ratio of Signal-to-Interference-and-Noise power at the receiver

$$SINR = \frac{S}{\sum I_j + N}$$

Because interference can be higher than noise so that noise can be neglected (Shannon's law).

Because interference affects the receiving signal

1.5.3. What defines an interference-limited system? What happens there? How can we increase the system capacity in these systems? And how not?



What defines interference-limited systems? (из тестов)

The addition of new transmitters to the system introduces so much interference to others, that there is no benefit for the overall system capacity; The increase of the transmit power of a single terminal does increase its capacity.

How can we increase the system capacity in these systems? (из тестов)

Add more base stations to reduce the pathloss, Add more sectors to the given base station infrastructure to decrease pathloss while decreasing interference, Add more frequency bands.

Further ideas: Use D2D communication where applicable, i.e. between terminals that are close together, Use interference-cancellation techniques, Use directional antennas at the terminals

Mobile station is within coverage zone

S is sufficient, but too much interference I at the receiver

SINR is too low

No more resources / capacity left

Ratio of Signal-to-Interference-andNoise power at the receiver

$$SINR = \frac{S}{\sum I_j + N}$$

to improve system capacity SINR should be increased or bandwidth

1.6. Multiplexing – Splitting up the Radio Resource

1.6.1. What is the purpose of multiplexing?

Goal: multiple use of a shared medium to provide it to multiple entities rather than just a single one

Separate multiple parallel communications from each other, Separate channels to minimize interference between them

Multiplexing can be used in 4 dimensions

space (si)

time (t)

frequency (f)

code (c)

1.6.2. What kind of multiplexing is used for public FM radio?

Frequency multiplexing, Spatial multiplexing

1.6.3. Which kind of multiplexing is used for GSM?

Frequency multiplexing, Time multiplexing, Spatial multiplexing

1.6.4. What is the problem of multiplexing in time, what is its advantage?

A channel gets the whole spectrum for a certain amount of time

Advantages:

- + Flexible scheme to quickly adapt to the traffic demand,
- + No guard band needed
- + only one carrier in the medium at any time
- + throughput high even for many users

Disadvantages:

- precise synchronization needed

1.7. Modulation and Coding

1.7.1. How can we add information to a periodic signal? What are the choices?

By modulation: Amplitude mod, Phase mod and Frequency mod

1.7.2. Why is amplitude modulation in its pure form rarely used in practice? When does it make sense? Where is the idea applied and what is the benefit of its application?

Highly sensitive to multipath problems, Receivers are highly sensitive to changing channel conditions - из теста

Amplitude Shift Keying (ASK): very simple

low bandwidth requirements

very susceptible to interference

Am is used in Quadrature Amplitude Modulation

(QAM) combines amplitude and phase modulation

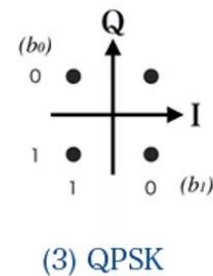
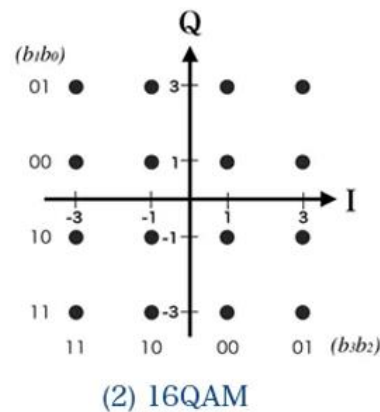
QAM is widely used, e.g. in LTE, GPRS, WLAN

1.7.3. What is the advantage of 16QAM over QPSK?

Преимущество - в QPSK 2 бита на символ, в 16qam - 4, значит, можно впихнуть больше информации в сигнал

Но из-за шума бывают отклонения от идеального значения, и чем больше плотность инфы, тем больше шанс, что что-то отклонится настолько, что будет интерпретировано неправильно, поэтому нужно более высокое S/N ratio

advantage of QPSK over 16-QAM - Lower error rate, Fewer problems with multipath propagation



- 1.7.4. What limits the use of higher order modulation schemes, as e.g. 16QAM or 64QAM?

Highly sensitive to distortions in the amplitude of the signal,
 Highly sensitive to distortions in the phase of the signal,
 Highly sensitive to multipath problems,
 High SINR needed for proper transmission

- 1.7.5. How can we optimize the throughput of a single transmitter-receiver pair? How can we optimize it under given bandwidth constraints as well as fixed signal and interference values?

Adapt modulation scheme, Adapt coding scheme, Use multiple antennas

1.8. Spectrum Spreading

- 1.8.1. What is the benefit of spectrum spreading? What is needed for spreading? *What is the advantage of spectrum spreading?*

Problem of radio transmission: frequency dependent fading can wipe out narrow band signals for duration of the interference

Solution: spread the narrow band signal into a broadband signal using a special code

Benefits: Reduced sensitivity to narrowband interference and noise, coexistence of several signals without dynamic coordination, tap-proof

Requirement: having an available broadband I guess

1.8.2. What is the benefit of frequency hopping? What are its problems?

Distribution of the signal over a wider bandwidth, Reduced sensitivity to narrowband interference and noise

Two versions

Fast Hopping: several frequencies per user bit

Slow Hopping: several user bits per frequency

Advantages

frequency selective fading and interference limited to short period

simple implementation

uses only small portion of spectrum at any time

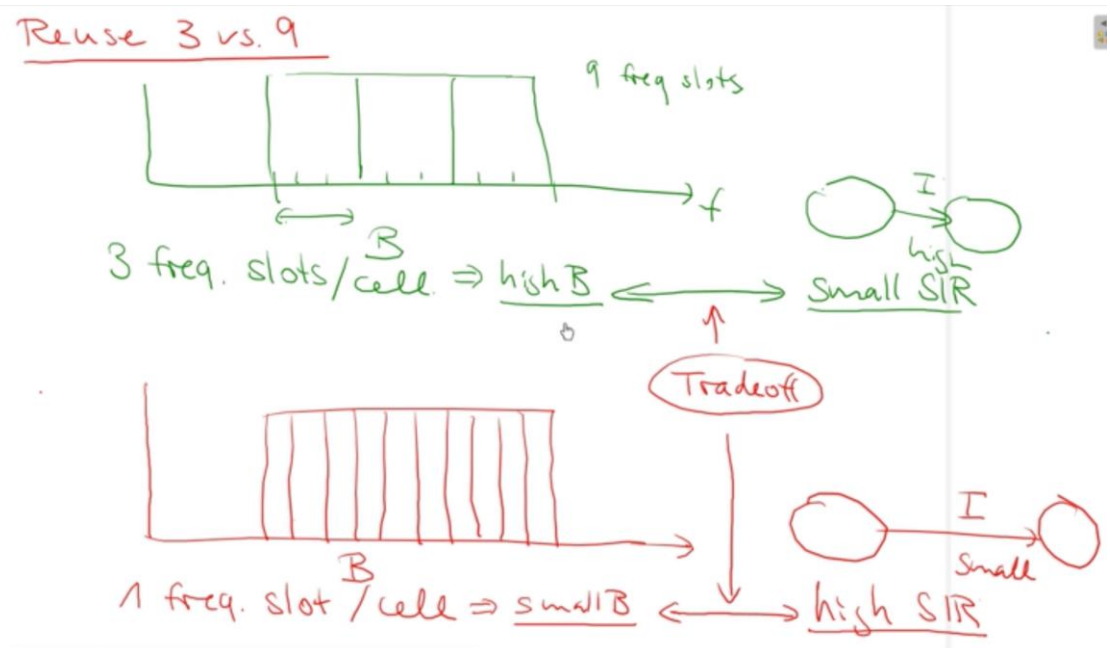
Disadvantages

not as robust as DSSS(Direct Sequence Spread Spectrum))

simpler to detect

1.9. Cellular Networks – Capacity and Interference

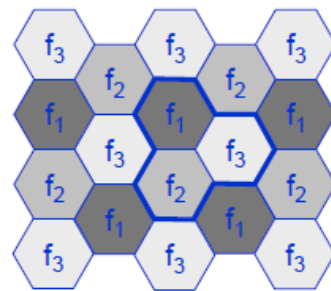
1.9.1. Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate). Use GSM and GPRS/EDGE as example systems



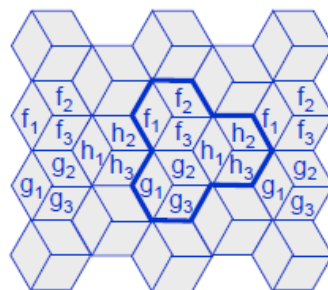
Capacity in case of 3 cell in cluster is less than in the case of 9 cell in cluster due to smaller distance between cell with similar

frequency and respectively that bigger interference between them.

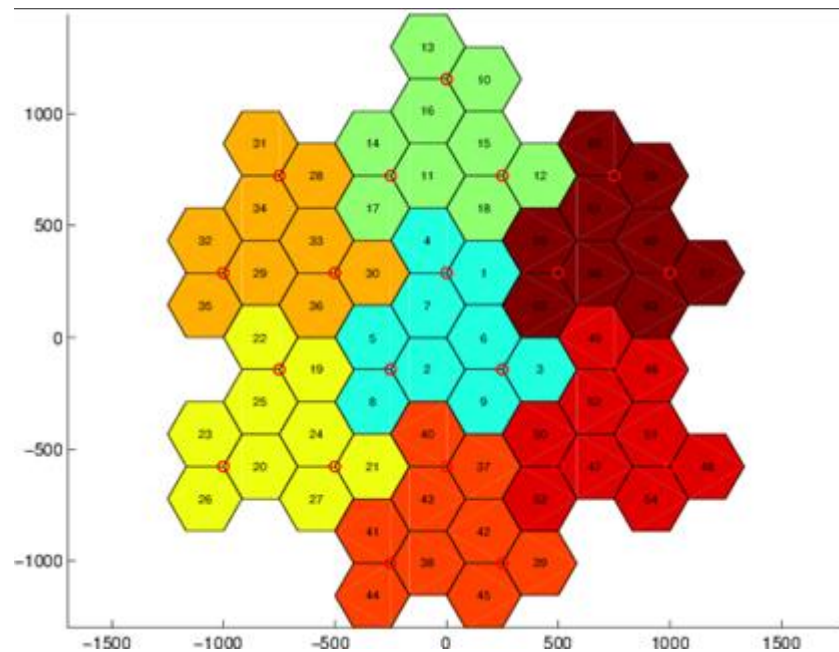
Reuse 3 provides higher raw throughput rate but with higher error rates (so, high bandwidth and low SINR)



3 cell cluster



3 cell cluster
with 3 sector antennas



- 1.9.2. What is the effect of increasing the transmit power of all base stations in interference-limited systems?

Interference gets worse?

Наверное, да. Т.к. у них итак есть интерференция друг на друга, а с увеличением мощности передачи она станет только больше

- 1.9.3. How can we increase the system capacity in interference-limited cellular systems? And how not? Discuss the effects of different approaches.

The correct answers are: Increase the number of base stations, Prefer users at the cell center instead of users at the cell edge, Increase the spatial diversity using multi-sector antennas, Use better receivers that can tolerate a lower signal-to-interference ratio, can't be through increasing the transmit power of the base stations and the users

- 1.9.4. What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available? What are the measures that need to be taken to handle this?

The correct answers are: Adapt the frequency reuse scheme, Adapt the transmit power of base stations, Adapt the transmit power of mobiles

- 1.9.5. 5. What are the effects of increasing the spatial separation, e.g. employing 6 instead of 3 sectors per base station?

Interference gets better?

SINR is higher, but the capacity of each sector is smaller

- 1.9.6. 6. Discuss how direct communication between mobiles (D2D) is a solution to improve the network capacity of cellular networks? What are the gains of this?

It's direct communication between two mobile users without traversing the Base Station (BS) or core network

Gain due to one instead of two wireless communications (sender to base station, base station to receiver)

Proximity gain due to smaller distances between sender and receiver

2. Medium Access Control

2.1. Basics of Wireless Access Control

2.2. - What are the problems with sensing in wireless environments?

- hidden terminal
- exposed terminal

2.2.1. What is the responsibility of the Media Access Control sublayer in communication systems? •

The correct answers are: Control for accessing the transmission medium, Addressing, Arbitration of access from different devices

2.2.2. What are the new problems wrt MAC in wireless systems? •

The correct answers are: Collisions happening at the receiver may not be noted by the sender, The sender and receiver do not have a consistent view on the radio resource

2.2.3. What are the causes of the problems in wireless systems? • Why does CSMA/CD known from Ethernet fail? •

What are the causes of the extra problems in wireless MAC?

Answer: Huge variation of the signal strength over distance, Much higher signal loss than wired, Changes of the signal strength due to mobility

Why does CSMA/CD known from the Ethernet fails in wireless networks?

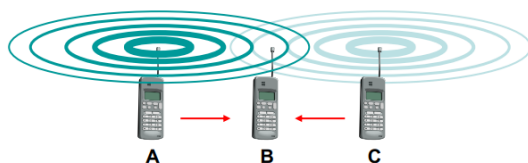
The correct answers are: Because Carrier Sensing would be needed at the receiver and not at the sender, Carrier Sensing may not work because of the Hidden Terminal problem, Carrier Sensing (CS) and Collision Detection (CD) may not work because of high signal attenuation in wireless networks, Because Collision Detection would be needed at the receiver and not at the sender side

2.2.4. What is the hidden terminal problem? When does it happen?

What happens?

- A sends to B, C cannot receive A
- C wants to send to B, C senses a "free" medium -> CS fails
- collision at B, A cannot receive C -> CD fails
- A is "hidden" for C

Three wireless terminals are given: A, B and C. A can communicate with B only. A and C are unaware of each other. If they send something to B simultaneously, collision at B will happen. Solution from IEEE - A sends RTS to B, B sends CTS to A, C hears them and shuts up for a period of time written in CTS.

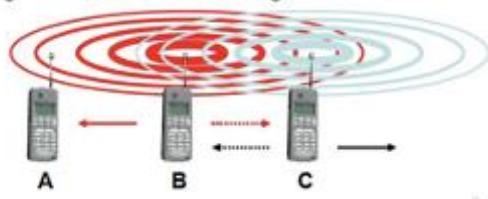


2.2.5. *How can the hidden-terminal problem be solved? Does your proposal solve the problem in any case? Name the limits of distributed approaches?*

The hidden terminal problem can be solved by medium reservation. Terminal tries to reserve medium for a time (RTS (right to send) packet), needed for packet transmission and starts to transmit it after acknowledgement (CTS a.k.a clear to send), while other terminals stay silent for this time. It doesn't solve the problem in any case, collision can still occur but only for RTS messages.

2.2.6. *What is the exposed terminal problem? When does it happen? What happens?*

Three wireless terminals are given: A, B and C. A can communicate with B only. C can communicate with B only. A is hidden from C. B sends to A. C hears them and thinks he can't send data to anyone because of it. This is wrong. Solution from IEEE - if terminal hears RTS and doesn't hear CTS, he thinks that he's an exposed one and can transfer data freely.



2.2.7. *What is the near far problem? What happens there?*

cause of the near-far terminal problem - Due to the decrease of signal strength with distance the receiver will receive more power from the nearest transmitter, The nearest transmitter "drowns out" the signal from the farthest transmitter

2.2.8. *What are the dimensions in which we can separate the access?*

The correct answers are: Space Division Multiple Access, Frequency Division Multiple Access, Time Division Multiple Access

2.3. Multiplexing and Duplexing

2.3.1. *What is simplex communication? What are example systems for this?*

Each terminal needs an uplink and a downlink channel

Types of communication links:

- Simplex
 - unidirectional link transmission

Examples: A communication channel that is used in one direction only, Streaming applications, TV

2.3.2. What are the pros and cons of TDD over FDD?

Frequency Division Duplex (FDD)

Separate frequency bands for up and downlink
+ separation of uplink and downlink interference
- no support for asymmetric traffic

Examples: LTE, UMTS, GSM, IS-95, AMPS

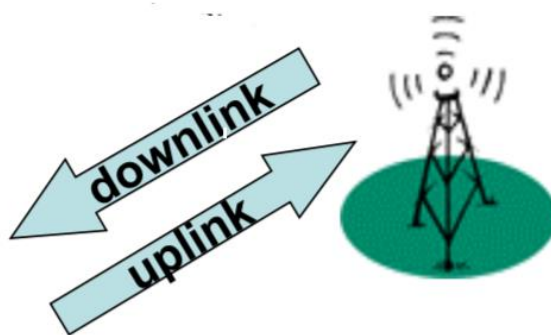
Time Division Duplex (TDD)

Separation of up- and downlink traffic on time axis
+ support for asymmetric traffic
- mix of uplink and downlink interference on single band
Examples: DECT, WLAN, UMTS (TDD), TD-LTE

2.3.3. For which of the given cases is FDD preferable? Why?

(TDD systems are often used in scenarios where short distances are required, with the possibility of unbalanced data traffic. FDD schemes are **better** over greater distances and where the traffic is balanced, i.e. similar in both directions.)

2.3.4. What does uplink and downlink mean?



Each terminal needs an uplink and a downlink channel
Downlink - from base station to mobile station, uplink - vice versa

2.4. Time Division Multiple Access

2.4.1. What is the idea of ALOHA? •

Aloha is a random access protocol. Any station can send data at any time. If multiple stations send data at the same time, a collision can happen. Therefore, the efficiency rate is 18 percent

2.4.2. Why does slotted ALOHA outperform ALOHA with respect to throughput? • Why does slotted ALOHA outperform ALOHA with respect to throughput?

In Slotted ALOHA, time is divided into discrete intervals called slots, corresponding to a frame. In slotted ALOHA, two transmissions can collide only if they begin at the start of the same time slot. Therefore, the efficiency rate is 36 percent.

2.4.3. What does implicit reservation mean? •

The correct answers are: After successful timeslot reservation, the station continues to use the same timeslot without competition in the following frame, The station releases the timeslot implicitly when it does not have any more data to send

2.4.4. How does explicit reservation work? •

The correct answer is: The sender and receiver explicitly agree on a resource prior to its use

2.4.5. What is the difference between random access and reservation based access? What are the pros and cons of the two strategies? •

random access protocol - Any station can send data at any time.
reservation-based - each station has to make a reservation. If there are N stations in the system, there are exactly N reservation minislots in the reservation frame. When a station needs to send a data frame, it makes a reservation in its own minislot and then can send its data frame after the reservation frame

random access: • fast, risk of collisions
reserved access: • slow, high capacity

The random-access strategy has lower latency under low load, The reservation-based access strategy may experience low latencies for data transmitted over given reservations

2.4.6. Compare the access latencies of both access schemes. Consider the latency under low and high loads. •

random access and reservation-based access strategies

The random-access strategy has lower latency under low load, The reservation-based access strategy may experience low latencies for data transmitted over given reservations

2.4.7. Compare the two access strategies with respect to the maximum load they can handle. •

The ALOHA and CSMA family(random access) of protocols are attractive for low to medium load conditions but suffer from stability problems at high load
Reservation is more stable, though less effective on low load(slow)

1. What are the two main strategies to organize access to the medium?
What are the pros and cons of the two strategies? Which one performs better under high load, which one under low load? Compare the access latency of both schemes.

Random access: - fast, risk of collisions - performs better on low load - minimal latency

Reserved access:- slow, high capacity - performs better on high load

2.4.8. Name the limits of distributed approaches to media access? •

Central vs. decentralized control

- central control:
 - consistent view on resource usage simplifies coordination
 - registration or static configuration of system needed
- decentralized control:
 - disjoint & inconsistent views on shared medium (hidden & exposed terminal)
 - distributed arbitration scheme needed to minimize collisions

2.4.9. How can the hidden-terminal problem be solved? Does the RTS/CTS scheme (MACA) solve the problem? In any case?

The correct answers are: Use of short signaling packets to reserve for a single data transmission, A sender initiates the transaction with a request of the right to send data to a receiver, The receiver grants the right to transmit data to the sender, Both sender and receiver inform their neighbors of an upcoming communication

No. During RTS packet transmission, During CTS packet transmission collision is possible

2.5. Code Division Multiple Access

2.5.1. What is the trick with CDMA to avoid collisions?

The correct answer is: All users share same frequency, but each user has own “chipping” sequence (i.e. code) to encode data, which allows multiple users to coexist and transmit simultaneously with minimal interference

2.5.2. Why is power control important for the correct functioning of CDMA?

The correct answers are: It is important to reduce the impact of the Near-Far Problem, Unbalanced received power levels combined with imperfect spreading codes with limited orthogonality will cause errors in detection

2.5.3. What happens if the keys are not orthogonal?

Orthogonal codes have a cross-correlation equal to zero; in other words, they do not interfere with each other

2.5.4. What is the advantage of CDMA with respect to the handling of noise?

The correct answers are: Noise is not coded and thus, its impact on the signal is reduced, CDMA supports useful communication with a much lower SINR compared to other techniques

2.5.5. Does CDMA require the same, higher or lower SINR for correct transmission? Why?

Noise is not coded and thus, its impact on the signal is reduced, CDMA supports useful communication with a much lower SINR(signal-to-interference plus noise ratio) compared to other techniques

может работать с более низким sinr

1. What are the two main strategies to organize access to the medium? What are the pros and cons of the two strategies? Which one performs better under high load, which one under low load? Compare the access latency of both schemes.

Random access: - fast, risk of collisions - performs better on low load - minimal latency

Reserved access:- slow, high capacity - performs better on high load

3. IEEE 802.11

3.1. Basics

- 3.1.1. • What is the difference between infrastructure and ad-hoc mode?

Infrastructure: (has central entity (AP) - centralized, uses reservation)

- Station (STA) – terminal with access mechanisms to the wireless medium and radio contact to the access point

- Basic Service Set (BSS) – group of stations using the same radio frequency
- Access Point – station integrated into the wireless LAN and the distribution system
- Portal – bridge to other (wired) networks
- Distribution System – interconnection network to form one logical network (ESS: Extended Service Set) based on several BSS

Ad-hoc: (no reservation - random access, decentralized)

- Direct communication within a limited range
 - Station (STA): terminal with access mechanisms to the wireless medium
 - Independent Basic Service Set (IBSS): group of stations using the same radio frequency

3.1.2. What is the purpose of the distribution system? •

Distribution System – interconnection network to form one logical network (ESS: Extended Service Set) based on several BSS

3.1.3. Which protocol layers does the 802.11 standard comprise? •

PHY layer, MAC layer

- MAC – access mechanisms, fragmentation, encryption
- MAC Management – synchronization, roaming, Management Information Base (MIB), power management
- PLCP (Physical Layer Convergence Protocol) – clear channel assessment signal (carrier sense)
- PMD (Physical Medium Dependent) – modulation, coding
- PHY Management – channel selection, MIB
- Station Management – coordination of all management functions

3.1.4. What are the main differences between the different PHY variants?

Radio implementations, Modulation schemes, Data rates

3.2. MAC

3.2.1. Exponential backoff: what does it mean? What is its purpose?

According to this algorithm the delay for the next trying of connection is exponentially increasing. It needs to avoid the blocking when too many users try to get access to the medium. In this case exponential backoff algorithm allows to spread the users' trying in time.

Purposes:

- 1) delays retransmissions after collisions to reduce the risk of subsequent collisions,
- 2) reduces the risk of collisions compared to ALOHA,

3) translates high network load in delay to avoid collisions,

4) tries to maximize the handled network load
exponential backoff time is computed from a uniformly distributed pseudorandom number from the interval $[0, CW]$ multiplied by the slot time.

In case of a repeated collision, the stations double the backoff time

3.2.2. Describe DCF and PCF. What are the differences, pros and cons of each approach?

Distributed coordination function. (~random access)(utilizes a contention-based MAC mechanisms)

Using this function means that the user before the start of transmission listens to the carrier and, if it is not busy, starts the session.

Pros: • Faster access if not many users

Cons: • Hidden terminal problem can be solved only by RTS/CTS

• Low QoS if too many users

In PCF (**point coordination function**) (~ pooling, ~reservation) certain point decides who will use medium (provides some support for time-bounded data services, PCF utilizes a contention-free MAC mechanism)

Pros: • Can work with a lot of users

• no collisions

• Guarantee QoS for time-dependent application

Cons: • Slower than DCF

3.2.3. DCF: what is the purpose of different interframe spaces?

The interframe spaces are needed to delay the beginning of the session to avoid collisions. Also size of IFS states the priority of messages.

Differentiate the access priority between different types of messages, ensure the non-delayed transmission of CTSs, ensure the non-delayed transmission of ACKs

Types of IFS: • SIFS (Short IFS) Highest priority. Applies for ACK, CTS and polling response

• PIFS (PCF IFS) Medium priority. Applies for an access point in PCF.

• DIFS (DCF IFS) Lowest priority. For users

3.2.4. Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?

No, typically only a single RTS/CTS cycle is required

3.2.5. For which messages does the SIFS apply? For what reason?

- SIFS (Short IFS) Highest priority. Applies for ACK, CTS and polling response

Ensures the non-delayed transmission of CTSSs, ensures the non-delayed transmission of ACKs

3.2.6. What is the purpose of the NAV?

NAV(network allocation vector) indicates the duration of the reservation of the medium, tries to solve the inconsistency problem in wireless systems existing between different stations and reduces the number of collisions

3.2.7. How does DCF handle situations with high load? What are the strategies?

Main problem with highload is collisions. It would occur if many stations start to transmit data immediately after DIFS. Therefore, there is a contention window (CW) defined. It is a random integer, which means the time the station has to wait after DIFS before transmission. If another station occupies the channel this CW timer stops, and this station will have the higher priority at the next iteration. Since the CWs are different due to randomness, collisions are avoided. We use RTS and CTS to decrease the number of collisions.

3.3. MAC Management

3.3.1. How does beaconing work in ad-hoc mode?

Beacon may be sent by any station (sending of beacon employs a random delay to avoid collisions), contains a timestamp and other management information used for power management and roaming, local clock is only updated when received timestamp is more recent

3.3.2. • How are collisions among beacons avoided in ad-hoc mode?

Sending of beacon employs a random delay to avoid collisions

3.3.3. • What's the idea behind energy saving in 802.11? How does it work? Can it be applied in adhoc mode as well? What is the disadvantage?

Idea: switch the transceiver off if not needed

- States of a station: sleep and awake

In order to know when to wake up, we need a TSF.

- Timing Synchronization Function (TSF)

– stations wake up at the same time

- Infrastructure

– AP stores frames intended for sleeping stations

- AP transmits indication about stored frames in periodic beacons (Indication Maps) sent during awake interval
- Traffic Indication Map (TIM): List of unicast receivers
- Delivery Traffic Indication Map (DTIM): List of broadcast/multicast receivers
- Ad-hoc
 - Ad-hoc Traffic Indication Map (ATIM)
- announcement of receivers by stations buffering frames
- more complicated - no central AP
- collision of ATIMs possible (scalability?)

from test - Station has two states: sleep and awake, Stations need to buffer frames for sleeping receivers

3.3.4. • Does power saving work in ad-hoc mode as well?

Yes, but there might be collisions between beacons, stations need to buffer frames for sleeping receivers

3.3.5. • What information does a TIM contain? What a DTIM?

- Traffic Indication Map (TIM): List of unicast receivers, TIM is used to notify the stations about frames buffered in the AP
- Delivery Traffic Indication Map (DTIM): List of broadcast/multicast receivers

3.3.6. • How does a handover work in 802.11? Why is it so slow, compared to GSM or UMTS?

No or bad connection? Then perform:

- Scanning
 - scan the environment, i.e. listen into the medium for beacon signals or send probes into the medium and wait for an answer
=> time-consuming (scan all channels)
- Association Request
 - station sends a request to an AP
- Association Response
 - success: AP has answered, station can now participate
 - failure: continue scanning
- AP accepts Reassociation Request
 - signal the new station to the distribution system
 - the distribution system updates its data base (i.e. location information)
 - the distribution system may inform the old AP so it can release resources
 - 802.11f standard (IAPP – Inter AccessPoint Protocol)

In 802.11 all operations for handover are initialized by users. Also there is no controller like in GSM or UMTS which prepared the new AP. Connection with new stations occurs like a new connection.

In test:

- during scanning for another AP, a station needs to wait in each channel for beacons,
- because it is not possible to monitor several channels with a single band receiver at the same time,
- the station does not start the scanning before the old association is lost.

- 3.3.7. • 802.11e: what are the extensions for QoS provisioning? How does the extended version of DCF work? Which main QoS strategy does it implement? What are the parameters to differentiate traffic? What are its limits?

In 802.11 for QoS provisioning EDCF is used. In EDCF the priority of traffic is set by choosing the IFS. Then less the delay then traffic have more priority, because transmission occurs earlier than others types of data. With EDCF traffic with high priority has more chances to be received then the lower priority traffic. STation with higher priority traffic waits a little bit less time before transmitting. The delay for priority traffic can not be less than SIFS.

3.4. Physical Layer Variants and Extensions

- 3.4.1. What are advantages and disadvantages of using higher frequencies?

Advantages:

- availability of unused spectrum,
- higher bandwidth available,
- typically less crowded bands.

Disadvantages:

- higher loss for non-line-of-sight transmissions,
- higher loss for penetrating obstacles

- 3.4.2. Why is the transmit power limited to 100 mW?

To reduce interference inside the ISM band, to reduce the risk of collisions

4. Mobility Management

4.1. Mobility Management in the Reference Models

- 4.1.1. What are the tasks of mobility management?

Tasks of mobility management

- Find (also called paging) and address a mobile away from home
- Support “continuous” communication after change of location
- Security: avoid misuse of mechanism
- Privacy: hide location from others

4.1.2. Which protocol layer should handle the mobility problem? What are the implications of selecting one or another layer?

If the old as well as the new AP belong to the same subnet, the problem is handled locally in the IP subnet

- Link (and PHY) layer mobility management is responsible for the establishment of a radio link between the Mobile Node (MN) and the new Access Point (AP)

If the new AP belongs to a different subnet (i.e. uses different IP addresses), we need

- Global mobility support either
 - in the network layer, i.e. the Internet layer of the TCP/IP reference model,
 - in the transport layer or
 - in the application layer

implications of mobility management by the link layer - Mobility may result in changes in the radio link, Mobility is transparent to the transport layer, Mobility management is local to the current subnet

implications of mobility management by the network layer - Mobility may result in changes of the IP address, Mobility may result in changes of the radio link, Mobility is transparent to the transport layer

implications of mobility management by the transport layer - Mobility may result in changes of the IP address

4.1.3. What are the performance criteria for handovers?

Handover (HO) performance

- Delay to update the route
- Packet loss due to handover

4.1.4. Why can the cooperation between layers improve the handover performance?

The correct answers are: Lower layers may inform higher layers of upcoming radio problems (decrease of SINR) before the transport connection actually fails due to errors, Higher layers may temporarily freeze its transmission during the handover in order to reduce the number of transmission errors, Higher layers may temporarily freeze its transmission during the handover in order to reduce the number of packets transmitted over an obsolete route, Higher layer may instruct PHY layer on which band to search for channels

- On which protocol layer can mobility be handled? What are the pros, cons and limits of different approaches?

- **Link layer mobility management** is responsible for the establishment of a radio link between the Mobile Node (MN) and the new Access Point (AP)
 - If the old as well as the new AP belong to the same subnet, no more procedures are required, i.e. the problem is handled locally in the subnet
 - If the new AP belongs to a new subnet, we need **global mobility support**
 - in the **network layer**, i.e. the Internet layer of the TCP/IP reference model,
 - in the **transport layer** or
 - in the **application layer**

*in other words if mobile node connect to AP of the same subnet - on Link layer
but if ap from other subnet - on layers from 2 to 7.*

4.2. Link Layer Mobility Management (Layer 2 and PHY)

4.2.1. Which mobility problems can be handled on the link layer? Which not!

The correct answers are: Mobiles moving within a single IP subnet, Mobiles moving within the switched layer 2 network, Mobility within the same AP, e.g. changing the band from 2.4 to 5 GHz

Can't be handled: Mobiles moving between different IP subnets

4.2.2. What happens if a mobile moves from one WLAN coverage area to another? What is the main source for handover delay?

The mobile searches for a new AP after identifying that no communication is possible anymore

Main sources are: The time needed to go through all channels with possible APs, The time required to receive beacons on all channels to identify APs

4.2.3. How could this be accelerated?

The correct answers are: Scan the medium for other APs before the actual radio link breaks (possibly using extra HW), Change to a new channel before the radio link on the current channel breaks, Provide the AP with radio information so that it can initiate the handover before the current link fails, Try to stay connected to multiple radio links (APs) at the same time so that there is a redundant link whenever another one fails

4.3. Network Layer Mobility Management

4.3.1. How can mobility be supported by the network layer? •

Adapt the routing tables of all routers of the network to ensure proper forwarding of the packets to the location of the mobile, Provide the mobile with a new IP address whenever it moves to a new subnet, Forward all data to all possible locations of the mobile, Install some central instance in the network that always knows the location of the mobile and where the destination address of packets is adapted so that it can be forwarded to the current location of the mobile

4.3.2. When does the sole update of routing tables make sense and when not? Why does it not scale? •

The correct answers are: In strictly tree-structured networks where all traffic passes through a central gateway, When the number of involved nodes is small

Won't work for MAC addresses in subnets, In mesh-structured networks, When the application takes care of mobility

When there're a lot of nodes, updates will be more costly (Maybe because we need to distribute and update tables on each router?)

4.3.3. What is the problem with changing the IP address for mobiles due to mobility?

The correct answers are: Typical terminal applications cannot handle changing IP addresses and fail, TCP connections cannot handle changing IP addresses and fail, Servers typically do not associate packets with a new source address (due to the mobiles new location) to the same connection and session

4.3.4. What is the purpose of tunneling?

The correct answers are: Keep the old IP address while using a new one for location purposes, Separating the terminal locator from the terminal identifier, Provide the original packet to the mobile node so that it is able to reply in the same way as when staying in its home network, Maintaining a static IP address while moving among subnets, Transparency of the mobile location to the sender (server)

4.3.5. What is the advantage of this over specific routes?

individual routes for each mobile do not scale - It requires updates in all routers (routing table) for every mobile that changes its subnet to ensure the mobiles reachability

4.4. Mobile IP

4.4.1. What is the overhead imposed on data traffic with Mobile IP?

The correct answers are: Overhead for additional IP header for any data packet send towards the mobile, Overhead for finding the current IP from the static IP in the HAs table

4.4.2. What is the overhead for signaling with Mobile IP?

Overhead for updating of the home agent upon changes of the mobiles IP address

4.4.3. When is triangular routing especially harmful? Can we solve this and how? What is meant by triangular routing? Can it be avoided?

Triangle routing" Problem The Communication Host (CH) has to send packets to the Mobile Host (MH) via the Home Agent (HA), while the MH sends packets directly to the CH. As the communication in the two directions follows different routes, the problem of "triangle routing" arises, which leads to low efficiency especially when the MH is far away from the HA and the CH is near to the MH.

it is esp harmful When the mobile terminal and the corresponding node are close to each other but the home agent is far from both of them

How can we solve the triangular routing problem?

Signal the CoA (care-of address) to the corresponding node for direct tunneling to the mobile terminal.

Idea of triangular routing is to send data to a MN via HA, what increases latency and loads the network. HA can send binding updates to CN to let him tunnel directly to the FA. This is the way to avoid triangular routing.

4.4.4. What are the differences between the CoA-handling in the FA vs. handling in the MN itself?

The correct answers are: In the first case the CoA is assigned by the FA, In the second case, no special support is needed in the foreign network

4.4.5. Which optimizations can be made in order to smooth data transport after a handover?

The correct answers are: New FA informs old FA of the handover, Old FA tunnels remaining packets for the mobile to the new FA

4.4.6. What is the purpose of reverse tunneling? In which case is it helpful?

The correct answer is: It solves the problem that firewalls block outbound traffic with nonlocal IP source addresses

4.4.7. What is the purpose of DHCP? What problem does it solve?

The correct answers are: Dynamically assign IP addresses to network entities, Provide mobile IP addresses to MNs to support global mobility

4.5. Network Layer MM – Advanced solutions

4.6. - How does Mobile IP work? What is the purpose of a CoA?

Mobile IP enables routing of IP datagrams to mobile nodes. The mobile node's home address always identifies the mobile node, regardless of its current point of attachment to the Internet or an organization's network. When away from home, a care-of address associates the mobile node with its home address by providing information about the mobile node's current point of attachment to the Internet or an organization's network. A Care-of Address is a temporary IP address for a mobile device. This allows a home agent to forward messages to the mobile device. The care-of address splits the dual nature of an IP address, that is, its use is to identify the host and the location within the global IP network.

4.6.1. Identify the limitations of Mobile IP. Which optimizations can you think of?

(IP v4) The correct answers are: Limited address space of IPv4 makes it hard to serve every MN on earth, High distances between MN and HA result in high delays even for local handovers, MIP tunnels may conflict with VPN tunnels, Tunneling complicates the provision of end-to-end QoS, Overhead for tunneling

(?) e.g. increase address space to IPv6

4.6.2. What is the purpose of these optimizations?

By increasing address space we can serve more MN

4.6.3. Which problem is addressed by Hierarchical MIP?

Reduces the delays for handovers for the case of local movement,
Reduces the signaling load of the HA

4.6.4. What is the idea of HMIP (Hierarchical mobile IP)? ? What is its advantage over MIP?

It was designed to reduce the amount of signaling required and to improve handoff speed for mobile connections.

MIPv6 defines a means of managing global (between-site) mobility, but doesn't address the issue of local (within-site) mobility separately. Instead, it uses the same mechanisms in both cases, which is an inefficient use of resources in the case of local mobility. HMIPv6 adds another level, built on MIPv6, that separates local from global mobility. In HMIPv6, global mobility is managed by the MIPv6 protocols, while local handoffs are managed locally.

How it works:

MN registers in its HA a Regional CoA (RCoA), i.e. a CoA in the Mobility Anchor Point (MAP)'s subnet, as the MN's CoA; the location of the MN inside the domain is defined by the local CoA (LCoA), which is a CoA in the current AR subnet. HA is not aware of the MN's mobility as long as the RCoA is not changed.

- What are the differences with MIPv6 over v4?

- A set of mobility options to include in mobility messages
- A new Home Address option for the Destination Options header
- A new Type 2 Routing header
- New Internet Control Message Protocol for IPv6 (ICMPv6) messages to discover the set of home agents and to obtain the prefix of the home link
- Changes to router discovery messages and options and additional Neighbor Discovery options

- Foreign Agents are no longer needed

4.6.5. Which problem is addressed by Cellular IP?

Cellular IP combines the capability of cellular networks to provide smooth fast handoff and efficient location management of active and idle mobile users with the inherent flexibility, robustness and scalability found in IP networks. support of fast moving wireless hosts.

4.6.6. What is the idea of Cellular IP? How does it differ from HMIP? Why do updates of the routing table make sense here? Can Cellular IP be generalized to non-hierarchical systems?

Cellular IP provides local handovers without renewed registration by installing a single cellular IP gateway (CIPGW) for each domain, which acts to the foreign world as a foreign agent (FA). Inside this domain, each node collect routing information for accessing MNs based on origin of packets sent by the MNs toward the CIPGW. Soft handover is achieved by allowing simultaneous forwarding of packets destined for a mobile node along multiple paths

Difference: in HMIP a MN should make BU by itself. in Cellular IP routing cache updated hop by hop - a MN communicate to only it's home base station.

Cellular IP cannot be generalized to a non-hierarchical systems (restrictions on the network topology)

- What is a colocated Foreign Agent? (FA which is co-located with the MN)– System in the current foreign network of the MN, typically a router

– Forwards the tunneled datagrams to the MN, typically also the default router for the MN

The correct answers are: Exploit the advantages of tree-structured cellular networks, Manage mobility locally in the network, Learn the downlink routes automatically from the MNs uplink data packets

4.6.7. Why do updates of the routing table make sense here?

Why do mobility-related updates of the routing tables make sense for Cellular IP?

The update is limited to a single path in the network, The number of routing entries to be handled by each router is limited to the nodes served by its respective subtree, Due to the

specific network structure, the number of required routing table updates is much smaller as in other networks

4.6.8. **Can Cellular IP be generalized to non-hierarchical systems?**

Cellular IP cannot be generalized to a non-hierarchical systems (restrictions on the network topology)

4.6.9. **Which problem is addressed by Proxy MIP?**

Helps with mobility. With PMIP the host can change its point-of-attachment to the Internet without changing its IP address. Contrary to Mobile IP approach, this functionality is implemented by the network, which is responsible for tracking the movements of the host and initiating the required mobility signalling on its behalf.

4.6.10. **What is the idea of Proxy MIP?**

The correct answers are: Support mobility management for mobiles without any specific mobility management support, Provide a transparent access to mobiles that looks like they are in their home network

or

Proxy MIP introduces a MAG (Mobility Access Gateway) into the Mobile IP architecture which interacts with the HA, termed a LMA (Local Mobility Anchor) in Proxy MIP parlance, on behalf of the Mobile Node.

- What is the idea of Proxy IP?

Proxy Mobile IP is a network-based mobility management protocol for building a common and access technology independent of mobile core networks. It introduces a Mobility Access Gateway (MAG) into the Mobile IP architecture which interacts with the Home Agent (HA), termed a Local Mobility Anchor (LMA) in Proxy MIP parlance, on behalf of the Mobile Node.

5. Transport over Wireless

5.1. Internet Protocol Suite

5.1.1. **What is the purpose of ARP?**

ARP: Address Resolution Protocol (RFC 826) – Address resolution from 32-bit IP addresses to hardware addresses (e.g. 48-bit)

functionality: Resolves the MAC address from a given IP address,

Uses a local lookup where available, Issues a broadcast request to the network in case the IP address is not known locally

5.1.2. What is the difference of ARP from DNS?

Applications use the Domain Name Service (DNS) to convert hostnames (e.g. "www.nokia.com") into IP addresses (88.221.214.58) and vice-versa

DNS Resolves the IP address from a given URL address, Issues a request to a known DNS server in case the IP address is not known locally

5.1.3. How does ARP differ from DHCP?

DHCP Provides an IP address, ARP resolves from IP to hardware

5.1.4. • What is the purpose of encapsulation and decapsulation?

purpose of encapsulation in the TCP/IP stack - Stepwise addition of headers from the different layers, Stepwise addition of information needed for the transmission over a network to the application data, Stepwise addition of control information needed for hop-by-hop as well as end-to-end transmission

5.1.5. • Why do we need multiple headers? Why not a single one?

To denote different concerns, e.g. addressing, error correction, flow control, avoidance of infinite forwarding loops, etc., To denote different aspects of the communication, To separate different concerns, e.g. hop-by-hop and end-to-end aspects

5.1.6. • What does the MTU mean? What is its purpose of the path MTU?

Maximum size of the IP packet

Path MTU: Used to avoid IP fragmentation

IP fragmentation: used on any link with $MTU < \text{original datagram length}$:

- Duplicates IP header for each fragment and sets flags for reassembly
- Re-assembly at the receiving host only, never in the network

5.1.7. • What is the purpose of the different header fields in IP? What is the purpose of the checksum and the protocol field?

IP packets consist of a **header** for addressing and routing, and a payload for user data. The **header** contains information about **IP** version, source **IP** address, destination **IP** address, time-to-live, etc.

16-bit one's complement sum of the IP header only checksum error =>

discard datagram & try to send ICMP message

Protocol field: Higher layer identifier, e.g.: ICMP=1 TCP=6 UDP=17

- 5.1.8. • What is the purpose of ports used by TCP and UDP? How does it differ from IP addresses?

To identify the application instance in charge of the connection, To identify the application-specific protocol handling the segment
Port numbers are used for application multiplexing

- “Stream sockets” use TCP
 - “Datagram sockets” use UDP
- “socket” = IP address + port number

- 5.1.9. • How are errors handled with UDP? Why is UDP not considered reliable?

Errors are ignored, Errors could be handled by the protocols run on top of UDP if needed

- 5.1.10. • Why does the TCP/IP model provide different transport layer protocols?

Different applications have different needs with respect to reliability of data, Different applications have different needs with respect to error control, UDPs transparency allows to solve transport issues in the application layer

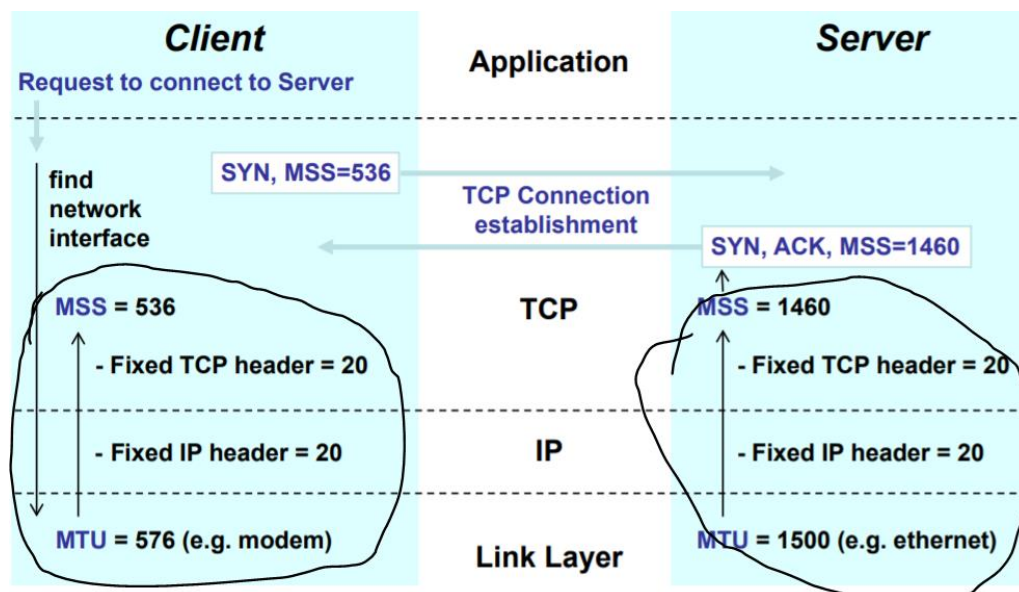
5.2. TCP (Transport Control Protocol)

- 5.2.1. How does the MSS differ from the MTU?

The MTU is the Maximum IP packet size for a given link .

MSS is Maximum TCP segment Size.

$MSS = MTU - \text{IP header} - \text{TCP header}.$



5.2.2. • What is the purpose of the MSS?

Specifies the largest amount of data, specified in bytes, that a computer or communications device can receive in a single TCP segment.

* MSS is optionally announced (not negotiated) by each host at TCP connection establishment. The smaller value is used by both ends, i.e. 536 in the above example.

Note that “real” TCP payload is smaller if TCP options are used.

5.2.3. • What is the purpose of the different flags?

URG: Urgent Pointer field significant - urgent data are outstanding

ACK: Acknowledgment field significant

PSH: Push Function - push to indicate prompt transmission of data

RST: Reset the connection

SYN: Synchronize sequence numbers

FIN: No more data from sender

5.2.4. • What is the purpose of the checksum field?

16-bit one's complement sum of TCP pseudo-header (12 bytes of the IP header) + TCP header + data (padded to 16-bit multiple)

checksum error

=> discard datagram silently!

=> using an erroneous header is dangerous; loss will be detected by other mechanisms

5.2.5. • What is the purpose of the window size field?

Advertised window size: number of bytes, starting with the one specified in the ack. **number field, that the receiver is willing to accept**

5.2.6. • What is the purpose of DUPACKs?

Indicates to the sender that a segment has arrived at the receiver, Indicates to the sender that the arrived segment is not in the right order, Indicates to the sender that a segment has arrived that is out of order

5.2.7. • How does a TCP sender learn about overload in the network? Which messages and mechanisms are used for this?

if No acknowledgement received at all within a specified time.

Receipt of duplicated acknowledgements

5.2.8. • How does the TCP congestion control algorithm in the sender react to overload problems? What are the options?

Reduce the congestion window

Half the data rate of the transmission

Reduce the congestion window to the size of a single segment

- How does a sender learn that there are problems with a connection? Which messages and mechanisms are used for this?

TCP sender starts a timer for a segment (only one segment at a time) If ack for the timed segment is not received before timer expires, outstanding data are assumed to be lost and retransmitted => go-back-N ARQ

=> TCP sender assumes that a (single) packet loss has occurred if it receives three dupacks consecutively

5.2.9. • Why does TCPs response to timeouts differ from the response for DUPACKs? What is the purpose of this?

The correct answers are: A raised timeout indicates a severe congestion problem, A timeout indicates that there has not been a successful transmission during the duration of the timeout

A DUPACK indicates that the arrived segment is not in the right order, A DUPACK implies a minor congestion problem

5.2.10. • Can we find out the direction in which the congestion occurs?

вроде можем)

с какой стороны пришел ACK в той стороне и пропал сегмент данных

- How does TCP react to connection problems?

Packet loss detected by timeout (=> **severe link problem**): Retransmit lost segments
Go back to Slow Start:

Packet loss detected by ≥ 3 dupacks (=> **single packet loss, but link is ok**): Fast Retransmit single missing segment Initiate Fast Recovery:

5.3. TCP over wireless

5.3.1. What are the problems of the use of TCP over wireless links?

DUPACKs are not necessarily a result of a congestion, Timeouts do not necessarily indicate a severe congestion problem, TCP may reduce the data rate of the transmission even though there is no overload at all

5.3.2. • Identify possible effects of high fluctuations of the channel capacity on TCP?

Increased delay jitter, Expiration of TCP transmission timeout, DUPACKs, Decreased throughput

- 5.3.3. • What is the difference in the impact of temporarily poor radio channels for the case that LLC is empty or uses some ARQ over the wireless link?

ARQ will turn packet drops in increased delay and delay jitter, ARQ may reduce the number of DUPACKs in certain cases, ARQ will reduce the load on the wired part of the transmission

- 5.3.4. • Can you think of mechanisms to alleviate the problem?
splitting TCP connection in a wired and a wireless part

- 5.3.5. • What is the advantage of splitting TCP connections in a wired and a wireless part, what's the disadvantage? *Are there less intrusive solutions to the TCP-over-wireless problem?*

- + The propagation of transmission problems on the wireless link towards the fixed part of the network is reduced,
 - + No changes for the TCP protocol on the wired host are necessary,
 - + Optimized TCP protocol can be used in a wireless part
- Acknowledgements received by the host in the fixed network do not indicate the reception of the packet by the wireless receiver,
- A handover between access points requires the migration of the TCP sockets and the TCP state (buffers, etc.)

- 5.3.6. • Are there less intrusive solutions to the TCP-over-wireless problem?
1. Using available bandwidth estimates (ABEs) to estimate transmit rates : Reduce its window size in a more intelligent fashion
 2. Loss discrimination algorithm: Additional information by an LDA about whether the loss was caused by congestion or link errors
 3. ZRW ACKs - zero size receive window

(?) Snooping TCP, Mobile TCP, Transmission/timeout freezing

- 5.3.7. • What benefit can TCP draw from lower layer information? How can it be used?

Information about deterioration of channel quality prior to link failure, Information about an upcoming handover are used by TCP transmission/timeout freezing to improve TCPs performance

5.4. Beyond TCP

5.4.1. What are the advantages of moving congestion control mechanisms into the user space?

The correct answers are: Flexibility, Adaptability to different needs, Adaptability to different network characteristics

5.4.2. • What does HoL blocking mean?

Happens if multiple objects are transmitted over a single TCP connection, Loss of a packet in a TCP connection blocks delivery of subsequent received packets of the same TCP connection

5.4.3. • What is the benefit of parallel multiple data streams?

Avoidance of head-of-line blocking, Support for redundant transmission, Increase of overall throughput

5.4.4. • What is a bufferbloat?

Persistent high occupancy of large buffers in network entities resulting in increased latency and jitter

5.4.5. • What are incast issues?

Multiple data servers sending data to a client surpass the ability of a clients' network entity to buffer packets, Synchronized data transmission from multiple data servers results in bursts of packet loss at a clients' network entity and consequently, poor TCP congestion control

5.4.6. • What are the benefits of SPDY?

Header compression is used to reduce required bandwidth, Solves head-of-line blocking problem by multiplexing data streams over multiple independent TCP connections, Packet retransmission is used whenever packets are received with bit errors from the lower layers

5.4.7. • What are the benefits of QUIC?

Congestion control is handled within the application layer, Solves head-of-line blocking problem by using UDP transport protocol, Simplified handshake procedure decreases the connection establishment latency in comparison to TCP

5.4.8. • What are the benefits of MPTCP?

Reduces load on congested links due to macro diversity, Increases throughput and reduces latency by using multiple TCP connections over different communication technologies, Packet retransmissions are handled independently per data flow, Head-of-line blocking

problem can be experienced per data flow over a given communication link

6. Quality of Service

6.1. QoS Basics

6.1.1. What are the QoS attributes of a connection?

- Throughput,
- Packet loss rate,
- Delay,
- Jitter

6.1.2. What are the basic functions for the provisioning of QoS?

Admission control,
Traffic classification,
Traffic conditioning,
Scheduling,
Overload control

6.1.3. Name the strategies for QoS provisioning

- reservation of „dedicated“ resources for a connection (e.g. CS voice, IntServ/RSVP)
- differentiation (e.g. prioritization) of the use of a shared resource by different connections (e.g. DiffServ)
- overprovisioning, i.e. dimensioning of the network such that all offered (or accepted) traffic can be handled

6.1.4. Which applications have the toughest QoS requirements?

Telnet, interactive games, distributed control

6.1.5. Name measures to minimize the error rate over wireless links

Which of the following PHY strategies are appropriate for the minimization of the error rate?
Higher FEC redundancy, Higher SINR, Increase of the transmit power,
Use of a directional antenna instead of an omnidirectional antenna

6.1.6. Name the measures to minimize the average delay over wireless links

(?) apply reservations, optimize scheduling schemes, use multiple radio links simultaneously

- 6.1.7. • Name the measures to maximize the throughput over wireless links

Higher-order modulation schemes, Higher channel bandwidth, Higher SINR, Higher transmit power (тест)

Maximize throughput, but with higher acceptable delay

Balance modulation, coding and SINR for max. throughput

Select (higher-order) modulation scheme and (lower) FEC redundancy that maximizes throughput (goodput) for given radio condition (rather than minimal delay or frame error rates)

- 6.1.8. • How can you compute the maximal throughput of an end-to-end connection from its constituents?

The maximal throughput is limited by the network element with the least throughput

- 6.1.9. • How compute the end-to-end delay for a connection for the case of zero transmission errors?

(?) В презентации (в таблицах), где указано отсутствие ошибок, для задержек везде стоит N.A.

- 6.1.10. • What are the reasons for delay jitter, what are the reasons for packet loss for wired and wireless communication?

Wired: Queueing delays

Wireless: Packet loss, System load, Varying channel quality

6.2. QoS Functions and Common Concepts

- 6.2.1. Which functions in a QoS-enabled network ensure that an application does not exceed its throughput requirements?

(?) overload control

- 6.2.2. • How does the token bucket scheme help in the limitation of the user throughput? How does the system support averaging?

Whenever a packet arrives, one token is removed from the bucket and the packet is marked to be “in profile”.

Whenever a packet arrives and no token is available in the bucket, the packet is marked to be “out of profile” .

- 6.2.3. • What is the purpose of explicit packet dropping in routers?

Test:

To avoid congestion in the network in a proactive rather than reactive manner,

To take advantage of the congestion control mechanisms of transport layer protocols,
To force congestion control mechanisms of the transport layer to reduce the data throughput

6.2.4. • How influences packet dropping the performance of TCP connections?

(?) Maybe, for TCP it will cause the same problems as packet loss, e.g. The TCP congestion window size will not permit an optimal throughput, dupacks

6.2.5. • What are the ingredients of the TOS field? Why are multiple fields needed?

Test: The priority of a packet, The requested tradeoff between throughput, delay, reliability and cost

6.2.6. • Why is the TOS field mostly relevant to the network as such rather than the end system?

(?) we use it as DiffServ field for prioritization of traffic class

6.2.7. • Why do networks differentiate between edge routers and core router? What's the purpose and what is the difference in behavior?

Edge routers

- accept customer traffic into the network
- characterize, police, and/or mark traffic, being admitted to the network
- may decline requests signaled by outside sources (admission control)

Core routers

- provide transit packet forwarding service between other core and/or edge routers
- differentiate traffic insofar as necessary to cope with transient congestion within the network

6.3. QoS Strategies: IntServ, DiffServ, Overprovisioning

6.3.1. What are the three main strategies for QoS provisioning in a network?

- reservation of „dedicated“ resources for a connection (e.g. CS voice, IntServ/RSVP)
- differentiation (e.g. prioritization) of the use of a shared resource by different connections (e.g. DiffServ)
- overprovisioning, i.e. dimensioning of the network such that all offered (or accepted) traffic can be handled

- How do reservation-based systems ensure that a certain level of QoS is provided to a customer?

applications ask to the network for an explicit resource reservation per flow. Network devices keep track of all the flows traversing the nodes checking if new packets belong to an existing flow and if there are enough network resources available to accept the packet.

- What is the main drawback of IntServ? What is the problem with individual reservations per user flow?

- Scalability problem: each flow, i.e. each IP packet, must be handled and maintained by each router on the data path individually even in the core network (consider that millions of flows have to be managed by a Gigabit router)
- Signaling overhead due to RSVP soft-state behavior
- Shortest path routing (OSPF) may not be optimal
- No fairness, i.e. fair distribution of limited resources among aspirants
- Violation of IP principle to keep individual states of connections in the edges (hosts) only

6.3.2. • Why does IntServ not scale? What is the problem of packet handling per individual flow? Hint: think about the implementation of rate control for individual flows.

- Because of the need for each flow being handled and maintained by each router on the data path individually and specifically,
- Because of the need to identify for each arriving IP packet the flow it belongs to

.....

6.3.3. • Why does RSVP use soft-states? What does it mean?

- Periodic Path messages are forwarded along the routing trees provided by the routing protocol (routing from source to sinks based on regular IP mechanism)
- Reservation refresh messages are forwarded along the sink trees (based on state information maintained by each router) to maintain current reservation state (identical to initial resv request)

Своими словами: Reservations should be updated periodically, because in other case, if the MN moves to another AP, the reservation will be forever => block of the resources

6.3.4. • How does RSVP handle the case that a router is not supporting it?

(?) The analog to the RSVP is NSIS and NSIS skips such routers. Maybe RSVP behaves the same way

- 6.3.5. • What are the common entities in routers that are present in both IntServ and DiffServ systems?
- 6.3.6. • What are the major differences between IntServ and DiffServ wrt – Provided QoS requirements – Separation of flows – Effort for edge and core routers
- 6.3.7. • How can we ensure that QoS works over multiple networks and multiple providers?

Providers need to agree on service level agreements between networks,

Providers need to agree on the data volumes and on how to handle different traffic classes

- 6.3.8. • What is the difference between metering and marking?

Marking builds on results from metering

- 6.3.9. • Under what circumstances can overprovisioning be applied?

appropriate where resources are abundant (typically not true for air interface) or maximum traffic volume is known (may hold for access network)

6.4. QoS Signaling

- 6.4.1. • What's the idea behind NSIS?

NSIS aims at providing a global model that supports several signaling applications by separating the protocol stack into two layers

- 6.4.2. • For which signaling purposes can NSIS be used?

Dynamic reservation of resources,
Mobility management,
Status information about network entities,
Firewall configuration

- 6.4.3. • Does NSIS support secured signaling?

- 6.4.4. • Does it use UDP or TCP for signaling?

да :))) еще SCTP

- 6.4.5. • What is the purpose of the GIST sublayer?

Establishment and support of a kind of signaling network that allows to address middle nodes

Routing of (per-flow) signaling messages:

1. Discovery of next node
2. Transport of signaling message
3. Reusing of existing transport and security protocols

6.5. QoS over Wireless

6.5.1. Why is QoS considered a cross-layer issue?

Multiple layers need to cooperate in order to provide a certain QoS, Lower layers provide feedback to higher layers to optimize QoS, Higher layers control the settings of lower layers to achieve given QoS requirements, QoS needs many protocol entities to be configured in a proper way, The weakest component in the communication system limits the achieved QoS

6.5.2. • What influences the QoS of a wireless link? Can the physical layer adapt or influence the QoS? How? Are there limits to this?

- Most QoS problems on wireless links due to fading, mobility, etc. caused by
 - increased path loss (due to none-line-of-sight, multipath, shadowing, etc.)
 - increased interference from others
 - high and fast variation of quality (SINR) of wireless link
 - fast variation of data throughput per radio resource (see Shannon)

How to combat this on the PHY layer?

- adapt modulation (e.g. QAM => QPSK => BPSK)
- adapt coding (increase redundancy for Forward Error Correction)
- reduce packet/frame size (smaller frames have lower probability for errors)
- increase transmit power – but increases interference to others!
- change link or use multiple links simultaneously
- wait for better link quality (consider fast changes due to mobility)

- Why is QoS considered a cross-layer issue?

QoS controls and manages network resources by setting priorities for specific types of data on the network. It can control physical layer parameters, such as coding/modulation scheme, but also establishes and controls link, so does both link and phy layer tasks.

6.5.3. • Which PHY strategies are appropriate for the maximization of the throughput?

Select the modulation and coding scheme that balances BLER and raw data rate such that the maximum throughput is achieved

- 6.5.4. • Which PHY strategies are appropriate for the minimization of the error rate? What are the drawbacks of the different approaches?

Higher FEC redundancy,
Higher SINR,
Increase of the transmit power,
Use of a directional antenna instead of an omnidirectional antenna
.....

- 6.5.5. • What is the benefit of ...

What is the benefit of reducing frame sizes in wireless networks?

Lower probability of frame errors, Reduced data volume needed for retransmissions

- 6.5.6. • What would be a good strategy to optimize the system capacity of the cellular system as a whole? On which layer would this apply?

Increase the bandwidth of the employed frequency band,
Prefer mobiles with better SINR over mobiles with low SINR,
Increase the number of base stations and base station locations,
Use highly directional antennas
(?) Physical layer

6.6. QoS in 802.11

- 6.6.1. DCF extensions for QoS provisioning:

- 6.6.1.1. – What is the idea of the DCF extension? Which QoS strategy does it implement?

– Replaces DIFS with different AIFS (Arbitration InterFrame Space), depending on traffic characteristics
– Adapts the contention window size to traffic characteristics
=> Different random backoff times and AIFSSs to provide differentiated services

- 6.6.1.2. – How does the extended version of DCF work?

- 6.6.1.3. – What are the parameters to differentiate traffic in EDCA?

- 6.6.1.4. – Provide an example for the parameters for 4 traffic classes

- 6.6.1.5. – With which parameter settings can an EDCA-enabled device outperform a non-QoS enabled device?

- 6.6.1.6. – What are the limits of the DCF extension with respect to QoS?

The relative performance is not easy to control

– The performance is NOT proportional to the backoff factor ratios

- It depends on the number of contending stations

6.6.1.7. – What is the purpose of TXOP?

Transmit opportunity (limit on access time to channel)

6.6.2. • HCCA – Which QoS strategy does it implement? – Which are the important functions for its implementation?

Provides policing and deterministic channel access by controlling the channel through the HC (Hybrid Coordinator)

HC derives schedule to provide the guaranteed QoS requirements

(?)

- Which QoS strategy does it implement? Reservation
- Which are the important functions for its implementation? PCF

6.6.3. • How do the extensions react to changing channel conditions?

6.7. QoS and Mobility

6.7.1. What is the conflict between mobility management and QoS?

Mobility may result in changes to the communication path which renders QoS reservations obsolete,

QoS management needs to be triggered after any handover,

New routes appearing after a handover may not support the required QoS

6.7.2. • What are possible solutions to this conflict?

- Coupling between QoS and mobility mechanisms (fast reservation and release of resources)

6.7.3. • What are the pros and cons of different approaches to the integration of QoS and mobility management

- No coupling
 - Protocols work separately and independently
 - For any change in the transmission path, i.e. due to mobility, the mobile and its communication peer need to trigger QoS management
 - High delays due to end-to-end handling of QoS
- Loose coupling
 - Separate protocols for mobility and QoS, but supporting interfaces for fast interaction
 - Example: Simple QoS
- Tight integration
 - Single joint protocol or mechanism for mobility and QoS

– Examples: • Wireless Lightweight Reservation Protocol (WLRP) • QoS-aware Mobile IP Fast Authentication (QoMIFA) • QoS-handling in UMTS, LTE and 5G

7. Self-Organization in Networking

7.1. Definitions and Properties

7.1.1. What's the idea of self-organized (SO) systems? •

Self-organizing autonomous systems Autonomously acting individual systems performing local programs and acting on local data but participating on a global task, i.e. showing an emergent behavior

7.1.2. Why is SO so popular for modern systems? •

Scalable by design, Adaptable to changes in the structure, Robust to failures of almost any number of entities, Intrinsically parallel

7.1.3. What are the properties of SO systems? •

May show oscillating behavior, Adaptive behavior of the system as a whole

Property	Description
No central control	No global control system No central information, information is distributed Subsystems perform completely autonomous
Emerging structures	Global behavior or functioning of the system emerges in form of observable pattern or structures
High resulting complexity	Even if the individual subsystems can be simple as well as their basic rules, the resulting overall system becomes complex and often unpredictable
High scalability	No performance degradation if more subsystems are added to the system System performs as requested regardless of the number of subsystems

- Absence of external control
- Global order and local interactions
- Fluctuations and instability
- Dissipation (waste of resources)
- Multiple equilibria and local optima
- Adaptation to changing conditions
- Redundancy
- Self-maintenance
- Complexity of understanding

7.1.4. What are the limits of SO systems? •

Systems are hard to predict and evaluate, Systems are hard to test, The impact of single entities on the system as a whole is limited, Local entities are limited in their knowledge of the global system state

7.1.5. What does emergence mean?

Sudden appearance of unforeseen behavior, Emergence of leaders in the structure, Appearance of global structures from local actions

- Provide examples of protocols that apply principles of self-organization

Applications of Self-organization to Networking

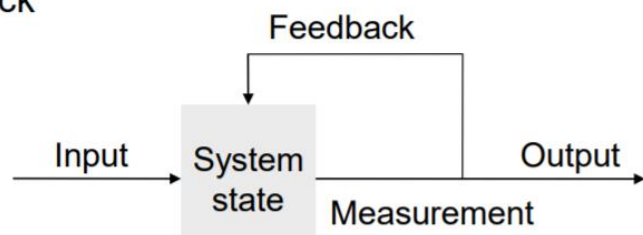
- Media Access Control:
 - CSMA/CD (802.3/Ethernet)
 - 802.11 DCF
 - Cognitive radio in interweaved mode with infrastructure network
- Load/congestion control:
 - Exponential backoff (CSMA systems)
 - TCP congestion control
- Routing: AODV route discovery and maintenance
- Election: cluster head selection in sensor networks (LEACH)
- Energy management: 802.11 ad hoc mode
- Network management:
 - LTE overload control
 - LTE energy saving
 - LTE mobility robustness optimization (HO parameter optimization)
 - Inter-system load balancing

7.2. Design of Self-organized Systems

7.2.1. Identify important design principles of self-organized systems •

- Positive and negative feedback

Simple feedback



- Interactions among neighbors and with the environment
 - Direct communication among neighboring systems
 - Indirect communication via the environment (stigmergy)
 - Interaction with (stimulation by) the environment
- Probabilistic techniques

7.2.2. Identify the design paradigms for algorithms of self-organized systems •

Paradigm #1: Design local behavior rules that achieve global properties

Paradigm #2: Do not aim for perfect coordination: exploit implicit coordination

Paradigm #3: Minimize long-lived state information

Paradigm #4: Design protocols that adapt to change

Paradigm #1: Design local behavior rules that achieve global properties

To design a network function that establishes a global property, this paradigm is to distribute the responsibility among the individual entities: no single entity is "in charge" of the overall organization, but each contributes to a collective behavior. Following this paradigm, localized behavior rules must be devised, if applied in all entities, automatically lead to the desired global property (or at least approximate it). "Localized" means that entities have only a local view of the network and interact with their neighbors and changes in the network have initially only local consequences.

Paradigm #2: Do not aim for perfect coordination: exploit implicit coordination

Implicit coordination means that coordination information is not communicated explicitly by signaling messages, but is inferred from the local environment. A node observes other nodes in its neighborhood; based on these observations, it draws conclusions about the status of the network and reacts accordingly.

Paradigm #3: Minimize long-lived state information

To achieve a higher level of self-organization, the amount of long-lived state information should be minimized. In general, the more localized the interactions are and the less coordination is needed, the less state information must be maintained.

Paradigm #4: Design protocols that adapt to changes

The need for the capability of nodes to react to changes in the network and its environment typically arises from changed resource constraints, changed user requirements, node mobility, or node failures. Since there are no centralized entities that could notify the nodes about changes, each node has to continuously monitor its local environment and react in an appropriate manner. Three levels of adaptation can be distinguished:

– Level 1: A protocol is designed so that it can cope with changes, such as failure and mobility.

- Level 2: A protocol is designed to adapt its own parameters (e.g., value of timers, cluster size) as a reaction to changes in order to optimize system performance.
 - Level 3: A protocol is designed so that it realizes if the changes are so severe that the currently employed mechanism is no longer suitable. To detect such situations the environment is observed, and significant changes in major parameters trigger a fallback or alternative solution. Typically, these levels of adaptation are combined using control loops
- Positive feedback, Negative feedback, Randomness, Interaction with neighbors

Test:

Design algorithms to achieve the desired global properties, Design algorithms that quickly adapt to changes, Design adaptive algorithms that can handle a multitude of situations

7.2.3. **How are these paradigms applied to communication protocols?**

A self-organizing network "will create its own connections, topology, transmission schedules, and routing patterns in a distributed manner" Some applications – Network management: Self-Organizing Networks (SON) in LTE – Protocol functions: TCP congestion control, 802.11 DCF, AODV route discovery, ... – Autonomous coordination of UAVs

7.3. Applications of Self-organization to Networking

7.3.1. **Why are the different examples considered examples of SO?**

Why is TCP congestion control considered as self-organized?

TCP congestion control detects network congestion indirectly through packet loss, TCP congestion control regulates the transmission rate dynamically using feedback from its communicating peer, TCP hosts only take local decisions, TCP congestion control utilizes negative feedback

Which of the given economic and social system is closest to a self-organized systems?

Societies following Kants categorical imperative

Which of the functions of 802.11 implements the principles of self-organization?

EDCA access strategy, DCF access strategy, Beaconing in ad hoc networks

Which function of TCP follows self-organization principles to achieve a global goal?

Congestion control

7.3.2. **Which aspect exactly is considered self-organizing**

Why is TCP congestion control considered as self-organized?

TCP congestion control detects network congestion indirectly through packet loss, TCP congestion control regulates the transmission rate dynamically using feedback from its communicating peer, TCP hosts only take local decisions, TCP congestion control utilizes negative feedback

DCF: random access? no centralized?

Beaconing: ну как б тоже децентрализовано всё

- 7.3.3. Identify how feedback is implemented in the different applications of SO; identify positive and negative feedback separately

ack, cts наверное позитив фидбек

What is negative feedback in 802.11 DCF?

Collision, Absence of ACK, Busy medium

What is negative feedback employed in TCP congestion control?

DupACKs, No feedback from receiver, Timeout raised at the sender

- 7.3.4. Identify the implementation of communication in the example applications •

- 7.3.5. Identify how stochastic aspects are integrated each of the examples • probabilistic = stochastic

What are the probabilistic aspects in 802.11 DCF?

Random backoff time, Randomness of distributed availability of packets, Randomness of occurrence of collisions, Packet loss due to varying propagation characteristics

- 7.3.6. Identify which level of the design paradigm 4 is implemented by the different examples

7.4. Challenges and Limits

- 7.4.1. What defines the Edge of Criticality? Why is it such an important issue?

The edge between

- order (i.e. coordination, patterns) and
- chaos (stochastics and variance)

Point of operation where properties of the system change dramatically

- 7.4.2. What are the advantages and disadvantages of operating close to this critical point? What are the properties of systems operating at that point?

- +Potentially very high efficiency of operation,
- +High risk for the case of unforeseen situations
 - unstable af

7.4.3. Identify the points of criticality for different applications as TCPs congestion control, RED, 802.11 DCF and the collision avoidance of UAVs

7.4.4. Identify ways to reduce the risk of criticality for different SO algorithms

design of the TCP congestion control algorithm and RED: Use a less aggressive increase of the data rate upon the receipt of ACKs, Quadruple the retransmission timer after a timeout, Reduce the packet drop probability thresholds to apply RED or WRED to TCP traffic, Always employ the congestion avoidance strategy instead of slow start

7.4.5. Discuss how the term “overengineering” influences the risk of criticality.

8. Ad Hoc Networks

8.1. Basics

8.1.1. What are the various forms of ad-hoc networks? •

Wireless Mesh Networks (WMNs),
Wireless Sensor Networks (WSNs),
Vehicular Ad Hoc NETWORKs (VANETs),
Delay-Tolerant Networks (DTNs)

8.1.2. How are ad-hoc networks formed? What are their constituent components? •

The wireless ad hoc networks consist of a collection of wireless nodes, that communicate over a common wireless medium. The nodes communicate without an infrastructure, such as base station, wired access point, etc. The establishment of the networks must be in a distributed and decentralized manner

8.1.3. What characteristics are exhibited by ad-hoc networks? •

Nodes organize themselves in a decentralized way, There is no fixed network infrastructure involved, The nodes can be mobile, Each node contributes to the establishment of routes

8.1.4. What advantages do ad-hoc networks offer? •

Neighboring nodes may communicate with lower transmit power compared to the case that an infrastructure is used, thus saving batteries, There is no “single point of failure”, High robustness towards single node failures

8.1.5. Which applications can benefit from the use of ad hoc networks?

Ad hoc networks can be very helpful during meetings or in any location where a network doesn't exist and where people need to share files.

Vehicle-to-vehicle communication, Communication among emergency forces in disaster scenarios like fire, earthquakes, etc., Extension of the coverage of cellular networks, Military communications between armed forces

Example applications:

- Military, e.g. battlefield communication
- Human gatherings, like conferences, exhibitions, meetings, lectures, gaming
- Car-to-car and car-to-X communication, e.g. accident warnings, traffic control
- Sensor applications, e.g. for fire detection, earthquake warnings, etc.
- Extension of cellular networks beyond infrastructure coverage
- Disaster communications, e.g. after earthquakes, tsunamis, terrorist attacks, ...
- Delay-tolerant networking, e.g. wildlife tracking, long-term data collection

8.1.6. Discuss the advantages and disadvantages of different routing strategies for ad hoc networks •

Proactive protocols have lower latency for data delivery due to prior setup of routing information, For low data traffic, proactive protocols consume more network resource than reactive protocol due to network-wide update of routing information

8.1.7. What is the idea of delay-tolerant networking? •

Idea: Nodes are not continuously connected to each other, rather use communication opportunities due to the mobility of nodes (message ferries)

- Message ferries
 - Message ferries may be cars, trains, UAVs, animals, smartphones
 - Message ferries mobility may be known (bus network with fixed schedule), partially known (animals or people following some routines) or just stochastically known
 - Mobility of nodes may also be controlled by the need for message transfers, e.g. special UAVs
- Message traffic
 - Message traffic may be totally random or follows some pattern
 - Messages are assumed to be tolerant to delay

- Forwarding between message ferries (and possibly other nodes) may be done as unicast or multicast/broadcast depending on requirements on reliability, average delay, etc

8.1.8. What are the advantages and disadvantages of geographical routing

geographical routing - a routing principle that relies on geographic position information. It is mainly proposed for wireless networks and based on the idea that the source sends a message to the geographic location of the destination instead of using the network address.

+ Geographic routing protocols greatly reduce the requirements of topology storage (no need for topology info) and provide flexibility in the accommodation of the dynamic behavior of mobile ad hoc networks

+ there is no need to send out route requests or periodic connectivity updates. This can save a lot of protocol overhead and consequently, energy of the nodes.

- in terms of latency, hierarchical and geographical routing performs the worst because it must send hello packets in order to acquire and transmit location information

- Name the multi-channel mac protocols in ad hoc networks.

- **Dedicated control channel**

- One channel for control messages and others for data traffic
- Needs usually two or more interfaces

- **Split phase**

- Communication in two phases
- Channel negotiation phase (a default channel is used by all nodes)
- Data transfer phase (all channels are used to transmit data during this phase including the default one)
- Works with one interface

- **Common hopping sequence**

- All nodes follow the same channel hopping sequence
- Works with one interface

- **Multiple rendezvous**

- Each node follows its own channel hopping sequence
- Works with one interface

- Why a node needs the seed number of a neighbor node to start communication in

multiple rendezvous approach?

Each node has a special hopping sequence generated by applying an equation on a certain seed. Notice that the equation is known to all nodes to switch to the same channel for transmission.

What is the idea of hopping sequence mac protocol? May a node leave the common sequence?

All nodes follow the same hopping sequence. In our example, the hopping sequence is as follows: Ch0Ch1Ch2Ch3Ch0

- *A wants to send data to B (AB)*
 1. *Node A checks the hopping sequence, i.e. (Ch0 in the example) and exchanges RTS/CTS with B*
 2. *A sends data to B*
 3. *There is no communication between nodes (x)*
- *C wants to send data to D as well (CD)*
 4. *Node C checks the hopping sequence, i.e. (Ch2 in the example) and exchanges RTS/CTS with D*
 5. *C sends data to D*
 6. *There is no communication between nodes (y)*

8.2. MANET Routing

8.2.1. What are the challenges of routing in multi-hop ad-hoc networks? •

if there're a lot of communication obstacles, part of traffic is likely not to reach destination

8.2.2. What are the differences between link state and distance vector routing protocols? •

Distance Vector Routing

– Each node computes for each destination in the network (stepwise starting with its neighbors) a matrix with an entry for each pair of destination and next hop with the total length of the path to the destination via this next hop

– Matrix (the complete routing table) is updated and sent to all neighbors whenever an update from neighbors arrives

– In conclusion, each node shares its complete (huge) routing table with its neighbors

– Examples • Wired networks: Routing Information Protocol (RIP) • Ad hoc networks: Destination-Sequenced Distance Vector Protocol (DSDV) •

Link State Routing

– Each node sends the distances to its neighbors to all nodes in the network

- Receiving this (small) information from all nodes each node knows the entire network topology
- Examples • Wired networks: Open Shortest Path First (OSPF) • Ad hoc networks: Optimized Link State Routing (OLSR)

8.2.3. What are different classes of ad hoc routing protocols? •

Proactive or table-driven (Examples: DSDV, GSR, WRP, OLSR)

- Routes are calculated periodically and before needed
- Keep routing information in all nodes up-to-date

Reactive or on-demand (Examples: AODV, DSR, LMR, ABR)

- Routes are only calculated when needed
- Do not keep routing information in all nodes up-to-date

Hybrid (Examples: ZRP, SHARP, Safari)

8.2.4. What are differences between reactive and proactive routing protocols? • *Pros and cons of reactive and proactive routing protocols?*

Topology-based routing:

- Proactive or table-driven (Examples: DSDV, GSR, WRP, OLSR)
 - Routes are calculated periodically and before needed
 - Keep routing information in all nodes up-to-date •

Reactive or on-demand (Examples: AODV, DSR, LMR, ABR)

- Routes are only calculated when needed
- Do not keep routing information in all nodes up-to-date

8.2.5. What are the motivations to have hop count and seq. number in RREQ of AODV? •

Hop count ensure that duplicates are eliminated (limits flooding)

SN prevents confusion between old and new RREQs (freshness)

Duplicate RREQ (in B and E) are detected by sequence numbers and discarded

Cycling RREQs (in B) are detected by hop count and discarded •

Outdated RREQs are detected by old sequence numbers and discarded

RREQ contains

- source and destination sequence numbers (to ensure "freshness" of routing information and to avoid forwarding loops)
- hop count (incremented by each intermediate node before forwarding RREQ)

sequence number changes Whenever a new route discovery is initiated by the originator

Duplicate RREQ are detected by sequence numbers and discarded

- Where the sequence number and hop count in RREQ packet of AODV are changed?

The hop count is incremented on each node. sequence number is always increased on the source node before sending.

- May we send a message immediately in AODV?

No. When a route to a new destination is needed, the node broadcasts a RREQ to find a route to the destination. A route can be determined when the RREQ reaches either the destination itself, or an intermediate node with a 'fresh enough' route to the destination. A 'fresh enough' route is a valid route entry for the destination whose associated sequence number is at least as great as that contained in the RREQ. The route is made available by unicasting a RREP back to the origination of the RREQ. Each node receiving the request caches a route back to the originator of the request, so that the RREP can be unicast from the destination along a path to that originator, or likewise from any intermediate node that is able to satisfy the request. After the RREP we can start sending a message.

8.2.6. Are RREQ and RREP broadcast messages or unicast? Why? •

In AODV, nodes discover routes in request-response cycles. A node requests a route to a destination by broadcasting an RREQ message to all its neighbors.

When a node receives an RREQ message but does not have a route to the requested destination, it in turn broadcasts the RREQ message. This process repeats until the RREQ reaches a node that has a valid route to the destination. This node (which can be the destination itself) responds with an RREP message.

This RREP is unicast along the reverse-routes of the intermediate nodes until it reaches the original requesting node. Thus, at the end of this request-response cycle a bidirectional route is established between the requesting node and the destination.

When a node loses connectivity to its next hop, the node invalidates its route by sending an RERR to all nodes that potentially received its RREP.

8.2.7. How does a node behave when it receives more than one RREQ? •

Duplicate RREQ (in B and E) are detected by sequence numbers and discarded

It is also possible that a destination node will receive more RREQ than it is allowed to reply to. This means that in some cases the destination may be forced to ignore RREQ packets with better routes if they reach the destination too late.

8.2.8. What is the idea of greedy geographical routing? •

Routing decisions are made on a hop-by-hop basis, Periodic or on-demand beaconing is required to detect any changes in the network topology, Geographical location information is needed for each node to achieve routing

8.2.9. What types of information are needed to apply geographical routing protocols in ad hoc networks?

Routing decisions are made on a hop-by-hop basis, Periodic or on-demand beaconing is required to detect any changes in the network topology, Geographical location information is needed for each node to achieve routing

8.3. Reactive Routing – AODV

8.3.1. Which routing table entries are established with the RREQ and RREP messages, respectively? •

Whenever a node receives a RREQ from a neighbor, it saves this routing information and generates a respective routing table entry for the construction of the reverse path from destination T to source S
Upon receipt of RREQ, receiving node updates its routing table

- Table entries:
 - Destination address
 - Dest. sequence number
 - Next hop address
 - Hop count
 - Time to live

RREP contains

- RREQ originator address and destination addresses
- Dest. sequence number
- Lifetime of a route

The information in the RREP is used for the genRREP updates the routing table of each node, thus establishing the forward path () from S to T

8.3.2. **How is the forwarding and how the reverse path established in the routing tables? •**

In AODV, nodes discover routes in request-response cycles. A node requests a route to a destination by broadcasting an RREQ message to all its neighbors.

When a node receives an RREQ message but does not have a route to the requested destination, it in turn broadcasts the RREQ message. This process repeats until the RREQ reaches a node that has a valid route to the destination. This node (which can be the destination itself) responds with an RREP message.

This RREP is unicast along the reverse-routes of the intermediate nodes until it reaches the original requesting node. Thus, at the end of this request-response cycle a bidirectional route is established between the requesting node and the destination.

When a node loses connectivity to its next hop, the node invalidates its route by sending an RERR to all nodes that potentially received its RREP.

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8.3.3. **What is the purpose of the sequence number and the hop count maintained in the routing tables? •**

Global repair of link failures

Hop count ensure that duplicates are eliminated (limits flooding)

SQLN prevents confusion between old and new RREQs (freshness)

8.3.4. **What is the purpose sequence number in RREQs and RREPs? •**

To monitor the freshness of a route towards a given destination, To prevent the formation of loops during the route discovery

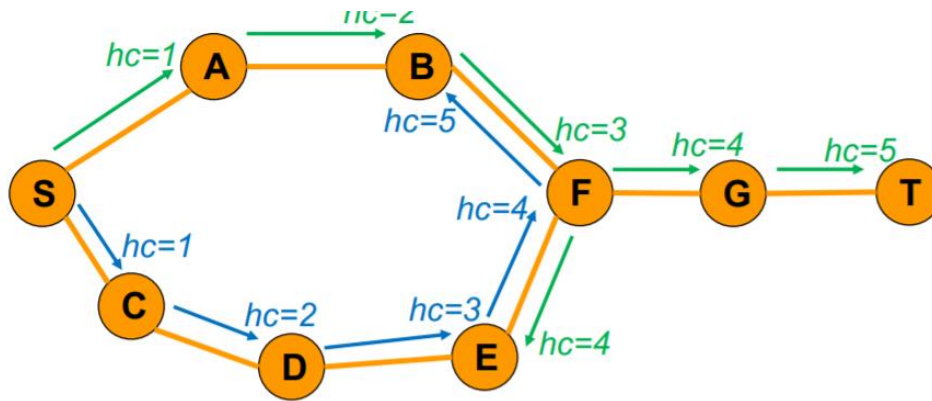
8.3.5. **How are multiple RREQs handled by the receiving node? What is the meaning of sequence number and hop count for this handling? •**

Network nodes may receive multiple RREQs, e.g. via loops, for the same source node SQLN

A node forwards a RREQ and updates its routing table only if:

- The RREQ has the same source SQLN and lower hop count (hc)
 - Nodes B and E discard subsequent RREQs due to higher hc
 - Node F forwards RREQ with lower hc
- The RREQ has a higher (newer) originator SQLN (update irrespective

of hc)



8.3.6. **How are link failures detected? What are the mechanisms to repair the link or route? •**

RREQ initiation for local repair

- Node X learns about link failure from missing ACKs or Hello messages
- Node X initiates RREQ broadcast with T as destination and X as originator, with X's SQN
- Node X invalidates all destinations in its routing table with Y as next hop (marks hop count as infinity)

Route ERROR (RERR) message initiation(global repair)

- Node X learns about link failure from missing ACKs or Hello messages
- Node X initiates RERR (broadcast or unicast) towards precursor nodes
- RERR contains all destinations in the routing table that are now unreachable (all destinations with Y as next hop), including their SQNs
- Node X invalidates all destinations in its routing table with Y as next hop (marks hop count as infinity)

Receipt of RERR (by any node)

- Invalidate all destinations in routing table (hop count – infinity) that are part of the RERR (unreachable destinations)
- If a routing entry has been invalidated, forward RERR (broadcast/unicast)

(• Global repair

– issuing RERR to the source S containing all destinations invalidated by broken link

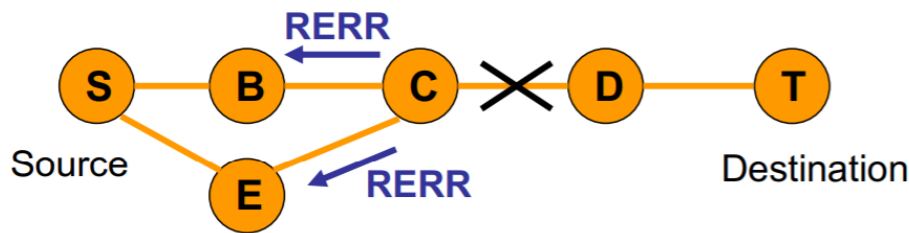
– applied when broken link is close to source or after local repair fails

• Local repair

– issuing RREQ to destination T from intermediate node X where forwarding is broken

– applied when the broken link is close to the destination)

8.3.7. What is the purpose of the RERR message? Which routing tables are modified with this message?



purpose: to modify routing tables?
S B E are modified?

9. Cognitive Radio Networks

9.1. Introduction

9.1.1. What is the idea of cognitive radio? Why is it important?

A cognitive radio is a radio that can change its transmitter parameters based on interaction with the environment in which it operates.

Required to enable better utilization of unused licensed spectrum

9.1.2. What defines a primary network, what a secondary network?

- **Primary networks**

- Networks with access right to certain spectrum bands, e.g. common cellular systems and TV broadcast networks
- Users of these networks are referred to as primary users. They have the right to operate in licensed spectrum.
- Users of a certain primary network do not care of other primary or secondary network users

- **Secondary networks**

- Do not have license to operate in the spectrum band
- Users of these networks are referred to as secondary users. They have no right to access licensed bands currently used.
- This is also called Opportunistic Spectrum Access or Dynamic Spectrum Access – DSA
- Additional functionalities are required to share licensed spectrum bands with other secondary or primary networks

9.1.3. What defines a spectrum hole?

The vacant **spectrum** band which is not in use by any user.

Secondary users opportunistically exploit the existence of *spectrum holes* to improve spectrum utilization. A spectrum hole is a licensed frequency band that is not being used for some time period usually in a specific area.

9.1.4. **What are the main tasks and challenges of cognitive radios?**
What are challenges?

Tasks:

- Determination of unused portion of spectrum (spectrum holes or white space)
- Selection of the best available channel
- Coordination of the access with other users
- Detection of the appearance of licensed users vacate the channel

Challenges:

• **Interference temperature measurement**

- Secondary users are aware of their locations and transmission power. They are not aware of primary users locations
- Currently, no practical way for a cognitive radio to measure or estimate the interference temperature at neighbor primary users

• **Spectrum sensing in multi-user networks**

- Multi-user environment makes it more difficult to sense primary users (secondary networks coexist with each other as well as with primary networks)
- Cooperative detection can be considered as possible solution for such environments

• **Detection capability**

- Detection of primary users in very short time is essential
- OFDM-based secondary users are best adequate (multi-carrier sensing can be exploited)

9.1.5. **Why do cognitive radios need to be configurable?**

To use the best wireless channels in its vicinity to avoid user interference and congestion.

Such a radio automatically detects available channels in wireless spectrum, then accordingly changes its transmission or reception parameters to allow more concurrent wireless communications in a given spectrum band at one location.

9.1.6. **What are the differences between cognitive radio networks and traditional wireless networks?**

Cognitive radio networks is a type of **networks** where the sensors have **cognitive** abilities, they can change and adapt their communication parameters **in** accordance to the change to the surrounding environment and this is major **difference** with the conventional **wireless networks**.

9.2. Spectrum Sensing

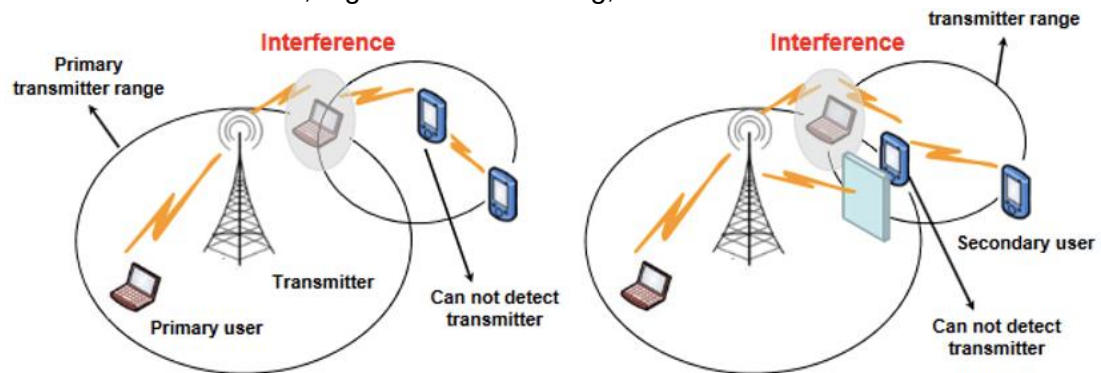
9.2.1. Why is accurate spectrum sensing for cognitive radios important?

The concept of **spectrum sensing** is **important** to secondary users in **cognitive radio** to operate without interfering with the primary users.

9.2.2. Explain the hidden terminal problem for cognitive radio networks. Draw a figure to show the hidden terminal problem in cognitive radio networks

Primary and secondary networks are physically separated but share the same radio resource, so transmitter detection can not avoid interference due to the lack of the primary receiver's information

Even if the secondary user has a line of sight to the primary receiver, the secondary user may not be able to detect the primary transmitter, e.g. due to shadowing, etc.



9.2.3. What are the advantages of cooperative sensing? How does cooperative sensing solve the hidden terminal problem? Describe a scenario which cooperative sensing solves the hidden terminal problem

Incorporating information from multiple secondary users to detect primary users.

It decreases the probability of misdetection and false alarm. And it is also capable of solving hidden terminal problem and it decreases sensing time. Also suitable for multi-users networks.

It helps secondary users to derive a better picture of the interference situation, It helps secondary users to avoid creating interference to primary users, It helps to utilize the spectrum more effectively

9.2.4. What is the idea of matched filter sensing? What are the prerequisites?

- Check of correlation of received signal with a given template, e.g. a pilot, that allows detecting the presence of a known signal (maximize SNR)

- Optimal detector when all characteristics of the primary user signal are known, otherwise poor performance
- High complexity due to need for carrier synchronization

Matched filter provides accurate detection of primary users if all characteristics of primary user signal are known, Matched filter works well for noisy input signals (тест)

9.2.5. What does energy detection mean? What is its advantage?

Energy detection looks for a specific energy level over a chosen period of time regardless of the shape of the signal, Energy detection can be used in case the characteristics of the primary user signal is unknown

9.2.6. How does cyclostationary feature detection work? •

- Mean and autocorrelation of modulated signals exhibit periodicity (modulated signals are coupled normally with sine wave carriers, pulse trains, repeating spreading, etc.)
- Differentiates noise from modulated signal (noise is a wide-sense stationary signal with no correlation)
- Computationally complex and requires long observation time

Test:

Autocorrelation of the signal with itself to identify its periodicity, Identification of primary network activity by differentiation of the signal from random noise

9.2.7. What is the idea of interference temperature management?

- Interference takes place at the receiver side (i.e. the primary receiver is disrupted), but is controlled by the transmitter side, e.g. using power control, etc.
- Interference temperature management regulates received power at primary user instead of transmitted power, typically approximated by maximum transmit power for secondary users

Test:

Ensure that secondary users do not exceed a predefined interference level at primary receivers, Allow a defined level of extra interference caused by secondary networks

9.3. Spectrum Decision, Sharing and Mobility

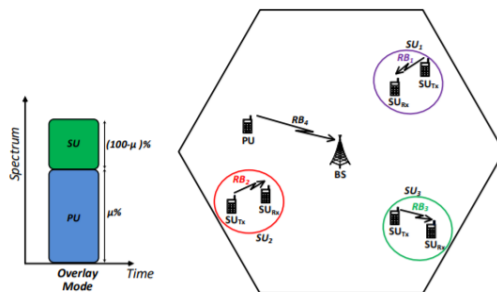
9.3.1. What are the advantages of a centralized secondary network architecture? •

Central entity manages, synchronizes and coordinates operations among secondary users

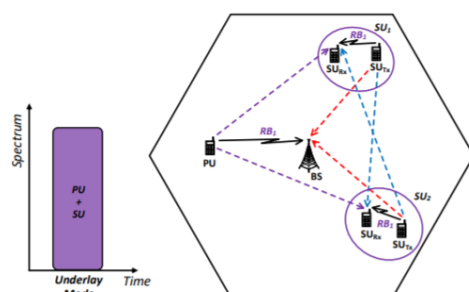
9.3.2. What are the differences between overlay and underlay? •

- Overlay spectrum sharing
 - Secondary users access unused spectrum holes
 - > Avoid interference to primary users
- Underlay spectrum sharing
 - Secondary users stay within acceptable interference temperature
 - Often based on CDMA and spread spectrum techniques

Overlay spectrum sharing



Underlay spectrum sharing



9.3.3. Which criteria are relevant for the effective selection of the spectrum hole? •

- Bandwidth of spectrum hole
- Path loss (interference generated) towards primary users
- Path loss between secondary users
- Characteristics of primary transmission
- Expected holding time of channel

9.3.4. What is spectrum mobility? Why is it needed?

The process when a secondary user changes its frequency of operation, also called spectrum HaNdoF

Reasons:

- Operating (secondary) channel becomes worse
- Primary User returns to the channel
- User movement (available spectrum bands change)

9.3.5. When can a secondary transmitter use a frequency band at the same time as a primary transmitter? What are the requirements for this? Can a secondary transmitter use a frequency band at the same time as primary transmitter?

- Underlay spectrum sharing
 - Secondary users stay within acceptable interference temperature
 - Often based on CDMA and spread spectrum techniques

9.3.6. For which kind of cognitive radio approaches is the application of spectrum spreading and CDMA especially useful? What is the advantage?

- Underlay spectrum sharing

- Secondary users stay within acceptable interference temperature
- Often based on CDMA and spread spectrum techniques

- What is the idea of cognitive engine? What are the tasks to be done in cognitive engine?

What is **Cognitive Engine** (CE) 1. It is known as the brain of CR which provides it intelligence to share the spectrum most efficiently. It consists of different types of interfaces like – User interface, Sensor interface, Radio interface and **CognitiveCore**.

Use paid

9.4. Cognitive Radio MAC Protocols

9.4.1. What are the benefits of having multiple radios for cognitive stations? •

Easy separation between common control channel and data channel,
Parallel operation of control channel and data channel, Faster
handling of control information

- What are different access schemes in cognitive radio networks?

- Overlay spectrum sharing
 - Secondary nodes access spectrum holes not used by primary networks → Interference to primary users is minimized
- Underlay spectrum sharing
 - Based on spread spectrum techniques developed for cellular networks
 - After acquiring spectrum allocation map, secondary users begin sending, so that their transmission power is regarded as noise by licensed users

9.4.2. What are the advantages of centralized MAC schemes? •

Central entity manages, synchronizes and coordinates operations among secondary users

9.4.3. What is the functionality of the common control channel? What are the challenges? - Why common control channel is needed in cognitive radio networks? What are the challenges to find one?

Connection establishment between secondary user transmitter and receiver, Exchange of sensing information among secondary users, Communication with a central entity organizing the spectrum allocation

– Tasks

- Exchange of sensing information
- Coordination (with a central entity) to organize the spectrum allocation

- Transmitter-receiver handshake

– Problems

- Fixed CCC is infeasible (CCC must be vacated when a primary user appears on it)
- CCC may vary over time and between users due to spatial distribution
- If no CCC is allocated, transmitter-receiver handshake becomes a challenge

9.4.4. What are the ideas of CSMA-MAC and SYN-MAC? How is the common control channel implemented in the two schemes? •

Idea CSMA-MAC:

- Operates in underlay (UL) or overlay (OL) spectrum sharing mode depending on PU activity
- Single transceiver required per user, CCC and data on same frequency
- Both PUs and SUs use CSMA to access the medium
- Extended sensing (waiting) period for SUs to prioritize PUs
- Power control for SUs to protect PUs from harmful interference (using CTS)
- SUs discriminate between 3 channel states: Busy, Partially busy, Idle

Idea SYN-MAC:

- Each channel has a time slot dedicated to control messages (control slot)
- The control slot (CCC) is multiplexed in time between the different channels
- Each node has two radios, one for listening to control messages and one for transmission of data

- Is CSMA in cognitive radio same as traditional wireless networks? What is the difference?

Carrier Sense Multiple Access has Carrier sense period in which it senses for primary user.

9.4.5. Identify the differences between CSMA-MAC and SYN-MAC and identify under which situations the individual schemes show their strengths

CSMA-MAC - CR centralized MAC and Single radio, that relates to Random Access MAC protocol.

- Operates in underlay (UL) or overlay (OL) spectrum sharing mode depending on PU activity •
- Single transceiver required per user, CCC and data on same frequency
- Both PUs and SUs use CSMA to access the medium

- Extended sensing (waiting) period for SUs to prioritize PUs
- Power control for SUs to protect PUs from harmful interference (using CTS)
- SUs discriminate between 3 channel states: (Busy, Partially busy, idle)

SYN-MAC - it's CR ad hoc hybrid MAC protocols uses in Multi Radio Syn-MAC:

- Each channel has a time slot dedicated to control messages (control slot) •
- The control slot (CCC) is multiplexed in time between the different channels •

Each node has two radios, one for listening to control messages and one for transmission of data

9.5. Software Defined Radio (SDR)

9.5.1. *What is the idea of SDRs? What is the idea of software defined radio? Give an example about an implementation (adaptive mac approach).*

- Reconfigurability by means of reprogrammable software
- Implement only most necessary things in HW
- Instead of HW components use DSPs, FPGAs, or GPPs
- Term Software Defined Radio also coined by Mitola

Test:

SDR can easily be reconfigured using software,
Most of processing logic is implemented in software,
SDR uses generic hardware components, such as DSPs, FPGAs or GPPs to facilitate the flexible use of radio spectrum

9.5.2. *What are the advantages of SDRs •*

High flexibility to implement different radio technologies

9.5.3. *What are the disadvantages and limits of SDRs? •*

Poorer performance with respect to receiver sensibility and noise,
Higher power consumption compared to fully customized approaches

9.5.4. *How do SDRs compare to traditional radio implementations*

- Designed with concrete applications in mind
- Tailored to fixed requirements and operating conditions
- Highly optimized off-the-shelf and application-specific components
- Optimized for cost, energy-efficiency, and performance
- Static design: modifications impossible after fabricated

10. Cellular Networks

Evolution from GSM to UMTS and LTE (важная инфа)

GSM: voice-dominated, dedicated channels, heavy states

GPRS: adds support for packet data on shared channels; adds IP-based core network

EDGE: increased packet data capacity employing adaptive modulation

UMTS: separate voice and packet data support; focus on dedicated channels and heavy states, complicated RAN architecture and protocols due to macro diversity (soft handover) and QoS requirements

HSPA: improved support for packet data; emphasis on shared channels and fast radio resource management

IMS: support for IP-based multimedia services, e.g. voice (VoIP) or video

LTE: strong packet data support (latency, throughput, control overhead), limited state; simplified protocols; PS only, i.e. no CS core network Evolution of technology • from circuit switching to packet switching • from slow, explicit setup and release of resources to fast channel-condition and demand-specific resource management

10.1. GSM, GPRS & EDGE

10.1.1. What is the idea behind GSM? What is its access scheme? Why is FDD preferred over TDD? Describe the mixture of FDMA and TDMA used for GSM. •

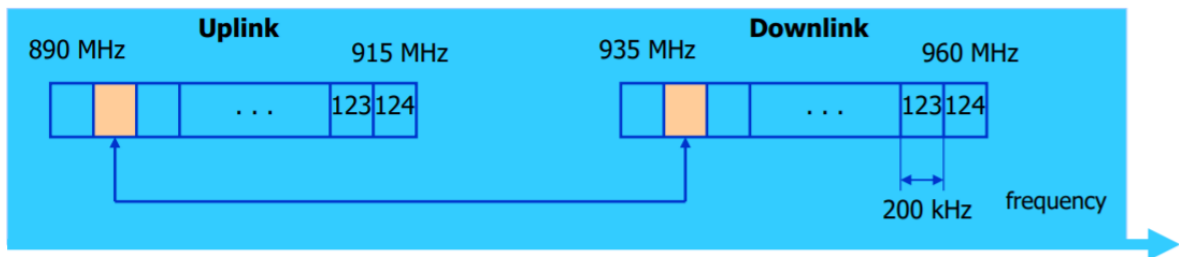
Segmentation of the area into cells. Frequency reuse only beyond a certain distance between base stations.

FDMA (Frequency Division Multiple Access)

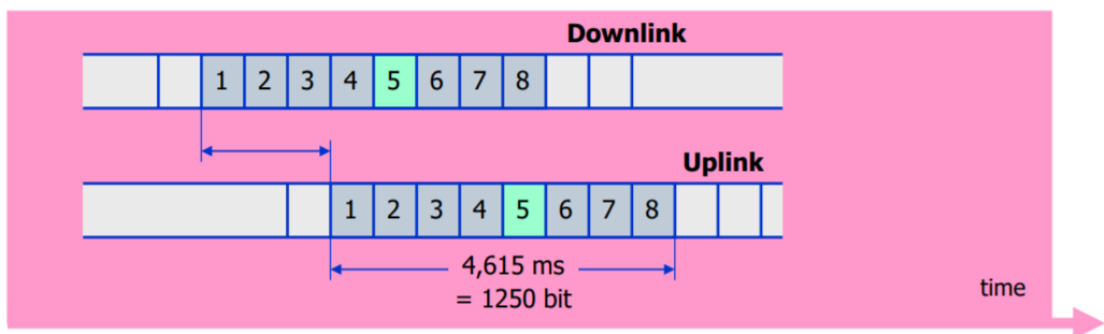
One problem with TDD is that different operators might need to synchronize their base stations, to avoid that one operates in the downlink and one in the uplink on adjacent frequency bands (due to the risk for strong out-of-band interference). Such problems are easily avoided in FDD by separating the uplink and downlink bands.

The FDMA part involves the division by frequency of the (maximum) 25 MHz bandwidth into 124 carrier frequencies spaced 200 kHz apart. One or more carrier frequencies are assigned to each base station. Each of these carrier frequencies is then divided in time, using a TDMA scheme.

FDMA (Frequency Division Multiple Access) / **FDD** (Frequency Division Duplex)



TDMA (Time Division Multiple Access)



10.1.2. GSM handover: what's the idea? What's the difference to a handover in a WLAN system? •

The problem: Change the cell while communicating

Reasons for handover:

- Quality of radio link deteriorates
- Communication in other cell requires less radio resources
- Supported radius is exceeded (e.g. Timing advance in GSM)
- Overload in current cell
- Maintenance

in WLAN (e.g. Wi-Fi) user equipment should disconnect and connect to BSs itself, even if connection with 1st BS getting weaker compared to 2nd BS, it will be still connected to the first one.

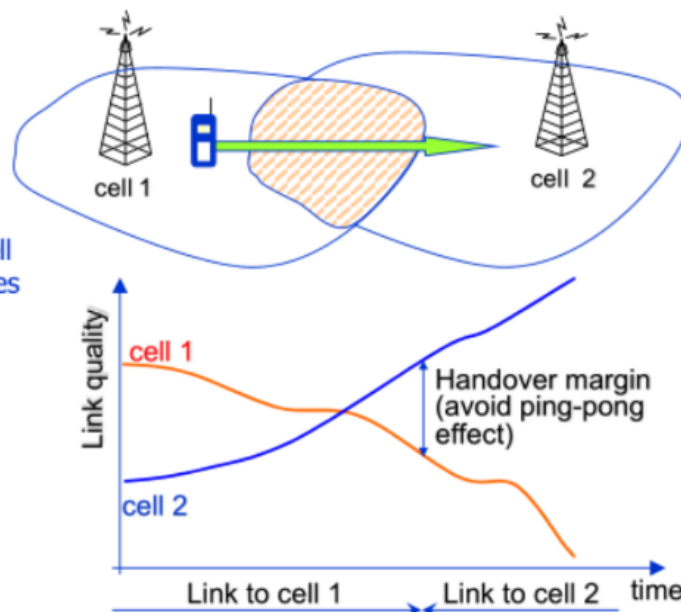
Handover

The problem:

Change the cell while communicating

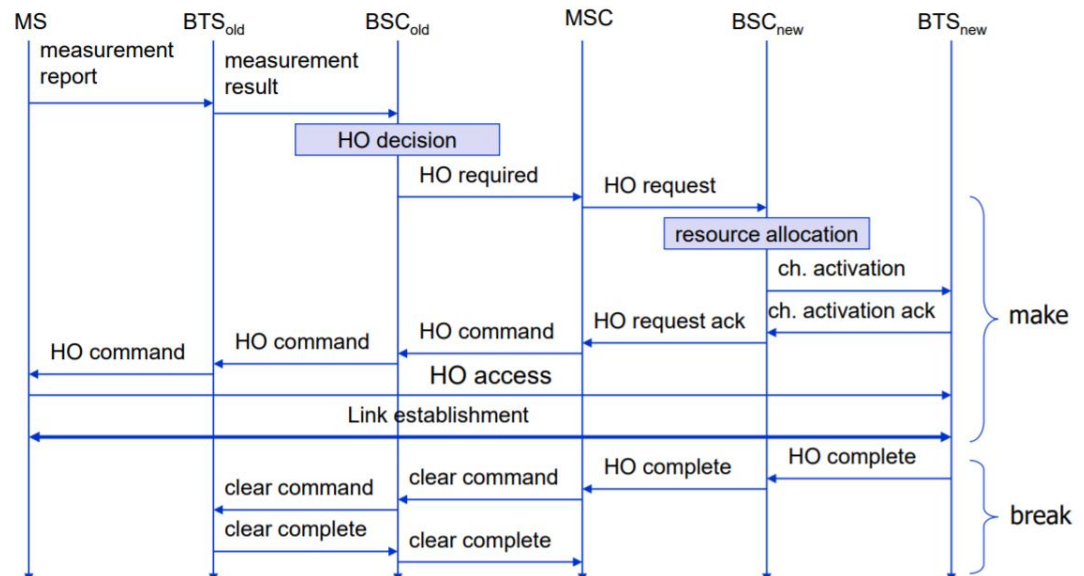
Reasons for handover:

- ◆ Quality of radio link deteriorates
- ◆ Communication in other cell requires less radio resources
- ◆ Supported radius is exceeded (e.g. Timing advance in GSM)
- ◆ Overload in current cell
- ◆ Maintenance



10.1.3. Is there a handover if a GSM mobile does not have an active call? Why or why not? •

yes, because mobile can change its place both with or without active call and network should manage it if it is in serving area

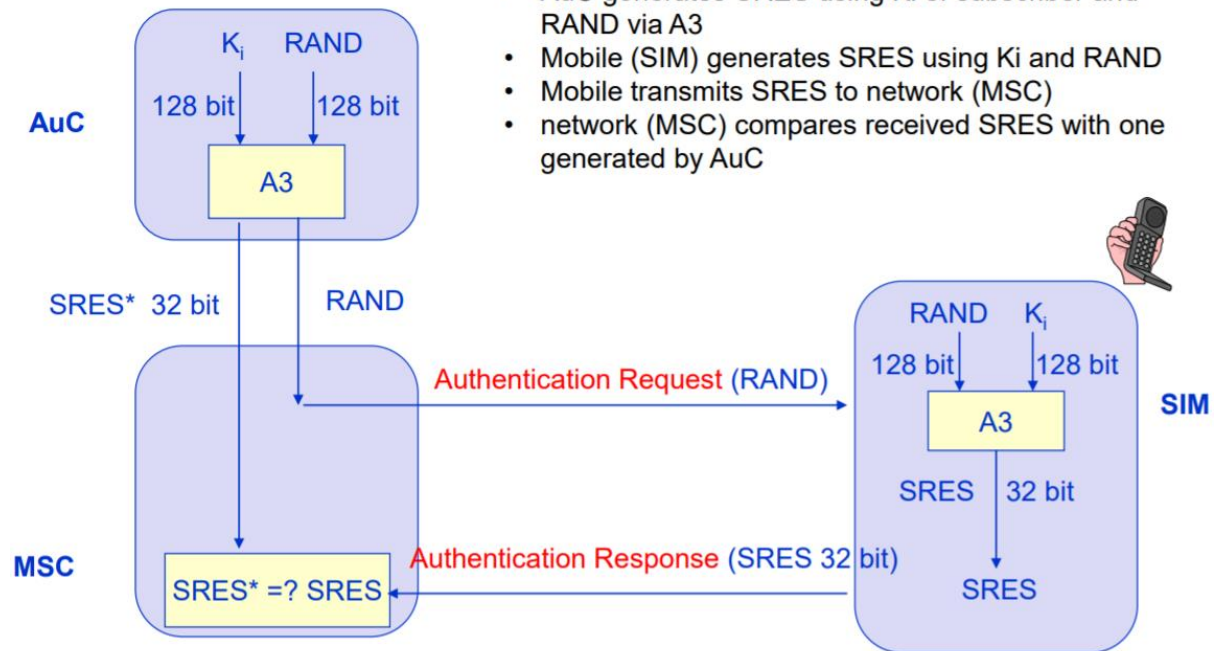


- 10.1.4. How does authentication work in GSM? Who or what is authenticated? How is the integrity of the shared key ensured? •

GSM - authentication

Challenge-Response:

- Authentication center provides RAND to Mobile
- AuC generates SRES using K_i of subscriber and RAND via A3
- Mobile (SIM) generates SRES using K_i and RAND
- Mobile transmits SRES to network (MSC)
- network (MSC) compares received SRES with one generated by AuC

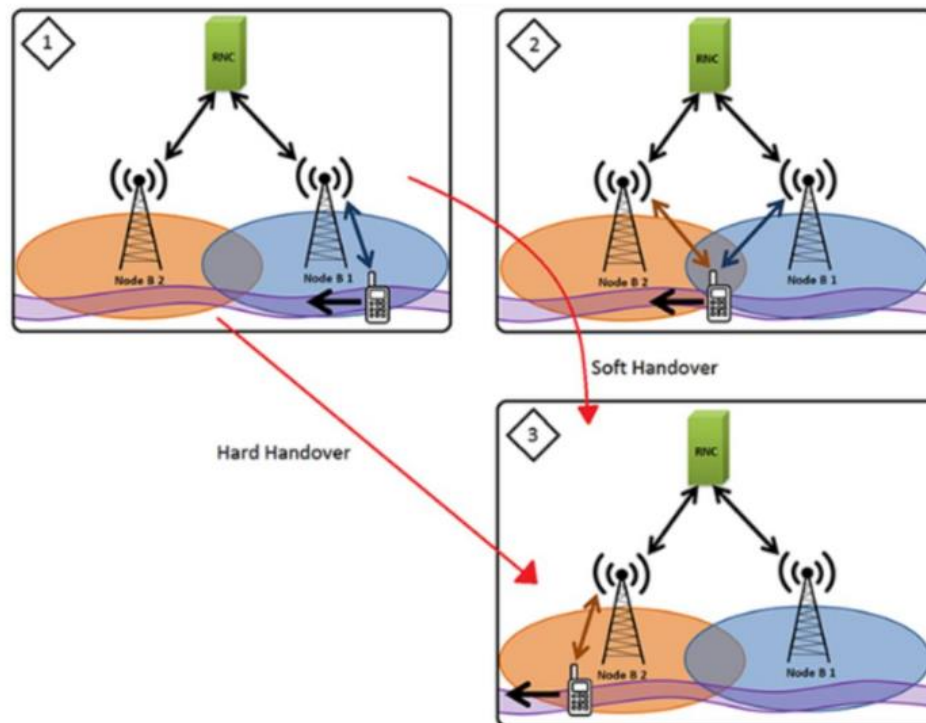


K_i : individual subscriber authentication key SRES: signed response

10.1.5. What is the difference between hard and soft handovers? •

Hard handoff: In GSM, it is general. we can say Break before Make. so in this case higher rates of call drops is found.

Soft Handoff: It means the radio links are added and removed in a way that the MS always keeps at least one radio link to the UTRAN. Make before Break



10.1.6. What is the difference between make-after-break and make-before-break? •

make-before-break - In a switching device, a configuration in which the new connection path is established **before** the previous contacts are opened. This prevents the switched path from ever seeing an open circuit.

make-after-break - In a switching device, a configuration in which the new connection path is established **after** the previous contacts are opened.

10.1.7. What are the main differences between a GSM and a WLAN system? Compare the two systems with respect to media access, radio resource management and handover execution.

Spectrum management and utilization

- coverage and interference management due to cell planning
- capacity and load management due to admission control and QoS support
- flexible cell size simplifies cost-efficient nationwide coverage

Mobility management

- fast, lossless HO due to make-before-break

General control structures and control philosophy

- high reliability and QoS guaranties due to centralized/infrastructurebased management and control of all resources

Energy

- high energy cost on network side, low cost on mobile due to passiv cell camping instead of active association, paging mode and sleep cycles

Customer relations

- monthly/bi-yearly contracts, pay per service
- security due to preshared credentials

Implementation

- simple implementation of TDMA, e.g. with GNUradio and SDR

10.1.8. Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GSM voice and GPRS/EDGE.

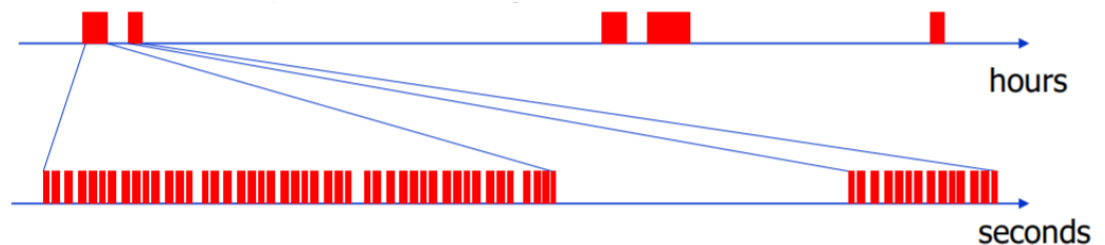
10.2. UMTS (Universal Mobile Telecommunication System)

10.3. Which multiplexing scheme is employed by UMTS? •

Packet session => supported by GPRS/UMTS core, HSDPA, LTE

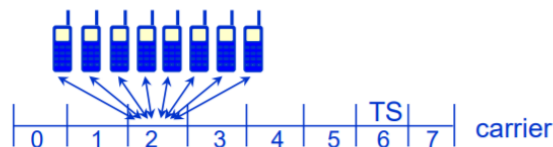
- packet arrival times are typically unknown to the system
- traffic is highly bursty

=> shared channels & packet scheduling

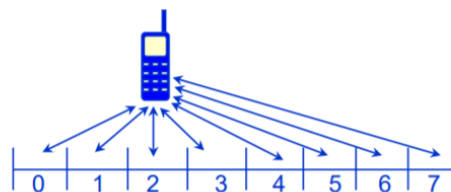


GPRS: Multiplexing and multislot allocation

Multiplexing



Multislot capability



10.4. What does macro diversity mean? •

In the field of wireless communication, macro diversity is a kind of space diversity scheme using several receiver antennas and/or transmitter antennas for transferring the same signal. The distance between the transmitters is much longer than the wavelength, as opposed to microdiversity where the distance is in the order of or shorter than the wavelength.

CDMA is a single frequency system

Mobile user may be served

- by multiple base stations
 - => soft handover
 - => combining in RNC
- by multiple cells from the same base station

- => softer handover
- => combining in base station

Due to increased redundancy, the combining of signals improves quality (BER) or allows to reduce required TX power to achieve given BER requirements

10.5. What does cell breathing in UMTS mean? •

In CDMA-based Cellular networks, cell breathing is a mechanism which allows overloaded cells to offload subscriber traffic to neighbouring cells by changing the geographic size of their service area. Heavily loaded cells decrease in size while neighbouring cells increase their service area to compensate. Thus, some traffic is handed off from the overloaded cell to neighbouring cells, resulting in load balancing.

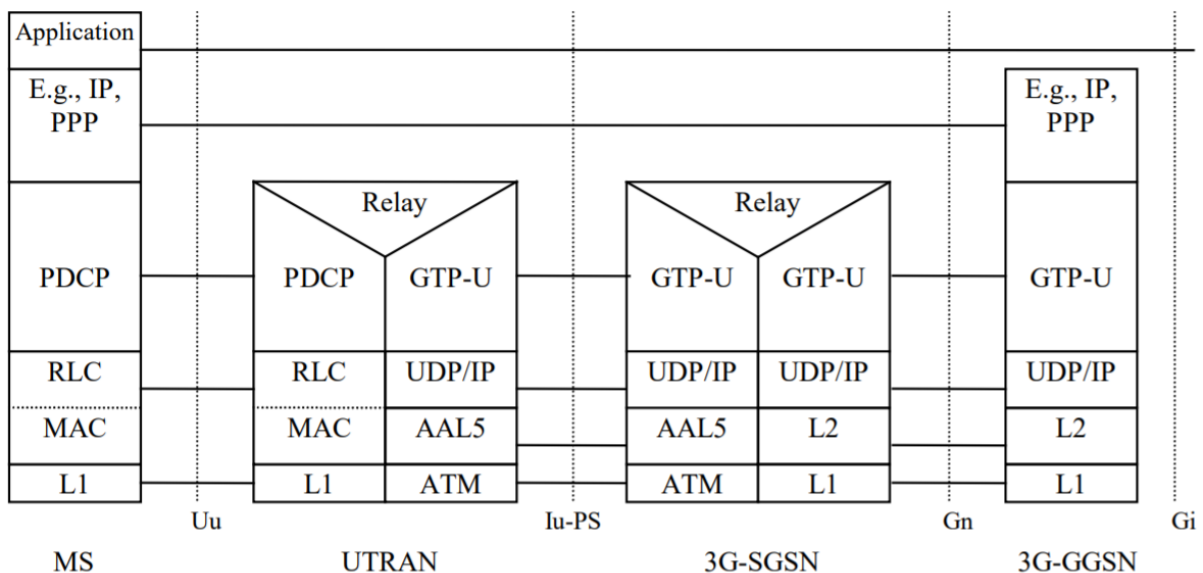
Issue in CDMA systems: cell size depends on the actual load

Effects:

- Additional traffic (i.e. more CDMA codes used) will cause higher interference to users of the same cell
- If the interference becomes too strong, users at the cell edge can no longer meet SINR requirements to continue communication
⇒ Effective coverage area of the cell reduces, i.e. cell shrinks

10.6. How does mobility management work in UMTS? •

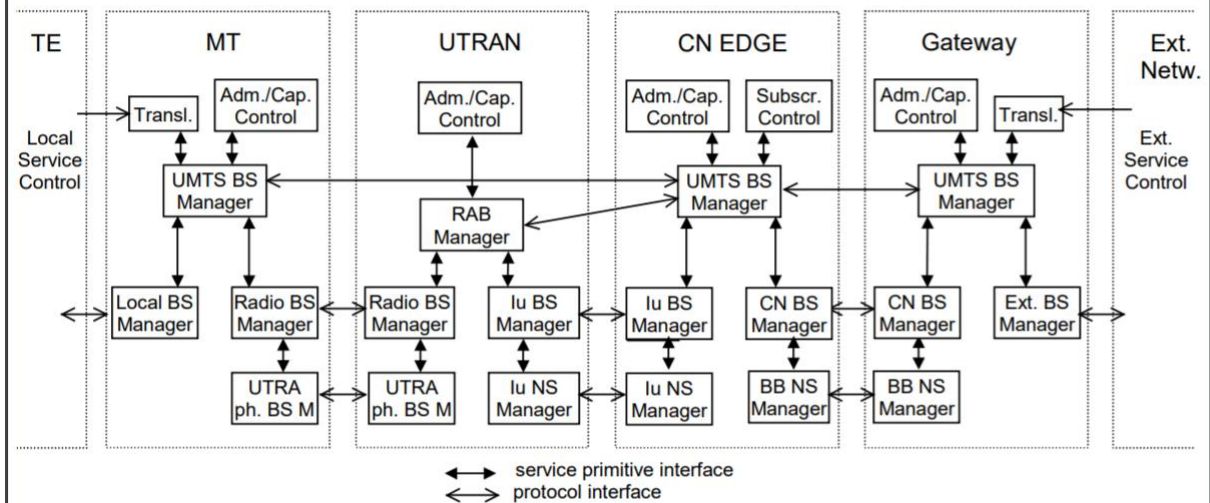
UMTS – Mobility Management in Packet Domain



10.7. How does QoS management work in UMTS?

End-to-End Resource Management in UMTS (contr. plane)

A sophisticated QoS architecture



11. LTE (Long-Term Evolution)

11.1. What is the advantage of a common IP-based packet core over separated networks for packet-switched and circuit-switched traffic? •

Packet-switched networks move data in separate, small blocks -- packets -- based on the destination address in each packet. When received, packets are reassembled in the proper sequence to make up the message. (data)

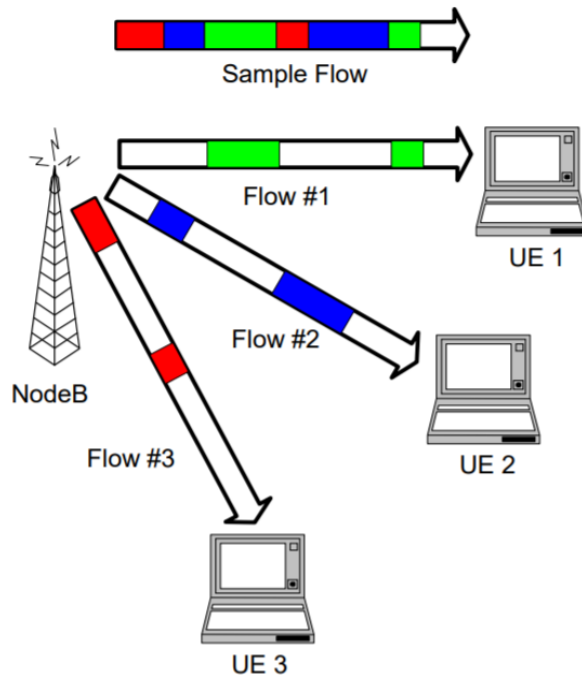
Circuit-switched networks require dedicated point-to-point connections during calls (voice)

Common core greatly simplifies design and implementation. VoLTE is cheaper and more versatile, that's why operators migrate to it.

11.2. What is the purpose of opportunistic scheduling? What is the disadvantage? •

Prefer mobiles with good channel quality => maximizes system throughput

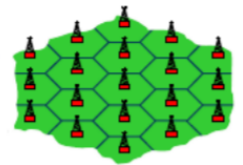
Ensure fairness for users with continuously poor channel quality



11.3. How does interference cancellation work? What is the idea? •

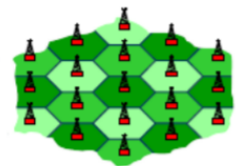
High data rates in limited spectrum allocations

- Entire spectrum must be available in each cell
- ➔ One-cell frequency reuse



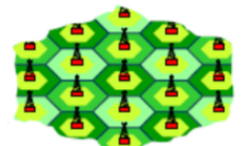
Reduced inter-cell interference with frequency reuse > 1

- Improved cell-edge SIR ➔ Higher cell-edge data rates



Adaptive reuse

- Cell-center users: Reuse = 1
- Cell-edge users: Reuse > 1



11.4. What is the goal of self-organization in LTE? •

Goal: minimize OPEX(operating expense) by automation of planning, optimization and repair of the network

11.5. What are the use cases for self-organized network management in LTE?

Use Cases for Self-organized Network Management

- Physical cell-ID automatic configuration (PCI)
- Automatic Neighbor Relation (ANR)
- Coverage and capacity optimization (CCO)

- Inter-cell interference coordination (ICIC)
- Random Access Channel (RACH) optimization
- Mobility load balancing optimization (MLB)
- Mobility robust optimization (MRO)
- Energy saving (power on/off)

Quizzes

Intro

1. Entry Quiz

1.1. Which of the following statements hold for the OSI model?

The correct answers are: It defines an architecture to structure communication systems, It divides functions needed for communications in different layers, The layers build on each other

1.2. Which of the following statements hold for the OSI model?

The correct answers are: It implements the idea of encapsulation and decapsulation, It defines a hierarchy of functionalities that build on each other

1.3. Which clauses hold for TCP?

The correct answers are: It is not a transport protocol, It controls overload in networks, It decides on the data rate at which data are transmitted, It corrects errors, It is connection-oriented

1.4. Which clauses hold for UDP?

The correct answer is: It is a transport protocol

1.5. What is a port?

The correct answers are: An entity defining and connecting an application to the TCP/IP stack, A part of the TCP SAP, An address to identify a host

1.6. Which of the clauses hold for the physical layer?

The correct answers are: It defines the rules to transmit data over a physically medium, It implements channel coding

1.7. Which functionality is provided by IP?

The correct answers are: Addressing of network elements, Forwarding of IP packets at routers, Segmentation of packets

1.8. What does VPN mean?

The correct answer is: Virtual Private Network

1.9. What are essential ingredients of an VPN system?

The correct answers are: Encryption of data, Authentication, Authorization, Gateway/Firewall

1.10. Which of the following networks connects multiple private networks located at different sites across public networks?

The correct answer is: Virtual Private Network

1.11. Which clauses hold for CSMA?

The correct answer is: It provides a function to control the load on the physical link

1.12. Which protocol layers provide end-to-end services?

The correct answers are: Application layer, Network layer, Transport layer

1.13. Which statements holds for a switch in a network?

The correct answers are: it uses MAC addresses for forwarding, it learns its forwarding table from the traffic

1.14. What is the purpose of DHCP?

The correct answers are: it provides a computer with an IP address, it is not needed where static IP addresses are used

1.15. _____ is used by applications that need a fast transport mechanism and can tolerate the loss of some data.

The correct answer is: UDP

1.16. Which of the following is not related to error detection?

The correct answer is: Burst error

Wireless Transmission

2. Quiz: Wireless Transmission - Frequency and Signals

- 2.1. How does the frequency spectrum of a sinusoidal signal with 100 MHz look like?

The correct answer is: There is a single peak at 100 MHz only

- 2.2. How does the frequency spectrum of a symmetric digital signal with a frequency of 100 MHz look like?

The correct answer is: There are peaks at 100, 300, 500, 700, ... MHz

- 2.3. How does the frequency spectrum of a symmetric triangular signal with a frequency of 100 MHz look like?

The correct answer is: There are peaks at 100, 300, 500, 700, ... MHz

- 2.4. What are the pros and cons of using higher frequencies for communications?

The correct answers are: More unused frequency space available,
Problems with penetration of obstacles as walls, doors, windows, etc.,
Smaller antennas

- 2.5. What is the wavelength of the GSM 1800 MHz band?

The correct answer is: 16,7 cm

- 2.6. What is the wavelength of the GSM 900 MHz band?

The correct answer is: 33,3 cm

- 2.7. What is the wavelength of WLAN at 2,4 band?

The correct answer is: 12,5 cm

- 2.8. What is the wavelength of WLAN at 5 GHz band?

The correct answer is: 6 cm

- 2.9. Which GSM frequency band typically provides better indoor coverage (from outdoor base stations)? What is the justification for this?

The correct answers are: GSM 900, Higher frequencies are easily absorbed by walls

2.10. Which of the following statements hold for the OSI model?

The correct answers are: It defines an architecture to structure communication systems, It divides functions needed for communications in different layers, The layers build on each other

3. Quiz: Wireless Transmission - Signal Propagation

3.1. In which scenarios would fast fading be experienced?

The correct answers are: Mobility of receiver, Mobility of objects in the transmission path

3.2. What are the reasons for fast fading?

The correct answers are: Movement of sender, Multiple transmission paths between sender and receiver

3.3. What exactly happens during fast fading?

The correct answer is: Multiple transmission paths from a single sender result in overlap of the signal at the receiver

3.4. What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 360° between two equal strength multipath signals result in:

The correct answer is: an increase in the signal strength

3.5. What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 180° between two multipath signals with equal strength result in:

The correct answer is: a cancelation of the signal as a whole

3.6. What is the effect of small phase shifts between multiple versions of the arriving signal? Phase shifts of 90° between two equal strength multipath signals results in:

The correct answer is: an increase in the signal strength

3.7. Why does fast fading depend on mobility?

The correct answer is: Because fast fading is due to a shift in the phases between multiple paths which requires some dynamic behavior

3.8. Why is fast fading frequency-dependent?

The correct answer is: Because the exact phase depends on the frequency

- 3.9. Why is it considered as the best case that the pathloss for line-of-sight connections is inversely proportional to the square of the distance between a transmitter and receiver?

The correct answer is: Assuming an isotropic radiator and zero absorption in space, the sum of the energy at any radius of the bowl can be assumed the same independent of the size of the surface of the bowl itself. However, as the surface of the bowl is proportional to the square of its radius, the energy per any fixed area reduces proportional to the inverse squared distance.

4. Quiz: Wireless Transmission - Antennas

- 4.1. Assume you have got 2 simple $\lambda/2$ antennas. How can you build a simple directional antenna from them?

The correct answer is: Separate the antenna by $\lambda/2$ and provide the same signal to both

- 4.2. What are the characteristics of an isotropic radiator?

The correct answers are: Emits same power in all directions, Is used as a reference for other antenna systems, Power is equally distributed in all three dimensions

- 4.3. What directional effects are shown by a simple dipole antenna?

The correct answer is: Dipole emits in two dimensions only

5. Quiz: Wireless Transmission - Channel Capacity (Signal vs. Interference)

- 5.1. From what aspects does Shannon abstract from?

The correct answers are: Modulation scheme, Receiver sensitivity, Receiver-induced noise

- 5.2. How can Shannon's law be used to estimate the capacity in interference-limited system?

The correct answers are: We assume interference to be similar to noise and apply the same formula replacing noise by interference, We assume that interference is much higher than noise so that noise can be neglected

- 5.3. How can the capacity in interference-limited systems be increased further?

The correct answers are: Add more base stations to reduce the pathloss, Add more sectors to the given base station infrastructure to decrease pathloss while decreasing interference, Add more frequency bands

5.4. **Impact of channel bandwidth, received signal strength and noise on the capacity of a channel: are there approximations for these factors?**

The correct answer is: Yes, Shannon's law

5.5. **The following factors affect the quality of reception:**

The correct answers are: Coding scheme, Signal bandwidth, Receiver-induced noise, Multipath propagation, Pathloss of radio channel from sender to receiver

5.6. **What are core results of Shannon's capacity law?**

The correct answers are: The channel capacity is proportional to the channel bandwidth, Signal and Noise are reciprocal of each other

5.7. **What defines interference-limited systems?**

The correct answers are: The addition of new transmitters to the system introduces so much interference to others, that there is no benefit for the overall system capacity, The increase of the transmit power of a single terminal does not increase its capacity

5.8. **What factors influence the quality of the reception at a given receiver's input?**

The correct answers are: Strength of the signal as such, Interference from others, Noise, Intersymbol interference, Modulation scheme

5.9. **What further ideas can be employed to increase the capacity in interference-limited systems?**

The correct answers are: Use D2D communication where applicable, i.e. between terminals that are close together, Use interference-cancellation techniques, Use directional antennas at the terminals

5.10. **What is not covered by Shannon's capacity law?**

The correct answers are: Intersymbol interference, Noise other than additive white Gaussian noise

6. Quiz: Wireless Transmission - Multiplexing (Splitting up the Radio Resource)

6.1. **What are the advantages/disadvantages of multiplexing in time?**

The correct answers are: Flexible scheme to quickly adapt to the traffic demand, No guard band needed

6.2. What is the purpose of multiplexing in communication?

The correct answers are: Separate multiple parallel communications from each other, Separate channels to minimize interference between them

6.3. What kind of multiplexing is used for public FM radio?

The correct answers are: Frequency multiplexing, Spatial multiplexing

6.4. Which multiplexing scheme(s) is applied by GSM?

The correct answers are: Frequency multiplexing, Time multiplexing, Spatial multiplexing

7. Quiz: Wireless Transmission - Modulation and Coding

7.1. How can we optimize the throughput of a single transmitter-receiver pair under given bandwidth constraints as well as fixed signal and interference values?

The correct answers are: Adapt modulation scheme, Adapt coding scheme, Use multiple antennas

7.2. How is direct communication between mobiles (D2D) a solution to improve the network capacity of cellular networks?

The correct answers are: Proximity gain due to smaller distances between sender and receiver, Gain due to one instead of two wireless communications (sender to base station, base station to receiver)

7.3. What are the available choices of adding information to a periodic signal?

The correct answers are: Phase modulation, Amplitude modulation, Frequency modulation

7.4. What is the advantage of QPSK over 16-QAM?

The correct answers are: Lower error rate, Fewer problems with multipath propagation

7.5. What is the benefit of frequency hopping?

The correct answers are: Distribution of the signal over a wider bandwidth, Reduced sensitivity to narrowband interference and noise

7.6. What is the benefit of spectrum spreading technology?

The correct answers are: Distribution of the signal over a wider bandwidth, Reduced sensitivity to narrowband interference and noise

7.7. What limits the use of higher order modulation schemes, as e.g. 64-QAM?

The correct answers are: Highly sensitive to distortions in the amplitude of the signal, Highly sensitive to distortions in the phase of the signal, Highly sensitive to multipath problems, High SINR needed for proper transmission

7.8. Which modulation schemes apply some sort of amplitude modulation?

The correct answers are: 16-QAM, 64-QAM

7.9. Why is amplitude modulation in its pure form rarely used in practice?

The correct answers are: Highly sensitive to multipath problems, Receivers are highly sensitive to changing channel conditions

8. Quiz: Wireless Transmission - Cellular Networks (Capacity and Interference)

8.1. Compare the spatial reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate). Assume the same total frequency spectrum and same modulation and coding schemes for both schemes.

The correct answer is: Reuse 3 provides higher raw throughput rate but with higher error rates

8.2. How can the capacity of a cellular system be improved assuming that we have an interference-limited system?

The correct answers are: Increase the number of base stations, Prefer users at the cell center instead of users at the cell edge, Increase the spatial diversity using multi-sector antennas, Use better receivers that can tolerate a lower signal-to-interference ratio

8.3. Under which circumstances is direct communication between two mobiles a solution to improve the network capacity of cellular networks?

The correct answers are: When the two mobiles are close to each other and reside in the same cell, When the two mobiles are close to each other and reside in different cells

- 8.4. What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available? What are the additional measures that need to be taken into account to handle this?

The correct answers are: Adapt the frequency reuse scheme, Adapt the transmit power of base stations, Adapt the transmit power of mobiles

Medium Access Schemes

9. Quiz: Medium Access Schemes - Basics of Wireless Access Control

- 9.1. Name the different dimensions in which we can separate the access to allow for multiple communications in parallel?

The correct answers are: Space Division Multiple Access, Frequency Division Multiple Access, Time Division Multiple Access

- 9.2. Three wireless terminals are given: A, B and C. A can communicate with B only. C can communicate with B only. A sends to B, C wants to send to B. Which terminal is “hidden” in this case? At which terminal the collision may happen in this case?

The correct answers are: A is “hidden” for C, C is “hidden” for A, Collision happens at B

- 9.3. Three wireless terminals are given: A, B and C. A can communicate with B only. C can communicate with B only. A is hidden from C. B sends to A. Which terminal is considered “exposed” to this specific communication?

The correct answer is: C is “exposed” to B

- 9.4. What are the causes of the extra problems in wireless MAC?

The correct answers are: Huge variation of the signal strength over distance, Much higher signal loss than wired, Changes of the signal strength due to mobility

- 9.5. What are the new problems with respect to MAC in wireless systems?

The correct answers are: Collisions happening at the receiver may not be noted by the sender, The sender and receiver do not have a consistent view on the radio resource

9.6. **What does Media Access Control deal with?**

The correct answers are: Control the dynamic access of senders to the medium, Avoid conflicts between multiple senders, Schedule a common radio resource dynamically

9.7. **What is the cause of the near-far terminal problem?**

The correct answers are: Due to the decrease of signal strength with distance the receiver will receive more power from the nearest transmitter, The nearest transmitter “drowns out” the signal from the farthest transmitter

9.8. **What is the responsibility of the Media Access Control sublayer in communication systems?**

The correct answers are: Control for accessing the transmission medium, Addressing, Arbitration of access from different devices

9.9. **What problems may happen with sensing and collision detection in wireless networks?**

The correct answers are: Sensing may report a free channel though it is busy, Sensing may report a busy channel though it is free, Collision detection may not report a collision though it happens, Collision detection may report a collision though none happens

9.10. **Why does CSMA/CD known from the Ethernet fails in wireless networks?**

The correct answers are: Because Carrier Sensing would be needed at the receiver and not at the sender, Carrier Sensing may not work because of the Hidden Terminal problem, Carrier Sensing (CS) and Collision Detection (CD) may not work because of high signal attenuation in wireless networks, Because Collision Detection would be needed at the receiver and not at the sender side

10. Quiz: Medium Access Schemes - Multiplexing and Duplexing

10.1. **Application of FDD and TDD: Please choose correct answers.**

The correct answer is: FDD is used in GSM

10.2. **What is simplex communication?**

The correct answers are: A communication channel that is used in one direction only, Streaming applications are an example, TV is another example of this

10.3. Which of the statements on TDD and FDD are correct?

The correct answers are: TDD is suitable for unsymmetrical services, FDD separates uplink and downlink interference

11. Quiz: Medium Access Schemes - TDMA (Scheduling the Access)

11.1. Compare the reserved-access and random access strategies with respect to the maximum load they can handle.

Random access schemes exhibit lower maximum throughput due to high number of collisions, Reservation-based schemes show higher network capacity due to the absence of data collision

11.2. How can the hidden-terminal problem be solved?

Use of short signaling packets to reserve for a single data transmission, A sender initiates the transaction with a request of the right to send data to a receiver, The receiver grants the right to transmit data to the sender, Both sender and receiver inform their neighbors of an upcoming communication

11.3. How does explicit reservation based on ALOHA work?

Stations compete during small reservation timeslots for data transmission timeslots, The station needs to compete for the timeslot in each frame

11.4. How does explicit reservation work?

The sender and receiver explicitly agree on a resource prior to its use

11.5. How does implicit reservation work?

After successful timeslot reservation, the station continues to use the same timeslot without competition in the following frame, The station releases the timeslot implicitly when it does not have any more data to send

11.6. How does reservation TDMA work?

Each station is provided a small reservation timeslot to reserve a number of data slots in a collision free way, Each station is provided with a dedicated timeslot to make a reservation, A station may also compete for un-reserved data slots in each TDMA frame, The mechanism ensures that each station has access to a guaranteed data volume in each frame

11.7. Name the pros and cons of distributed approaches to media access?

Inconsistent view on radio resource may result in collisions, No need for an infrastructure and management

11.8. What are the pros and cons of the random access and reservation-based access strategies? Compare the access latencies of both access schemes.

- The random-access strategy has lower latency under low load, The reservation-based access strategy may experience low latencies for data transmitted over given reservations
- 11.9. What is correct for pure ALOHA?
New packets are transmitted immediately, In case of packet collisions, the station waits a random time and retransmits the packet again
- 11.10. When are collisions possible in MACA if we assume a stationary wireless system?
During RTS packet transmission, During CTS packet transmission
- 11.11. Which information is contained in signaling packets in MACA?
Sender address, Receiver address, Packet size
- 11.12. Which of the following problems are solved by RTS/CTS scheme (MACA)?
Hidden Terminal problem, Exposed Terminal problem
- 11.13. Why does slotted ALOHA outperform pure ALOHA with respect to throughput?
In slotted ALOHA, two transmissions can collide only if they begin at the start of the same time slot

12. Quiz: Medium Access Schemes - CDMA

- 12.1. What are advantages of CDMA?
All terminals can use the same frequency, less planning needed, Huge code space compared to frequency space, Noise is not coded, Forward error correction and encryption can be easily integrated
- 12.2. What is the advantage of CDMA with respect to the handling of noise?
Noise is not coded and thus, its impact on the signal is reduced, CDMA supports useful communication with a much lower SINR compared to other techniques
- 12.3. What is the trick with CDMA to avoid collisions?
All users share same frequency, but each user has own "chipping" sequence (i.e. code) to encode data, which allows multiple users to coexist and transmit simultaneously with minimal interference
- 12.4. Which statement is correct for CDMA?
All terminals use the same frequency band at the same time, The code imposed on the data has a much higher rate as the data itself, Each user has a unique pseudo random code
- 12.5. Why is power control important for the correct function of CDMA?
It is important to reduce the impact of the Near-Far Problem, Unbalanced received power levels combined with imperfect spreading codes with limited orthogonality will cause errors in detection

IEEE 802.11

13. Quiz: IEEE 802.11 - Basics

- 13.1. What are the main differences between the different PHY variants?
Radio implementations, Modulation schemes, Data rates
- 13.2. What is the function of the PLCP sublayer in 802.11?
Clear channel assessment signal (carrier sense)
- 13.3. Which protocol layers does the 802.11 standard comprise?
MAC layer, PHY layer

14. Quiz: IEEE 802.11 - MAC

- 14.1. Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?
No, typically only a single RTS/CTS cycle is required
- 14.2. Consider the case that a new packet internally arrives from higher layers. For which cases does the backoff apply?
In the case that the medium is sensed as busy, In the case that the NAV has indicated a busy medium
- 14.3. Exponential backoff: what is its purpose?
It delays retransmissions after collisions to reduce the risk of subsequent collisions, It reduces the risk of collisions compared to ALOHA, It translates high network load in delay to avoid collisions, It tries to maximize the handled network load
- 14.4. How is the exponential backoff time computed?
It is computed from a uniformly distributed pseudorandom number from the interval $[0, CW]$ multiplied by the slot time, In case of a repeated collision, the stations double the backoff time
- 14.5. What is correct for DCF?
DCF utilizes a contention-based MAC mechanism
- 14.6. What is correct for DIFS?
Lowest priority for random access
- 14.7. What is correct for PCF?
PCF provides some support for time-bounded data services, PCF utilizes a contention-free MAC mechanism
- 14.8. What is correct for PIFS?
Priority for polled stations over PCF
- 14.9. What is correct for SIFS?
Highest priority: for ACK, CTS, polling responses
- 14.10. What is the purpose of different interframe spaces?
Differentiate the access priority between different types of messages, Ensure the non-delayed transmission of CTSs, Ensure the non-delayed transmission of ACKs

- 14.11. What is the purpose of the NAV?
NAV indicates the duration of the reservation of the medium, It tries to solve the inconsistency problem in wireless systems existing between different stations, NAV reduces the number of collisions
- 14.12. Which mechanism is preferable for high-load situations?
PCF ensures a better utilization of the medium

15. Quiz: IEEE 802.11 - MAC Management

- 15.1. Does power saving work in ad-hoc mode as well?
Yes, but there might be collisions between beacons, Yes, stations need to buffer frames for sleeping receivers
- 15.2. How are collisions among beacons avoided in ad-hoc mode?
Sending of beacon employs a random delay to avoid collisions
- 15.3. How does beaconing work in ad-hoc mode?
Beacons may be sent by any station, Beacons contain a timestamp and other management information used for power management and roaming, Local clock is only updated when received timestamp is more recent
- 15.4. What is correct for a DTIM?
DTIM contains a list of broadcast receivers, DTIM contains a list of multicast receivers
- 15.5. What is correct for a TIM?
TIM contains a list of unicast receivers, TIM is used to notify the stations about frames buffered in the AP
- 15.6. What's the idea behind energy saving in 802.11?
Station has two states: sleep and awake, Stations need to buffer frames for sleeping receivers
- 15.7. Which steps are required for a handover in 802.11?
Scanning of channels, Association to the new AP
- 15.8. Why is a handover in 802.11 slow, compared to GSM or UMTS?
During scanning for another AP, a station needs to wait in each channel for beacons, Because it is not possible to monitor several channels with a singleband receiver at the same time, The station does not start the scanning before the old association is lost

16. Quiz: IEEE 802.11 - Newer Developments and Physical Variants

- 16.1. What are advantages of using higher frequencies?
Availability of unused spectrum, Higher bandwidth available, Typically less crowded bands
- 16.2. What are typical disadvantages of using higher frequencies?
Higher loss for non-line-of-sight transmissions, Higher loss for penetrating obstacles

- 16.3. Which techniques are used in 802.11ac to maximize the throughput?
Higher modulation schemes, Higher bandwidth of channels, Parallel transmissions to multiple mobiles at the same time
- 16.4. Which techniques are used in 802.11g to adapt to varying channel quality?
adaptive modulation schemes, adaptive coding
- 16.5. Why does the maximum throughput in 802.11a depend on the distance between the stations?
Pathloss increases over distance, Received signal strength decreases over distance, SINR decreases over distance
- 16.6. Why is the transmit power typically limited to 100 mW?
To reduce interference inside the ISM band, To reduce the risk of collisions

Mobility Management

17. Quiz: Mobility Management - Basics

- 17.1. Identify performance goals for mobility management mechanisms?
Minimize the number of lost packets during the handover process, Minimize the overhead for data and signaling traffic, Scalability of the mobility management solution to large networks, Scalability of the mobility management solution to a large number of mobiles
- 17.2. What are the implications of mobility management by the application layer?
Mobility may result in changes of the TCP or UDP port, Mobility may result in changes of the IP address, Mobility may result in changes of the radio link
- 17.3. What are the implications of mobility management by the link layer?
Mobility may result in changes in the radio link, Mobility is transparent to the transport layer, Mobility management is local to the current subnet
- 17.4. What are the implications of mobility management by the network layer?
Mobility may result in changes of the IP address, Mobility may result in changes of the radio link, Mobility is transparent to the transport layer
- 17.5. What are the implications of mobility management by the transport layer?
Mobility may result in changes of the IP address
- 17.6. What are the tasks of seamless mobility management?
Enable that a mobile is reachable wherever it is located, Ensure that communication continues even after the mobile associates to a

different point of access, Ensure that the application is not interrupted due to mobility

- 17.7. Which is the highest layer in the TCP/IP model that is commonly implemented in routers?

IP

- 17.8. Which is the lowest layer protocol in the TCP/IP model to support end-to-end packet flow between communicating entities?

IP

- 17.9. Why can the cooperation between layers improve the handover performance?

Lower layers may inform higher layers of upcoming radio problems (decrease of SINR) before the transport connection actually fails due to errors, Higher layers may temporarily freeze its transmission during the handover in order to reduce the number of transmission errors, Higher layers may temporarily freeze its transmission during the handover in order to reduce the number of packets transmitted over an obsolete route, Higher layer may instruct PHY layer on which band to search for channels

18. Quiz: Mobility Management - Link Layer Mobility

- 18.1. Identify ideas how handovers could be accelerated:

Scan the medium for other APs before the actual radio link breaks (possibly using extra HW), Change to a new channel before the radio link on the current channel breaks, Provide the AP with radio information so that it can initiate the handover before the current link fails, Try to stay connected to multiple radio links (APs) at the same time so that there is a redundant link whenever another one fails

- 18.2. What are the main sources for handover delay in WLANs?

The time needed to go through all channels with possible APs, The time required to receive beacons on all channels to identify Aps

- 18.3. What happens if a mobile moves out of WLAN coverage?

The mobile searches for a new AP after identifying that no communication is possible anymore

- 18.4. Which mobility problems can be handled on the link and PHY layer?

Mobiles moving within a single IP subnet, Mobiles moving within the switched layer 2 network, Mobility within the same AP, e.g. changing the band from 2.4 to 5 GHz

18.5.

18.6.

19. Quiz: Mobility Management - Network Layer Mobility

- 19.1. How can mobility – at least in theory – be supported by the network layer?

Adapt the routing tables of all routers of the network to ensure proper forwarding of the packets to the location of the mobile, Provide the mobile with a new IP address whenever it moves to a new subnet, Forward all data to all possible locations of the mobile, Install some central instance in the network that always knows the location of the mobile and where the destination address of packets is adapted so that it can be forwarded to the current location of the mobile

19.2. Identify characteristics of network layer mobility solutions

Mobility may be handled by the network protocol in the mobile node, Mobility may be handled by routers in the network

19.3. What is the problem with changing the IP address for mobiles due to mobility?

Typical terminal applications cannot handle changing IP addresses and fail, TCP connections cannot handle changing IP addresses and fail, Servers typically do not associate packets with a new source address (due to the mobiles new location) to the same connection and session

19.4. What is the purpose of tunneling? What is the advantage of this over specific routes?

Keep the old IP address while using a new one for location purposes, Separating the terminal locator from the terminal identifier, Provide the original packet to the mobile node so that it is able to reply in the same way as when staying in its home network, Maintaining a static IP address while moving among subnets, Transparency of the mobile location to the sender (server)

19.5. When does the sole update of routing tables make sense?

In strictly tree-structured networks where all traffic passes through a central gateway, When the number of involved nodes is small

19.6. Which of the solution from the above is the most practical?

Install some central instance in the network that always knows the location of the mobile and where the destination address of packets is adapted so that it can be forwarded to the current location of the mobile

19.7. Why do individual routes for each mobile not scale?

It requires updates in all routers (routing table) for every mobile that changes its subnet to ensure the mobiles reachability

20. Quiz: Mobility Management - Mobile IP

20.1. How can we solve the triangular routing problem?

Signal the CoA to the corresponding node for direct tunneling to the mobile terminal

20.2. How differs CoA-handling in the FA from handling in the MN itself?

In the first case the CoA is assigned by the FA, In the second case, no special support is needed in the foreign network

- 20.3. **What is the functionality provided by DHCP?**
Dynamically assign IP addresses to network entities, Provide mobile IP addresses to MNs to support global mobility
- 20.4. **What is the overhead for signaling with Mobile IP (basic version)?**
Overhead for updating of the home agent upon changes of the mobile's IP address
- 20.5. **What is the overhead imposed on data traffic with Mobile IP?**
Overhead for additional IP header for any data packet sent towards the mobile, Overhead for finding the current IP from the static IP in the HAs table
- 20.6. **What is the purpose of reverse tunneling? In which case is it helpful?**
It solves the problem that firewalls block outbound traffic with nonlocal IP source addresses
- 20.7. **When is triangular routing especially harmful?**
When the mobile terminal and the corresponding node are close to each other but the home agent is far from both of them
- 20.8. **Which optimizations can be made in order to smooth data transport during and shortly after a handover?**
New FA informs old FA of the handover, Old FA tunnels remaining packets for the mobile to the new FA

21. Quiz: Mobility Management - Network Layer Mobility (Advanced Concepts)

- 21.1. **Identify the limitations of Mobile IP v4.**
Limited address space of IPv4 makes it hard to serve every MN on earth, High distances between MN and HA result in high delays even for local handovers, MIP tunnels may conflict with VPN tunnels, Tunneling complicates the provision of end-to-end QoS, Overhead for tunneling
- 21.2. **What are the ideas of Cellular IP?**
Exploit the advantages of tree-structured cellular networks, Manage mobility locally in the network, Learn the downlink routes automatically from the MNs uplink data packets
- 21.3. **What is the idea of localized enhanced routing schemes?**
Modified mobility handling in local domains, Hop-by-hop routing in the local domain, Updating of individual routing tables in the local domain
- 21.4. **What is the idea of Proxy MIP?**
Support mobility management for mobiles without any specific mobility management support, Provide a transparent access to mobiles that looks like they are in their home network
- 21.5. **What is the purpose of optimizations in Hierarchical MIP?**

- Reduces the delays for handovers for the case of local movement,
Reduces the signaling load of the HA
- 21.6. Which modifications make sense?
Handling local handovers locally rather than globally in the HA,
Adding one or more hierarchies to the mobility management entities
- 21.7. Which statements hold for Cellular IP?
Special function, beyond standard IP, is needed in each router,
Router failures result in failures of the respectively served subtrees,
Increased load for forwarding table lookups
- 21.8. Why do mobility-related updates of the routing tables make sense for Cellular IP?
The update is limited to a single path in the network, The number of routing entries to be handled by each router is limited to the nodes served by its respective subtree, Due to the specific network structure, the number of required routing table updates is much smaller as in other networks

Transport Protocols

22. Quiz: Internet Protocol Suite

- 22.1. How does error handling work for the case that UDP is used?
Errors are ignored, Errors could be handled by the protocols run on top of UDP if needed
- 22.2. What are the main reasons that the TCP/IP model provides UDP as an alternative to the TCP transport protocol?
Different applications have different needs with respect to reliability of data, Different applications have different needs with respect to error control, UDPs transparency allows to solve transport issues in the application layer
- 22.3. What does the MTU mean?
Maximum size of the IP packet
- 22.4. What is the functionality of ARP?
Resolves the MAC address from a given IP address, Uses a local lookup where available, Issues a broadcast request to the network in case the IP address is not known locally
- 22.5. What is the functionality of DHCP?
Provides an IP address
- 22.6. What is the functionality of DNS?
Resolves the IP address from a given URL address, Issues a request to a known DNS server in case the IP address is not known locally
- 22.7. What is the purpose of encapsulation in the TCP/IP stack?
Stepwise addition of headers from the different layers, Stepwise addition of information needed for the transmission over a network

- to the application data, Stepwise addition of control information needed for hop-by-hop as well as end-to-end transmission
- 22.8. What is the purpose of ports used by transport protocols?
To identify the application instance in charge of the connection, To identify the application-specific protocol handling the segment
- 22.9. What is the purpose of the checksum field in the IP header?
Compute the checksum of the IP header
- 22.10. What is the purpose of the protocol field in the IP header?
Identify the protocol used on top of IP, To support the interpretation of the IP data part, To support the proper decapsulation of the packet at the receiver side
- 22.11. Why do we need multiple headers? Why not a single one?
To denote different concerns, e.g. addressing, error correction, flow control, avoidance of infinite forwarding loops, etc., To denote different aspects of the communication, To separate different concerns, e.g. hop-by-hop and end-to-end aspects

23. Quiz: TCP and TCP Congestion Control

- 23.1. How does a TCP sender actually learn about overload in the network?
No acknowledgement received at all within a specified time, Receipt of duplicated acknowledgements
- 23.2. How does the TCP congestion control algorithm in the sender react to overload problems? Mark all correct options?
Reduce the congestion window, Half the data rate of the transmission, Reduce the congestion window to the size of a single segment
- 23.3. What is the purpose of DUPACKs?
Indicates to the sender that a segment has arrived at the receiver, Indicates to the sender that the arrived segment is not in the right order, Indicates to the sender that a segment has arrived that is out of order
- 23.4. What is the purpose of the ACK flag?
Indicate if the segment includes information about acknowledgements, Indicate if the 32-bit acknowledgement in the segment should be interpreted or not
- 23.5. What is the purpose of the MSS?
Prevent fragmentation of the TCP packet on the first and last hop, To avoid the fragmentation of packets on intermediate routers, To inform the peer about the limits of its link interface
- 23.6. What is the purpose of the TCP checksum field?
It is used to check for errors incurred during the transmission, It allows to detect errors of the data included in the IP data field, It allows to detect errors occurring to the TCP header fields, It allows

- to detect errors in the address fields of the IP header, To discard TCP/IP packets that have incurred errors during transmission
- 23.7. What is the purpose of the window size field? Which of the statements hold for all situations?
To control the transmitter data rate, To indicate to the receiver of the segment how many data octets the sender is willing to accept, beyond the ones acknowledged in the acknowledged number field
- 23.8. Which statements hold with respect to a timeout raised at the sender? (single answer)
It indicates that there is a transmission problem in at least one direction
- 23.9. Why does TCPs response to timeouts differ from the response for DUPACKs?
A raised timeout indicates a severe congestion problem, A timeout indicates that there has not been a successful transmission during the duration of the timeout, A DUPACK indicates that the arrived segment is not in the right order, A DUPACK implies a minor congestion problem

24. Quiz: TCP over Wireless

- 24.1. Forced fast retransmit approach: How is slow start avoided after the change of the FA in wireless networks?
MH sends DUPACKs after the registration with a new FA
- 24.2. How is packet loss on the wireless link detected by snooping TCP?
FA detects packet loss using timeouts, FA detects packet loss by tracing TCP sequence numbers
- 24.3. Identify all wireless improvement schemes that employ additional buffers on the downlink.
Indirect TCP, Snooping TCP, LLC ARQ mechanisms
- 24.4. Identify all wireless improvement schemes that maintain the end-to-end semantics of TCP.
TCP transmission/timeout freezing, Forced fast retransmit, Selective transmission, LLC ARQ mechanisms
- 24.5. Identify possible approaches to alleviate the problem with wireless links?
Employ fast local retransmission mechanisms for the wireless link, Freezing of TCP with its settings in case the channel quality deteriorating, Ignorance of timeouts for the case of a poor radio channel, Increase of the RTO settings
- 24.6. Identify possible effects of high fluctuations of the channel capacity on TCP?
Increased delay jitter, Expiration of TCP transmission timeout, DUPACKs, Decreased throughput
- 24.7. Identify statements that hold for snooping TCP

- FA employs fast retransmission on wireless link, FA buffers, analyzes and selectively filters TCP packets
- 24.8. Indirect TCP: What are the advantages of splitting TCP connections in a wired and a wireless part?
The propagation of transmission problems on the wireless link towards the fixed part of the network is reduced, No changes for the TCP protocol on the wired host are necessary, Optimized TCP protocol can be used in a wireless part
- 24.9. What are the disadvantages of splitting TCP connections in a wired and a wireless part?
Acknowledgements received by the host in the fixed network do not indicate the reception of the packet by the wireless receiver, For the case of handovers, forwarding of the TCP state to the new network is required
- 24.10. What are the problems of the use of TCP over wireless links?
DUPACKs are not necessarily a result of a congestion, Timeouts do not necessarily indicate a severe congestion problem, TCP may reduce the data rate of the transmission even though there is no overload at all
- 24.11. What is the difference with the use of an ARQ mechanism over the wireless link compared to the case that LLC of the wireless link is empty?
ARQ will turn packet drops in increased delay and delay jitter, ARQ may reduce the number of DUPACKs in certain cases, ARQ will reduce the load on the wired part of the transmission
- 24.12. Which lower layer information could be useful to improve TCP over wireless?
Information about deterioration of channel quality, Information about an upcoming handover, Information about a temporary non-availability of a radio link, Information about the used MCS
- 24.13. Which lower layer information is used by TCP transmission/timeout freezing to improve TCPs performance?
Information about deterioration of channel quality prior to link failure, Information about an upcoming handover

25. Quiz: Beyond TCP

What are the advantages of moving congestion control mechanisms into the user space?

Flexibility, Adaptability to different needs, Adaptability to different network characteristics

What defines the bufferbloat problem?

Persistent high occupancy of large buffers in network entities resulting in increased latency and jitter

What defines the incast issues?

Multiple data servers sending data to a client surpass the ability of a clients' network entity to buffer packets, Synchronized data transmission from multiple data servers results in bursts of packet loss at a clients' network entity and consequently, poor TCP congestion control

What does HoL blocking problem in TCP mean?

Happens if multiple objects are transmitted over a single TCP connection, Loss of a packet in a TCP connection blocks delivery of subsequent received packets of the same TCP connection

What is the benefit of parallel data streams over multiple independent TCP connections?

Avoidance of head-of-line blocking, Support for redundant transmission, Increase of overall throughput

Which statements hold for multipath TCP?

Reduces load on congested links due to macro diversity, Increases throughput and reduces latency by using multiple TCP connections over different communication technologies, Packet retransmissions are handled independently per data flow, Head-of-line blocking problem can be experienced per data flow over a given communication link

Which statements hold for QUIC?

Congestion control is handled within the application layer, Solves head-of-line blocking problem by using UDP transport protocol, Simplified handshake procedure decreases the connection establishment latency in comparison to TCP

Which statements hold for SPDY?

Header compression is used to reduce required bandwidth, Solves head-of-line blocking problem by multiplexing data streams over multiple independent TCP connections, Packet retransmission is used whenever packets are received with bit errors from the lower layers

Quality of Service

26. Quiz: QoS Basics and Requirements

How can you compute the maximal throughput of an end-to-end connection from its constituents?

The maximal throughput is limited by the network element with the least throughput

What are the basic functions for the provisioning of QoS?

Admission control, Traffic classification, Traffic conditioning, Scheduling, Overload control

What are the main reasons for packet loss in wired communication?

Finite buffering capacity in routers, Finite link capacity

What are the main reasons for packet loss in wireless communication?

Unexpected deterioration of the wireless link quality

What are the QoS attributes of a connection?

Throughput, Packet loss rate, Delay, Jitter

What are the reasons for delay jitter in wired communication?

Queueing delays

What are the reasons of delay jitter in wireless communication?

Packet loss, System load, Varying channel quality

What contributes to the end-to-end delay of a TCP connection?

Processing delays experienced by each router, Transmission delays across each link, Queueing delays, Number of transmission errors, Transport capacity of the links

What is correct for the differentiation strategy?

It prioritizes packets by traffic class, The expedited forwarding strategy ensures QoS attributes for defined traffic classes

Which of the given applications has the most stringent QoS requirements?

Interactive games

27. Quiz: QoS Functions and Common Concepts

What are the ingredients of the TOS octet?

The priority of a packet, The requested tradeoff between throughput, delay, reliability and cost

What is correct for the token bucket scheme?

Enforces an average rate limit over the long run, Allows a limited degree of burstiness

What is the purpose of explicit packet dropping in routers?

To avoid congestion in the network in a proactive rather than reactive manner, To take advantage of the congestion control mechanisms of transport layer protocols, To force congestion control mechanisms of the transport layer to reduce the data throughput

Which function does traffic policing perform?

It drops packets to control the output rate

Which function does traffic shaping perform?

It buffers and queues packets exceeding the traffic specification

Which of the following mechanisms could help to reduce or avoid packet loss for higher priority traffic?

Increasing the end-to-end bandwidth, Using WRED, Policing the traffic flows

Which of the following will typically be used by WRED to determine dropping priority? (single answer)

Precedence bits in the TOS field

Which of the statements hold for metering and marking?

Marking builds on results from metering, Marking and metering are both needed for traffic shaping

Which protocols react to actions of the RED mechanism?

TCP

28. Quiz: QoS Strategies and Signaling

Why is RSVP called a soft-state protocol?

Because it treats a QoS reservation as temporary information that needs to be explicitly refreshed periodically, Because RSVP-enabled routers time out and release unused reservations

Why is IntServ not scalable?

Because of the need for each flow being handled and maintained by each router on the data path individually and specifically, Because of the need to identify for each arriving IP packet the flow it belongs to

Which transport layer protocols can be used for signaling in NSIS?

UDP, TCP, SCTP

Which of the statements hold for IntServ?

It provides the best separation between individual flows, Supporting IntServ in routers is the most costly technique to implement QoS, Core router processing for IntServ is more complex than for DiffServ

Which network strategy provides explicit end-to-end guarantees for individual flows?

IntServ

Which mechanisms are used to differentiate between different traffic classes in the DiffServ strategy?

Metering, Marking, Shaping, Packet dropping

Which kind of traffic would you map on the Expedited Forwarding PHB in DiffServ?

Conversational voice, Distributed gaming

What is correct for NSIS?

Supports a variety of signaling applications, Decouples the signaling application from the transport of the signaling messages, Supports receiver- and sender-initiated reservation, Supports virtual signaling networks

Identify necessary conditions for overprovisioning to be applied?

The maximum traffic imposed on the network needs to be known, The maximum traffic in each router needs to be known, The maximum traffic over each link needs to be known

Identify characteristics of the DiffServ QoS strategy?

Low overhead in routers per IP packet due to the small number of states

How does RSVP handle the case that a router is not supporting it?

The RSVP messages are transparently passed through without being handled, The RSVP messages are forwarded by the router, but no QoS reservation takes place

How can we ensure that DiffServ works over multiple networks and multiple providers?

Providers need to agree on service level agreements between networks, Providers need to agree on the data volumes and on how to handle different traffic classes

For which of the following signaling purposes can NSIS be used?

Dynamic reservation of resources, Mobility management, Status information about network entities, Firewall configuration

29. Quiz: QoS over Wireless

30. Why is QoS considered a cross-layer issue?

Multiple layers need to cooperate in order to provide a certain QoS

Lower layers provide feedback to higher layers to optimize QoS

Higher layers control the settings of lower layers to achieve given QoS requirements

QoS needs many protocol entities to be configured in a proper way

The weakest component in the communication system limits the achieved QoS

31. Why is increasing the transmit power not always a good idea?

Because of increased interference within the network

Because of increased energy consumption

Because of the increased level of radiation

Because of a decrease of the SINR of other mobiles

32. Which of the following PHY strategies are appropriate to provide a constant bit rate with invariant error rate over a varying wireless channel?

Adapt the transmit power to maintain a constant SINR

Adapt the modulation and coding scheme along with the used channel bandwidth to ensure that QoS requirements are met

Adapt the time the channel is used along with the modulation and coding scheme to ensure that QoS requirements are met

33. Which of the following PHY strategies are appropriate for the minimization of the error rate?

Higher FEC redundancy, Higher SINR

Increase of the transmit power

Use of a directional antenna instead of an omnidirectional antenna

34. Which of the following PHY strategies are appropriate for the maximization of the throughput (goodput) for a given SINR?

Select the modulation and coding scheme that balances BLER and raw data rate such that the maximum throughput is achieved

35. Which of the following measures can be used to considerably increase the system capacity (throughput over all mobiles) of an interference-limited cellular system as a whole?

Increase the bandwidth of the employed frequency band

Prefer mobiles with better SINR over mobiles with low SINR

Increase the number of base stations and base station locations

Use highly directional antennas

36. What is the benefit of reducing frame sizes in wireless networks?

Lower probability of frame errors

Reduced data volume needed for retransmissions

37. Name measures that may help to maximize the throughput (goodput) over wireless links; please consider all cases!

Higher-order modulation schemes

Higher channel bandwidth

Higher SINR

Higher transmit power

38. Assuming an invariant PHY configuration, what may influence the QoS achieved over a wireless link?

Varying received signal strength

Varying pathloss to sender

Varying pathloss towards interferers

Varying interference

39. Quiz: QoS in 802.11

Does EDCA provide any hard QoS guarantees?

No

How do EDCA and HCCA react to changing channel conditions?

They do not take channel conditions into consideration

In the legacy DCF model, once a mobile station has access to the medium, it is able to keep sending frames as long as required. Which mechanism is designed to prevent this behavior in EDCA?

TXOP

What is the reason to introduce variable AIFSSs in EDCA?

Differentiation between traffic classes in a single mobile, Differentiate the buffer waiting time for different traffic classes, Differentiate the medium access times between different traffic classes, Provide a scheme to roughly differentiate traffic classes in a distributed system

What is TXOP?

A set period of time when a wireless station may send as many frames as possible without having to contend with other stations

Which MAC-layer technique for QoS provisioning would you select to provide guaranteed QoS in an 802.11 network?

HCCA

Which mechanisms are required to be implemented for HCCA on the AP side?

Admission control, Shaping, Scheduling

40. Quiz: QoS and Mobility

What are the conflicts between mobility management and QoS management?

Mobility may result in changes to the communication path which renders QoS reservations obsolete, QoS management needs to be triggered after any handover, New routes appearing after a handover may not support the required QoS

Which of the statements hold for the tight integration of QoS and mobility management?

Changes due to mobility do not require changes in the QoS settings of the complete path, Reduced communication gaps during handovers, Mobility mechanisms may inform QoS management in advance to prepare for a new path or parts of it before the handover, Violates the principles of the OSI model

41. Quiz: QoS over Wireless

Why is QoS considered a cross-layer issue?

Multiple layers need to cooperate in order to provide a certain QoS, Lower layers provide feedback to higher layers to optimize QoS, Higher layers control the settings of lower layers to achieve given QoS requirements, QoS needs many protocol entities to be configured in a proper way, The weakest component in the communication system limits the achieved QoS

Why is increasing the transmit power not always a good idea?

Because of increased interference within the network, Because of increased energy consumption, Because of the increased level of radiation, Because of a decrease of the SINR of other mobiles

Which of the following PHY strategies are appropriate to provide a constant bit rate with invariant error rate over a varying wireless channel?

Adapt the transmit power to maintain a constant SINR, Adapt the modulation and coding scheme along with the used channel bandwidth to ensure that QoS requirements are met, Adapt the time the channel is used along with the modulation and coding scheme to ensure that QoS requirements are met

Which of the following PHY strategies are appropriate for the minimization of the error rate?

Higher FEC redundancy, Higher SINR, Increase of the transmit power, Use of a directional antenna instead of an omnidirectional antenna

Which of the following PHY strategies are appropriate for the maximization of the throughput (goodput) for a given SINR?

Select the modulation and coding scheme that balances BLER and raw data rate such that the maximum throughput is achieved

Which of the following measures can be used to considerably increase the system capacity (throughput over all mobiles) of an interference-limited cellular system as a whole?

Increase the bandwidth of the employed frequency band, Prefer mobiles with better SINR over mobiles with low SINR, Increase the number of base stations and base station locations, Use highly directional antennas

What is the benefit of reducing frame sizes in wireless networks?

Lower probability of frame errors, Reduced data volume needed for retransmissions

Name measures that may help to maximize the throughput (goodput) over wireless links; please consider all cases!

Die richtigen Antworten sind: Higher-order modulation schemes, Higher channel bandwidth, Higher SINR, Higher transmit power

Assuming an invariant PHY configuration, what may influence the QoS achieved over a wireless link?

Varying received signal strength, Varying pathloss to sender, Varying pathloss towards interferers, Varying interference

Self-Organization

42. Quiz: Self-Organization - Definition, Properties and Design

Identify advantages of self-organized systems

Scalable by design, Adaptable to changes in the structure, Robust to failures of almost any number of entities, Intrinsically parallel

Identify characteristics of self-organized systems

May show oscillating behavior, Adaptive behavior of the system as a whole

Identify important design principles of self-organized systems

Positive feedback, Negative feedback, Randomness, Interaction with neighbors

Identify important paradigms for the design of self-organized mechanisms for communication protocols

Design each entity that it can cope with changes, Design each entity that it can adapt its parameters to changes, Design each entity that it employs some fallback in the case that there are major changes of circumstances, Design each entity to monitor its local environment and to adapt to it, Design each entity to coordinate with other entities sharing some notion of neighborhood

Identify limitations of self-organized systems

Systems are hard to predict and evaluate, Systems are hard to test, The impact of single entities on the system as a whole is limited, Local entities are limited in their knowledge of the global system state

Identify properties of self-organized systems?

Absence of an explicit global state, Distributed coordination, Communication limited to entities which share some notion of neighborhood, Some randomness of algorithms

Identify properties that are prohibitive in self-organized systems

The use of a single database, The use of master-slave structures, The use of hierarchies

Identify the design paradigms for algorithms of self-organized systems

Design algorithms to achieve the desired global properties, Design algorithms that quickly adapt to changes, Design adaptive algorithms that can handle a multitude of situations

What is emergent behavior?

Sudden appearance of unforeseen behavior, Emergence of leaders in the structure, Appearance of global structures from local actions

Emergent behavior of a system is provided by the apparently meaningful collaboration of components (individuals) in order to show capabilities of the overall system (far) beyond the capabilities of the individual components.

Which of the following signs indicate that the system is not self-organized?

Some entities direct the activities of the system by providing each entity with detailed instructions about what to contribute to the overall goal

43. Quiz: Self-Organization - Applications

Why is TCP congestion control considered as self-organized?

TCP congestion control detects network congestion indirectly through packet loss, TCP congestion control regulates the transmission rate dynamically using feedback from its communicating peer, TCP hosts only take local decisions, TCP congestion control utilizes negative feedback

Which of the given economic and social system is closest to a self-organized systems?

Societies following Kants categorical imperative

Which of the functions of 802.11 implements the principles of self-organization?

EDCA access strategy, DCF access strategy, Beaconing in ad hoc networks

Which function of TCP follows self-organization principles to achieve a global goal?

Congestion control

What is negative feedback in 802.11 DCF?

Collision, Absence of ACK, Busy medium

What is negative feedback employed in TCP congestion control?

DupACKs, No feedback from receiver, Timeout raised at the sender

What are the probabilistic aspects in 802.11 DCF?

Random backoff time, Randomness of distributed availability of packets, Randomness of occurrence of collisions, Packet loss due to varying propagation characteristics

What are the causes of randomness in the operation of CSMA/CA?

Distributed generation of MAC frames, Random backoff time, Movement of mobile terminals

Provide examples for self-organized human and biological systems

Discussion with your buddies in a pub, Ants building a colony

Identify the effects that introduce randomness into TCPs congestion control mechanism

Distributed generation of packets, Impact from other TCP connections, Impact from UDP connections, Queuing delay in routers, Delay by underlying lower layer communication

How can we limit the implosion effect for TCP traffic?

Apply RED in routers, Apply WRED in routers

44. Quiz: Self-Organization - Challenges and Limits

45. What are the properties of systems operating close to its critical point?

- Potentially very high efficiency of operation, High risk for the case of unforeseen situations
46. What are the challenges of the development of self-organized systems and algorithms?
Testing in the presence of stochastic behavior, High efforts for evaluation due to complex system behavior, Stability issues, Understanding of the distributed behavior, Understanding of the global effects of local algorithms
 47. Criticality: What are the effects of RED and the queue size of routers on criticality?
Early dropping smooths the operation of TCPs congestion control algorithm, RED and the router queues should be designed to support a fast but smooth reaction to overload
 48. Criticality: Identify changes in the design of the TCP congestion control algorithm and RED to minimize the risk of the appearance of critical points
Use a less aggressive increase of the data rate upon the receipt of ACKs, Quadruple the retransmission timer after a timeout, Reduce the packet drop probability thresholds to apply RED or WRED to TCP traffic, Always employ the congestion avoidance strategy instead of slow start
 49. Criticality: Identify changes in the design of the TCP congestion control algorithm to provoke instability (self-organized criticality)
Half the congestion window upon the receipt of a timeout, Eliminate the reduction of the congestion window upon the receipt of DupACKs, Leave the retransmission timer unmodified after a timeout, Always employ slow start instead of the congestion avoidance strategy

Ad-hoc Networks

50. Quiz: Ad-hoc Networks - Basics

Pros and cons of reactive and proactive routing protocols? Consider different application scenarios.

Proactive protocols have lower latency for data delivery due to priori setup of routing information, For low data traffic, proactive protocols consume more network resource than reactive protocol due to network-wide update of routing information

What are the advantages of the use of ad hoc networks compared to infrastructure-based networks?

Neighboring nodes may communicate with lower transmit power compared to the case that an infrastructure is used, thus saving batteries, There is no "single point of failure", High robustness towards single node failures

What is the idea of greedy geographical routing? What type of information is needed to apply geographical routing protocols in ad hoc networks?

Routing decisions are made on a hop-by-hop basis, Periodic or on-demand beaconing is required to detect any changes in the network topology, Geographical location information is needed for each node to achieve routing

Which of the following applications may make use of ad hoc networks?

Vehicle-to-vehicle communication, Communication among emergency forces in disaster scenarios like fire, earthquakes, etc., Extension of the coverage of cellular networks, Military communications between armed forces

Which of the following statements characterize mobile ad hoc networks?

Nodes organize themselves in a decentralized way, There is no fixed network infrastructure involved, The nodes can be mobile, Each node contributes to the establishment of routes

Which of the following statements hold for proactive routing in ad hoc networks?

Each node maintains routing entries for all nodes independent of its need, Nodes maintain routing table entries periodically

Which of the following statements hold for reactive routing in ad hoc networks?

Routes between any source and destination are only established when needed, Nodes update their routing tables based on demand, Routes are established by signaling packets prior to their use for data packets

Which statements hold for delay-tolerant networking?

DTNs allow nodes without a continuous connectivity to exchange data, DTNs may employ broadcasting to improve reliability, DTNs may employ broadcasting to reduce average delivery latency

51. Quiz: Ad-hoc Networks - AODV

How are link failures detected when using the AODV protocol?

Lack of MAC-layer acknowledgements, Absence of 'hello' messages from some nodes along an established route, Upon the receipt of RERR messages

How does a node behave when it receives more than one identical RREQ?

It only forwards the first RREQ

Identify correct statements about the routes established using AODV?

Routes are discovered and maintained when necessary, Multiple routes are explored between a source and a destination, but only a single route is maintained for data exchange, Established routes are bidirectional (same route in forward and return direction)

Self-organization: What is the positive feedback employed by AODV route discovery and maintenance with respect to a specific route?

Receipt of RREP, Receipt of ACKs on link layer, Receipt of data packets

What happens when a node receives multiple RREPs for a given destination during route discovery?

The node forwards the first RREP it receives, The node forwards another RREP if it has a greater destination sequence number

What is the purpose of sequence numbers in the route discovery process of AODV?

To monitor the freshness of a route towards a given destination, To prevent the formation of loops during the route discovery

What is the purpose of the RERR message?

Notifies neighbors of a node about non-availability of a destination for which a route has been available so far, Notifies the source node of the need to reinitiate the route discovery process, Distributes a list of all destinations that are unreachable due to a link failure

What is true for RREQ and RREP messages?

RREQ is a broadcast message while RREP is a unicast message, For a given network topology, RREQs can take several routes towards the destination while RREPs take a specific route, RREQ is initiated by the source node while the RREP can be initiated by any other node along the route to the destination node.

When is the hop count in the RREQ message of AODV changed?

With every hop the RREQ traverses

When is the originator sequence number in the RREQ message of AODV changed?

Whenever a new route discovery is initiated by the originator

Which of the following events result in the initiation of the route discovery process?

A source node having data for a given destination but without any routing information in its routing table for the destination node, Expiration of a routing entry in any of the constituent nodes along the route between source and destination node while the source node has data to send to the destination node, A source node not being able to deliver data to a given destination node

Which statement holds for the establishment of the routing entries in the nodes' routing tables with AODV?

Entries for the forwarding path from originator to destination are established with the receipt of RREPs, Entries for the reverse path from destination to originator are established by RREQs

Cognitive Radio Networks

52. Quiz: Cognitive Radio Networks - Basics

What are the main tasks of cognitive radios?

Coordination of the medium access with other users, Determination of unused portions of the spectrum, Detection of the activity of licensed users

What is meant by "spectrum hole" in CR?

Unused portion of the radio spectrum

What is meant by the primary network?

The network with priority to utilize the radio resources, The network operated by the owner of the radio spectrum

What is the motivation for cognitive radios?

Improvement of the utilization of radio spectrum, Increase of the overall capacity for radio communication due to the temporary reuse of otherwise unused spectrum, Reuse of unused TV bands or military bands

Why do Cognitive Radios need to be reconfigurable?

To be able to transmit data over a variety of frequencies, To be able to use various radio technologies, To adapt to the current radio environment situation

53. Quiz: Cognitive Radio Networks - Spectrum Sensing

Identify correct statements about energy detection sensing technique for detection of primary users?

Energy detection looks for a specific energy level over a chosen period of time regardless of the shape of the signal, Energy detection can be used in case the characteristics of the primary user signal is unknown

Identify correct statements about matched filter sensing technique for detection of primary users?

Matched filter provides accurate detection of primary users if all characteristics of primary user signal are known, Matched filter works well for noisy input signals

In which of the following cases can primary and secondary users simultaneously access the same frequency band?

If the secondary user transmitter is very far from the primary users, If the interference created for the primary user is below a specified threshold

What are the advantages of cooperative sensing?

It helps secondary users to derive a better picture of the interference situation, It helps secondary users to avoid creating interference to primary users, It helps to utilize the spectrum more effectively

What is the idea of cyclostationary feature detection?

Autocorrelation of the signal with itself to identify its periodicity, Identification of primary network activity by differentiation of the signal from random noise

What is the idea of interference-based detection and interference temperature management?

Ensure that secondary users do not exceed a predefined interference level at primary receivers, Allow a defined level of extra interference caused by secondary networks

Which of the following primary user signal characteristics are essential for the matched filter transmitter detection scheme?

Operational frequency, Modulation scheme

Why is the Hidden Terminal Problem relevant for CR networks?

Because the interference created towards primary users needs to be minimized, Because the secondary user may not detect active primary user transmitters

54. Quiz: Cognitive Radio Networks - Spectrum Decision, Sharing and Mobility

For which kind of cognitive radio approaches is the application of spreading and CDMA especially useful? (single answer)

For underlay networks

In which of the following situations does a secondary user need to perform spectrum handoff?

Degraded channel conditions for the secondary user, Higher QoS requirements by the secondary user application than can be provided on the current channel, Secondary user detects the appearance of the primary user on the current channel, Secondary user enters a new area, where available spectrum bands are different

Is it possible to provide strict QoS guarantees for secondary user applications in CR networks?

No, because the primary user has priority over a licensed spectrum

What are the advantages of the application of spreading technology to cognitive radios?

Spreading increases the used channel bandwidth which in turn allows to increase the user throughput without an increase in the interference level, Spreading allows for a reduction of the interference level without a reduction of the user throughput, Spreading for a given data rate allows to minimize the impact from and on low bandwidth primary users

What is the difference between overlay and underlay spectrum access technology in CR?

Assuming perfect sensing, interference to primary users is minimized in overlay spectrum access technology, In overlay secondary users access only unused spectrum holes, In underlay secondary users stay within acceptable interference raise at primary users

Which statements are true for overlay spectrum sharing?

Secondary users wait for the availability of spectrum holes to access spectrum, Ideally, there should not be interference between primary and secondary users during spectrum sharing, Increase in the number of active primary users typically decreases the spectrum available for secondary users

Which statements are true for underlay spectrum sharing?

Secondary users may simultaneously share spectrum with primary users as long as their resultant interference is within the interference temperature

55. Quiz: Cognitive Radio Networks - CR MAC Protocols

What are the advantages of multiple radios supported by a cognitive radio device?

Easy separation between common control channel and data channel, Parallel operation of control channel and data channel, Faster handling of control information

What are the contents of Information Event message in SYN-MAC protocol?

List of channels available to the secondary user

What is the architectural difference between CR centralized MAC and CR ad-hoc MAC?

CR ad-hoc MAC protocols rely on cooperation between neighbors to gain access to available channels, CR centralized MAC protocols use a central entity, which manages and coordinates operations among secondary users

What is the difference in the CSMA-MAC protocol for cognitive radio networks?

Secondary users can still transmit RTS even when the channel is sensed as busy, The secondary user's transmit power is adapted to stay within the allowed interference temperature

What is the functionality of a common control channel in CR?

Connection establishment between secondary user transmitter and receiver, Exchange of sensing information among secondary users, Communication with a central entity organizing the spectrum allocation

What is the reason of not operating the common control channel on a fixed frequency?

The channel should be vacated when a primary user appears on the channel, The channel quality variations can result in connection establishment problems

Why does SYN-MAC require two radios?

To listen to control messages and to send data simultaneously, To be able to detect the appearance of primary user while transmitting

56. Quiz: Cognitive Radio Networks - SDR

Identify advantages and disadvantages of SDRs

Poorer performance with respect to receiver sensibility and noise, Higher power consumption compared to fully customized approaches, High flexibility to implement different radio technologies

What is the idea behind Software-Defined Radio (SDR)?

SDR can easily be reconfigured using software, Most of processing logic is implemented in software, SDR uses generic HW components, such as DSPs, FPGAs or GPPs to facilitate the flexible use of radio spectrum

Which components of SDR are typically realized in software?

Modulation and demodulation, Baseband signal processing

Cellular Networks

57. Quiz: Cellular Networks - GSM, GPRS, EDGE

Why is the same frequency channel typically not used in neighboring cells in GSM?

Because it would result in too much interference at mobile receivers, Because it would result in too much interference at base station receivers, Because it would increase the error rate of transmitted data

Which scheme for the separation of uplink and downlink traffic is used by GSM and what are the reasons for this?

FDD because voice calls generate symmetric traffic in uplink and downlink

Which multiplexing techniques are used in GSM networks

TDMA, FDMA, SDMA

Which kind of access scheme is employed for GSM voice communication?

Reservation of fixed time slots for the duration of a complete call

What is the purpose of different modulation and coding schemes in EDGE?

Provide higher data rates to mobiles with excellent radio conditions, Provide mobiles with poor radio conditions with a robust communication, Adapt the modulation and coding to the specific radio conditions of the mobile, Maximize the spectral efficiency of the cell

What is the difference between make-after-break and make-before-break?

Make-before-break means that the network tries to initiate and complete the handover before the old link to the mobile fails, Make-before-break means that the network acquires the resources in the new base station or cell before the old resources are actually released

What is the difference between circuit and packet switched networks?

Circuit switched networks guarantee fixed data rates, Circuit switching does not rely on buffers, Circuit switched systems has less end-to-end delay jitter than packet switched systems

What are the reasons to use two different core networks in order to support voice and data?

Each network is optimized for its specific use, Both networks differ considerably in the their signaling procedures and the handling of packets and QoS requirements

Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GPRS/EDGE

With reuse factor of 3, more robust modulation schemes will be combined with robust coding to combat higher interference compared to reuse 9 setups, With reuse factor of 9, higher transmit power may be used to maximize system capacity

Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GSM voice

With reuse factor of 3, the voice capacity is higher compared to reuse of 9

How is the integrity of the shared key ensured?

The key is not exchanged at all

How does GSM deal with movements of mobiles between cells in the case that there is no active call?

There is no interaction with the infrastructure as long as the mobile does not move out of its location area

How does authentication work in GSM? Who or what is authenticated?

The mobile authenticates itself with the system, The authentication employs a challenge-response protocol

Handovers in GSM: What's the difference to handovers in WLAN systems?

In GSM handovers are controlled by base stations rather than mobiles, GSM handovers are initiated earlier than WLAN handovers, GSM supports seamless handovers while WLAN does not, GSM handovers are initiated proactively

Data over GPRS: Identify correct statements

Several TDMA slots may be combined to increase the data rate for single mobiles, Each TDMA slot is multiplexed between multiple mobiles in order to maximize the channel usage

58. Quiz: Cellular Networks - UMTS

Which multiplexing techniques are used in UMTS networks?

CDMA, SDMA

What does cell breathing in UMTS mean?

Increase of the interference due to high load causing the effect of reduced cell size, Reduced effective cell size due to drop of required SINR below minimum required threshold at cell edges, Cell breathing is present in systems where the broadcast channel interferes with data channels

What defines macro diversity?

A mobile receives the same data from two or more different base stations at the same time, The data transmitted by the mobile is received and processed by more than one base station

UMTS QoS management...

... supports QoS within the UMTS eco system, ... provides multiple QoS management entities to support end-to-end QoS in a hierarchical manner

Mobility management in UMTS

Employs bidirectional tunnels (joint up- and downlink traffic), GTP-U is responsible for mobility and QoS management, UMTS employs a tunneling solution to mobility management, UMTS supports macro and micro mobility using a hierarchy of tunnels

Identify possible benefits of macro diversity in UMTS?

Reduction of the transmit power of the mobile without loss in quality, Combining multiple received signals to achieve a gain in quality, Improved reliability in the uplink, Improved reliability in the downlink

How does macro diversity work in UMTS?

Reuse of the same frequency in neighboring cells, A mobile receives downlink signals from two or more different base stations at the same time, Receipt of the same uplink signal by more than one base station at the same time

59. Quiz: Cellular Networks - LTE

Identify statements that hold for interference cancelation in LTE

Replaces the standard reuse of 1 for LTE by a more flexible scheme for different subareas of the cell, Increases the system capacity, Reduces interference for neighbor cells

Identify the benefits of opportunistic scheduling

Increased system capacity

Identify use cases for self-organized network management in LTE

Energy saving, Mobility robustness optimization, Coverage and capacity optimization

What are the benefits of beamforming?

It may increase the area covered by a base station, It allows to increase the capacity of the cell, It allows to reduce the interference to other users, It may increase the throughput for a user

What is the advantage of a common IP-based packet core compared to the use of separated networks for packet-switched and circuit-switched traffic?

Reduced CAPEX due to economy of scale, Reduced number of network elements to be configured and managed

What is the benefit of SU-MIMO?

It allows to increase the capacity of the cell, It increases the maximum throughput for a user